Reinforcement-Learning Framework for Robust and Universal Analog Quantum Control on Superconducting Qubits

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Chapter 1

Introduction

1.1 Quantum control for near-term quantum computing

Quantum computing is ultimately a control problem as much as it is a computational one. At the abstract level, a quantum algorithm is formulated as a sequence of logical unitary operations acting on qubits; at the hardware level, however, those operations must be realized through continuous-time dynamics generated by experimentally accessible Hamiltonians. The practical power of a quantum device therefore depends not only on the existence of an algorithmic advantage, but also on the ability to engineer control protocols that implement the required gates with sufficiently high accuracy, sufficiently short runtime, and sufficient robustness against physical imperfections [**?**, **?**]. This issue is especially acute in near-term and pre-fault-tolerant quantum processors, where limited coherence times, calibration uncertainty, waveform distortions, and multilevel effects strongly constrain the depth and reliability of executable circuits [**?**, **?**]. In this regime, quantum control is not a secondary layer of fine-tuning but a central bottleneck linking the abstract circuit model to the actual computational capacity of the hardware. For this reason, understanding how gate fidelity, total runtime, leakage, and robustness compete with one another is essential for assessing the practical viability of analog gate design, as will be discussed in Sections 2.1.2, 2.2.2, 2.2.3, and 2.3.3. Within this broader context, the reference work analyzed in this thesis is particularly relevant because it proposes a unified framework in which these competing objectives are encoded into a single optimization problem and solved through reinforcement learning on a realistic superconducting-qubit platform [**?**]; its main ideas and results will be introduced in Sections 3.2.3 and 3.4.3.

1.2 Direct analog gate implementation versus compiled gate synthesis

A central question in quantum gate design is whether a target unitary should be implemented through a compiled sequence of elementary gates or realized directly through a single analog control protocol. In the standard circuit-model view, universality is achieved by expressing a desired operation as a product of gates drawn from a finite universal set, an approach that is indispensable in fault-tolerant architectures and remains the dominant abstraction in quantum algorithm design. However, on near-term hardware this decomposition is not neutral from a physical perspective: every additional elementary gate consumes runtime, increases exposure to decoherence and control noise, and may introduce further leakage or calibration error. As a result, a circuit that is optimal or near-optimal at the logical level may still be inefficient at the hardware level once translated into pulses and executed on a concrete device [**?**, **?**, **?**]. For pre-fault-tolerant platforms, this observation motivates a different perspective in which one seeks to implement the target unitary directly through continuous-time Hamiltonian evolution, thereby

exploiting the native interaction structure of the hardware rather than forcing the operation through a fixed discrete gate alphabet.

This distinction between compiled synthesis and direct analog realization is particularly relevant for the family of two-qubit gates considered in the reference paper. There, the authors explicitly argue that unrestricted analog control can reduce the resource overhead associated with long synthesized circuits, and they benchmark their learned controls against optimal gate-synthesis counterparts precisely to quantify this hardware-efficiency advantage [**?**]. The significance of this comparison is not only algorithmic but physical: if a target unitary is well aligned with the natural couplings of the device, then a pulse-level implementation may achieve the same logical transformation with substantially shorter runtime than a decomposition into standard one- and two-qubit primitives. This idea will be developed more systematically in Sections 2.1.3, 3.1.1, and 3.4.1, and will reappear in the numerical part of the thesis when the reproduced analog controls are compared against gate-synthesis baselines in terms of total gate time; see in particular Sections 3.4.2 and 3.4.3. In this sense, the present thesis is motivated not only by the general problem of gate optimization, but also by the more specific question of when direct analog quantum control can outperform compiled gate implementations on realistic superconducting hardware.

1.3 Practical limitations: leakage, stochastic errors, and robustness

The promise of direct analog control is accompanied by equally important practical difficulties. Real quantum devices are not ideal few-level systems manipulated by perfectly calibrated controls, but open and imperfect physical platforms subject to multilevel dynamics, stochastic fluctuations, and partial model uncertainty. In superconducting circuits, in particular, the computational qubit subspace is embedded in a larger Hilbert space associated with higher oscillator levels, so fast or strongly modulated controls can drive population outside the logical sector and generate leakage errors. At the same time, the implemented control waveforms are affected by amplitude noise, parameter drift, transfer-function distortions, and calibration mismatch, all of which can substantially degrade the quality of the final gate if they are not accounted for during optimization [**?**, **?**, **?**, **?**]. These non-idealities imply that achieving high nominal fidelity alone is not sufficient: a practically useful control protocol must also remain stable under realistic perturbations and must suppress leakage while preserving short runtime.

The reference paper is motivated precisely by the lack of a unified framework capable of treating these competing requirements simultaneously. Its starting point is that leakage and stochastic control errors should not be regarded as secondary corrections to an otherwise solved nominal-control problem, but as structural features of the optimization landscape itself [**?**]. This observation is central for the present thesis as well. Indeed, much of the background developed later, from the multilevel description of superconducting qubits to the definition of computational and leakage subspaces, from the speed–leakage tradeoff to the discussion of robustness under uncertain perturbations, serves to make this point precise; see Sections 2.2.1, 2.2.2, 2.2.3, and 2.3.3. It is exactly because these physical limitations are intertwined that the paper introduces an analytic leakage treatment, a multi-objective cost function, and a stochastic training environment within a reinforcement-learning framework. These ingredients, which will be discussed in Sections 3.2.2, 3.2.3, and 3.3.4, provide the conceptual basis for the experiments of this thesis, including the reproduction of the original robustness results and the out-of-distribution robustness tests proposed in the extension part of the work.

1.4 The reference framework: *Universal Quantum Control through Deep Reinforcement Learning*

The present thesis takes as its main point of departure the framework introduced in *Universal Quantum Control through Deep Reinforcement Learning* by Niu *et al.* [**?**]. The central idea of that work is that realistic quantum gate design should be formulated as a genuinely multi-objective control problem, in which gate accuracy, runtime, leakage suppression, and robustness against stochastic control errors are optimized jointly rather than sequentially or in isolation. Instead of treating leakage and control noise as secondary effects to be evaluated only after a nominal pulse has been found, the paper incorporates them directly into the optimization target through a unified control objective. This is a conceptually important step, because it shifts the emphasis from finding merely high-fidelity nominal solutions to finding controls that remain physically meaningful under the non-ideal conditions relevant to superconducting quantum hardware.

To realize this program, the paper combines three main ingredients. First, it introduces an analytic treatment of leakage based on the time-dependent Schrieffer–Wolff transformation, allowing the dominant leakage mechanisms to be organized into a perturbative bound that can be used during optimization. Second, it defines the so-called Universal cost Function control Optimization (UFO) objective, which augments gate infidelity with explicit penalty terms for leakage, boundary-condition violation, and total runtime. Third, it casts the resulting control problem into a continuous-action reinforcement-learning framework and trains the control policy with trust-region policy optimization in a stochastic environment designed to mimic experimentally relevant control fluctuations [**?**]. The framework is implemented on a two-qubit superconducting gmon architecture and benchmarked on a structured family of two-qubit gates that are relevant both from the viewpoint of hardware-native interactions and from applications in quantum simulation.

This reference framework is particularly suitable as the foundation of the present thesis for two reasons. On the one hand, it is physically rich: it does not rely on an idealized two-level approximation, but explicitly addresses multilevel dynamics, leakage channels, and stochastic perturbations in a realistic superconducting-qubit setting. On the other hand, it is methodologically modular: its physical model, leakage analysis, cost function, reinforcement-learning formulation, and evaluation strategy can each be reproduced, tested, and extended independently. For this reason, the work developed in this thesis is organized around a two-stage objective. The first is to reconstruct and validate the main ingredients and key numerical results of the original framework. The second is to extend that framework along directions not fully explored in the original paper, including out-of-distribution robustness tests, comparisons against more standard quantum optimal-control baselines, and a systematic study of the role played by the different terms of the UFO objective. The technical content of the framework itself will be developed in detail in Sections 3.1.2–3.4.3, while its reproduction and extension constitute the core of the empirical chapters of this thesis.

1.5 Objectives and scope of this thesis

The aim of this thesis is to study, reproduce, and extend a recent reinforcement-learning-based framework for robust analog quantum control in superconducting qubits. More specifically, the work takes the control strategy proposed in Ref. [**?**] as its main methodological reference and investigates both its physical content and its empirical performance on a realistic multilevel control problem. The scope of the thesis is therefore twofold. On the one hand, it is a *reproduction study*: the physical model, the leakage-aware control objective, the reinforcement-learning formulation, and the principal numerical benchmarks of the original paper are reconstructed in a consistent simulation pipeline. On the other hand, it is an *extension study*: once the original framework has been validated, additional experiments are carried out to test its robustness beyond the

specific setting considered in the reference work.

The first objective is thus to reproduce the main results of the original paper as faithfully as possible. In practical terms, this means implementing the gmon control model, the UFO-inspired optimization objective, and the learning environment described in Ref. [**?**], and then verifying whether the principal reported trends can be recovered. The core reproduction targets are the comparison of analog gate times against synthesized-gate baselines, the behavior of average gate fidelity as a function of control-noise strength, and the corresponding fidelity variance under stochastic perturbations. These results are particularly important because they are the main empirical support for the central claims of the paper, namely that noisy-environment reinforcement learning can improve both robustness and hardware efficiency relative to simpler baselines [**?**]. In the structure of this thesis, this first objective motivates the material developed in Sections 3.1.1–3.4.3 and culminates in the reproduction experiments reported later in the empirical part.

The second objective is to examine the limits and generality of the original framework. While Ref. [**?**] demonstrates strong performance under Gaussian amplitude noise, it leaves open several broader questions about the nature of the learned robustness and about the comparison with more standard quantum optimal-control techniques. This thesis therefore extends the original study in several directions. In particular, it asks whether a controller trained under one stochastic perturbation model can generalize to qualitatively different perturbations, including quasi-static drift, colored noise, parameter mismatch, and decoherence. It also asks how the reinforcement-learning strategy compares with more traditional quantum optimal-control methods when the Hamiltonian, constraints, target family, and evaluation metrics are kept fixed. Finally, it investigates how the different weights in the UFO objective influence the tradeoff between fidelity, leakage, boundary behavior, and runtime. These questions define the scope of the thesis beyond reproduction: the goal is not merely to restate the original results, but to evaluate how robust, transferable, and interpretable the framework remains when tested outside its original domain of validation.

It is also important to state what the thesis does *not* attempt to do. The purpose is not to provide a complete survey of all reinforcement-learning methods for quantum control, nor to derive a universally optimal alternative to the reference framework. Likewise, the thesis does not aim to perform a full experimental hardware implementation on a superconducting device. Instead, its contribution is more focused: it develops a physically grounded simulation-based assessment of the framework of Ref. [**?**], verifies its main claims in a reproducible setting, and then probes its behavior under controlled extensions that are scientifically motivated by open questions on robustness and control generalization. In this sense, the thesis is positioned at the intersection of reproduction, validation, and targeted methodological extension.

1.6 Research questions

The general objectives outlined above can be formulated more precisely in terms of a small number of concrete research questions.

- The first question concerns *reproducibility*: can the main physical and numerical claims of the original UFO framework be recovered within an independently implemented simulation pipeline based on the same effective Hamiltonian, target gate family, and control metrics? This question is fundamental because the scientific value of any proposed control framework depends not only on its conceptual elegance, but also on the reproducibility of its reported gains in gate speed and robustness. In the context of the present thesis, this question is addressed through the reconstruction of the original control setting and through the experiments devoted to reproducing gate-time comparisons, average-fidelity curves, and fidelity-variance behavior under stochastic control noise.

- A second question concerns *robustness generalization.* The original paper optimizes control policies in the presence of Gaussian amplitude fluctuations and evaluates them under the same family of perturbations [**?**]. This naturally raises the question of whether the resulting robustness is specific to that training distribution or whether it generalizes to qualitatively different perturbation mechanisms. Put differently, if a policy is trained to be robust against one stochastic error model, does it also remain reliable under quasi-static drifts, temporally correlated noise, Hamiltonian parameter mismatch, or simple decoherence processes? This question is important both physically and methodologically. Physically, real devices are affected by several non-identical error sources simultaneously. Methodologically, the answer helps determine whether noisy-environment RL is merely learning distribution-specific compensation or whether it is capturing a broader form of control regularity. This question motivates the out-of-distribution robustness experiments introduced later in the thesis.

- A third question concerns *methodological comparison.* Reinforcement learning is not the only available approach to analog quantum control, and the original paper itself positions its method against simpler direct optimization baselines [**?**]. It is therefore important to ask how the RL-based framework compares with more standard quantum optimal-control strategies, such as GRAPE- or GOAT-like approaches, when the comparison is made under identical physical assumptions. More precisely, if one fixes the Hamiltonian model, amplitude constraints, target gate family, and evaluation metrics, does the RL-based method provide a measurable advantage in nominal performance, robust performance, or runtime, and under which conditions? This question is central to the scientific interpretation of the framework, because it determines whether the observed gains should be attributed to the physics of the cost function, to the RL optimization strategy, or to some interaction between the two.

- A fourth question concerns *objective-function sensitivity and interpretability.* The UFO objective is built as a weighted combination of gate infidelity, leakage penalties, boundary penalties, and runtime. While this scalarization is one of the strengths of the framework, it also raises a natural question: how sensitive are the learned controls to the relative choice of these weights? In particular, how do the coefficients associated with fidelity, leakage, boundary conditions, and runtime modify the qualitative behavior of the optimized pulse? Do they reveal smooth tradeoff curves, or do they induce sharp transitions between qualitatively different control regimes? This question is scientifically relevant because it connects the empirical performance of the controller to the structure of the underlying physical objective. It also bears directly on the interpretability of the framework, since a robust optimization method should ideally not depend in an opaque way on finely tuned scalar weights.

Taken together, these research questions define the logical trajectory of the thesis. The first establishes whether the original framework can be reconstructed reliably. The second probes whether the learned robustness extends beyond the perturbation model used during training. The third evaluates the distinct contribution of the reinforcement-learning approach relative to more traditional control methods. The fourth examines how the multi-objective structure of the control cost shapes the resulting solutions. The empirical chapters of the thesis are organized around these questions, so that each numerical study addresses a clearly defined scientific issue rather than serving as an isolated computational exercise.

1.7 Main contributions

The main contributions of this thesis can be divided into three closely connected parts. The first contribution is a structured reconstruction of the physical and methodological framework

introduced in Ref. [**?**]. Rather than treating the reference paper only as a source of final numerical claims, this thesis develops its logic step by step: from the multilevel superconducting-qubit model and the subspace structure of the gmon architecture, to the time-dependent Schrieffer–Wolff treatment of leakage, the construction of the UFO objective, and the continuous-action reinforcement-learning formulation used for control optimization. In this sense, the thesis provides not only a numerical reproduction attempt, but also a systematic theoretical exposition of the framework, organized so as to make explicit the physical role of each component and the way in which they fit together within a realistic analog quantum-control problem.

The second contribution is an empirical validation of the original framework through a dedicated reproduction study. In particular, the thesis implements a consistent simulation pipeline for the target gate family considered in Ref. [**?**] and uses it to reproduce the main classes of results reported in the paper, namely the comparison of analog gate times with synthesized-gate baselines, the dependence of average fidelity on stochastic control-noise strength, and the corresponding fidelity-variance behavior. This part of the work is important because it tests whether the main claims of the original framework remain observable under an independent implementation, thereby providing a concrete assessment of its reproducibility. The associated numerical studies also serve as the reference point against which all subsequent extensions are interpreted.

The third contribution is the extension of the original framework along directions that are scientifically relevant but not fully explored in the reference paper. These extensions include, first, an out-of-distribution robustness analysis in which a controller trained under Gaussian amplitude noise is evaluated under qualitatively different perturbation models, including quasi-static drift, colored noise, parameter mismatch, and decoherence; second, a comparative analysis between the reinforcement-learning-based controller and more standard quantum optimal-control baselines under matched Hamiltonian, target-family, and evaluation conditions; and third, an ablation study on the coefficients of the UFO cost function in order to clarify the tradeoffs induced by the different objective terms. Taken together, these extensions aim to move the discussion beyond the question of whether the original method works in its native setting, toward the broader question of how robust, transferable, and interpretable the framework is when viewed as a general strategy for analog quantum control.

A further contribution of the thesis lies in the interpretive perspective adopted throughout the work. The goal is not only to report whether one optimizer outperforms another on a given metric, but also to connect those outcomes to the underlying physics of the control problem. For this reason, the empirical results are consistently interpreted through the conceptual themes developed in the background chapters, including the distinction between compiled and analog gate realization, the multilevel origin of leakage, the speed–leakage tradeoff, and the multi-objective structure of realistic control costs. In this way, the thesis aims to contribute not only a set of numerical results, but also a clearer understanding of how reinforcement-learning-based control interacts with the structure of superconducting quantum hardware.

1.8 Thesis outline

The remainder of this thesis is organized as follows. Chapter 2 develops the theoretical background required for the study, including quantum computation as a controlled dynamical problem, the role of gate fidelity and runtime as performance metrics, the distinction between compiled universal gate synthesis and direct analog control, the multilevel nature of superconducting qubits, the notions of computational subspace and leakage, the speed–leakage tradeoff, and the main open-loop, closed-loop, and reinforcement-learning approaches relevant to quantum control. This chapter establishes the general physical and methodological language used throughout the rest of the thesis.

Chapter 3 introduces the paper-specific framework and its physical implementation on the

gmon architecture. It presents the effective Hamiltonian and control channels, the computational and leakage subspaces in the notation of the reference paper, the motivation for the time-dependent Schrieffer–Wolff treatment of leakage, the resulting leakage bound, and the construction of the Universal cost Function control Optimization objective. The same chapter then formulates the control problem as a reinforcement-learning task, explains the use of trust-region policy optimization in a stochastic training environment, introduces the target gate family $N(\alpha, \alpha, \gamma)$, describes the baseline comparison strategy of the original paper, and summarizes the main findings that the present work seeks to reproduce and extend.

Chapter 4 contains the empirical assessment of the framework. It first presents the numerical protocol and implementation choices used in this thesis, including the evaluation metrics and the selected hyperparameter regime. It then reports the reproduction experiments on representative targets and on the original benchmark figures. After this, the chapter develops the extension studies proposed in this work: out-of-distribution robustness tests, comparisons against standard quantum optimal-control baselines under matched assumptions, and an ablation analysis of the coefficients entering the UFO objective. Where appropriate, the numerical results are interpreted in light of the theoretical considerations developed in the previous chapters.

Finally, Chapter 5 collects the main conclusions of the thesis. It summarizes the extent to which the original framework has been successfully reproduced, discusses what the extension experiments reveal about the generality and limitations of the method, and outlines possible future directions, including broader robustness studies, alternative control objectives, and possible hybrid open-loop / closed-loop refinements. In this way, the thesis progresses from conceptual background, to paper-specific framework, to empirical validation and extension, with the aim of providing both a faithful reconstruction of the original method and a critical assessment of its broader significance.

1.9 AI usage policy

As noted by the world-renowned mathematician Terence Tao, the emergence of large language models has already begun to reshape the modern research pipeline, not only in writing and technical implementation, but also in the broader organization of exploratory work. In particular, AI systems have significantly reduced the cost of idea generation, making it possible to produce, inspect, and compare a much larger number of tentative directions than was previously practical. In this sense, their main impact is not necessarily to replace depth of reasoning, but rather to expand breadth of exploration, while shifting the main epistemic bottleneck from generation to verification, selection, and validation [**?**, **?**]. This perspective is especially relevant to the present thesis, where AI-based tools were used as auxiliary instruments in both the implementation and writing workflow.

More specifically, AI assistance was used on the coding side to generate limited code snippets, support debugging, clarify the usage of specific Python libraries, and accelerate minor implementation or refactoring tasks. On the writing and research side, it was used to assist the discovery of potentially relevant literature through semantic search and query refinement, to identify related work, and to support the drafting, rephrasing, and stylistic revision of selected passages. In all cases, however, AI remained a supportive and non-autonomous tool. The mathematical derivations, physical interpretations, modeling assumptions, numerical methodology, experimental design, source verification, and final presentation of results were independently checked and validated by the author. No scientific claim, citation, or conclusion was included in this thesis solely on the basis of AI-generated output.

Chapter 2

Background

2.1 Why quantum control matters

2.1.1 Quantum computation as controlled dynamics

In the abstract circuit model, a quantum algorithm is described as a sequence of unitary gates acting on an n-qubit register. From the hardware point of view, however, these gates are not primitive objects: they arise from the real-time evolution of a physical quantum system driven by externally programmable fields. Quantum control is therefore the discipline that bridges the algorithmic layer of quantum information processing and the dynamical laws governing a concrete device. In this perspective, the implementation of a gate is naturally formulated as an inverse problem: given a desired logical operation, one seeks a family of control signals that steers the system so that its physical evolution reproduces the target operation with high accuracy on the computational degrees of freedom [**?**, **?**, **?**, **?**].

For an n-qubit processor, the logical information is encoded in the computational subspace

$$\mathcal{H}_{\text{comp}} \cong (\mathbb{C}^2)^{\otimes n} \subseteq \mathcal{H}_{\text{phys}}, \tag{2.1}$$

where $\mathcal{H}_{\text{phys}}$ denotes the full Hilbert space of the device. The inclusion in Eq. (2.1) is essential: in realistic platforms, the physical Hilbert space is generally larger than the ideal qubit space. This point is conceptually important already at the beginning of the thesis, because it clarifies that a quantum gate is not merely a map on $\mathcal{H}_{\text{comp}}$, but the effective restriction of a higher-dimensional physical evolution to the logical sector. In superconducting architectures, for instance, this enlarged Hilbert space is associated with higher oscillator levels beyond the two-level qubit approximation, an observation that will later motivate the discussion of leakage errors [**?**, **?**].

A broad and standard model for controlled quantum dynamics is a time-dependent Hamiltonian of the form

$$H(t; \mathbf{u}) = H_{\text{d}} + \sum_{k=1}^{m} u_k(t)\, H_k, \tag{2.2}$$

where H_{d} is the drift Hamiltonian, fixed by the underlying hardware, while the operators $\{H_k\}_{k=1}^{m}$ specify the controllable interaction channels and the real functions

$$\mathbf{u}(t) = \big(u_1(t), u_2(t), \ldots, u_m(t)\big) \tag{2.3}$$

are the externally applied control fields. Depending on the platform, the components of $\mathbf{u}(t)$ may represent microwave amplitudes, phases, detunings, flux biases, or other tunable experimental knobs. The state vector then obeys the time-dependent Schrödinger equation

$$i\hbar \frac{d}{dt} \left|\psi(t)\right\rangle = H(t; \mathbf{u}) \left|\psi(t)\right\rangle, \tag{2.4}$$

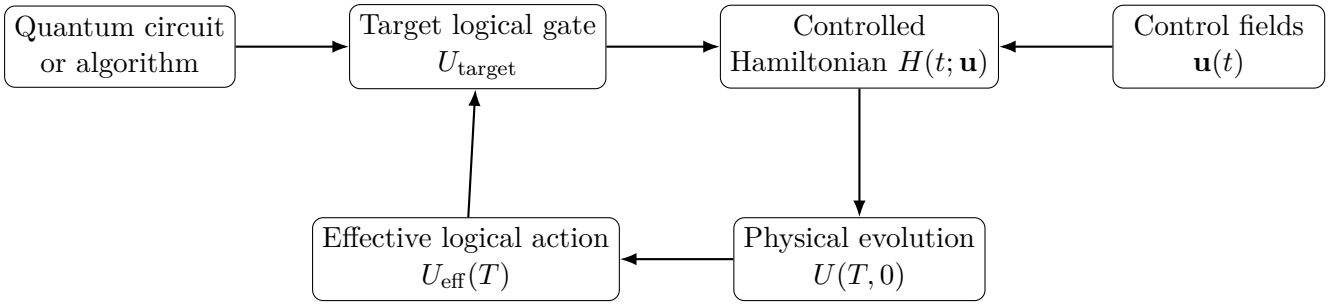

Figure 2.1: Conceptual relation between the algorithmic description of a quantum task and its physical realization through controlled dynamics. The central inverse problem of quantum control is to determine the control fields $\mathbf{u}(t)$ such that the induced propagator $U(T,0)$ reproduces the target logical gate on the computational subspace.

or, equivalently, the propagator $U(t,0)$ satisfies

$$i\hbar\frac{d}{dt}U(t,0) = H(t;\mathbf{u})\,U(t,0), \qquad U(0,0) = \mathbb{I}, \tag{2.5}$$

with formal solution

$$U(T,0) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^T H(t;\mathbf{u})\;dt\right], \tag{2.6}$$

where $\mathcal{T}$ denotes the time-ordering operator [**?**, **?**, **?**].

The physical task of gate synthesis can now be stated precisely. Let Π_{comp} be the projector onto $\mathcal{H}_{\text{comp}}$, and let $U_{\text{target}} \in U(2^n)$ be the desired logical gate. The effective action induced on the computational sector is

$$U_{\text{eff}}(T) := \Pi_{\text{comp}}\,U(T,0)\,\Pi_{\text{comp}}, \tag{2.7}$$

and the control problem consists in selecting the fields $\mathbf{u}(t)$ and, in many cases, the gate duration T, so that

$$U_{\text{eff}}(T) \approx e^{i\phi}U_{\text{target}}, \qquad \phi \in \mathbb{R}, \tag{2.8}$$

up to a physically irrelevant global phase. A convenient generic mismatch functional is therefore

$$\varepsilon_U(T) = \min_{\phi\in\mathbb{R}}\left\|U_{\text{eff}}(T) - e^{i\phi}U_{\text{target}}\right\|, \tag{2.9}$$

where the choice of norm depends on the application. In practical quantum-control problems, the abstract objective in Eq. (2.9) is usually replaced by experimentally meaningful figures of merit, such as gate fidelity, leakage penalties, smoothness constraints, and runtime penalties. These quantities will be introduced in the following paragraphs and then specialized, later in the thesis, to the UFO framework studied in this work [**?**, **?**].

For numerical optimization, it is often useful to discretize the controls on a grid $0 = t_0 < t_1 < \cdots < t_N = T$ and use a piecewise-constant parameterization,

$$u_k(t) = u_{k,\ell}, \qquad t \in [t_\ell, t_{\ell+1}), \quad k = 1,\ldots,m, \quad \ell = 0,\ldots,N-1. \tag{2.10}$$

Equation (2.10) turns the search for an admissible control field into a finite-dimensional optimization problem over the amplitudes $\{u_{k,\ell}\}$. This viewpoint is especially convenient because it connects naturally both with gradient-based pulse engineering and with reinforcement-learning formulations, where the control trajectory is built sequentially over discrete time steps [**?**, **?**, **?**].

2.1.2 Gate fidelity and runtime as central performance metrics

Once a target logical operation has been formulated as a controlled dynamical task, the next question is how to assess the quality of a candidate control protocol. In quantum control, two figures of merit play a particularly central role: the *fidelity* of the implemented gate and its *runtime*. The first quantifies how accurately the physical evolution reproduces the desired logical transformation, while the second measures how long the quantum system must remain under active control in order to realize that transformation. These quantities are not merely technical diagnostics: they directly determine the practical computational power of a hardware platform. In particular, the paper studied in this thesis explicitly identifies fidelity and runtime as the key measures governing the usefulness of a control strategy, and builds its optimization framework around their joint treatment [**?**].

Let $U_{\text{target}} \in U(d)$ denote the desired logical gate acting on a d-dimensional computational subspace, and let $U(T)$ denote the unitary evolution generated by a chosen control protocol at final time T. In the idealized closed-system setting, where the action of the control sequence remains unitary on the relevant subspace, a standard overlap-based figure of merit is the *gate fidelity*

$$F_{\text{gate}}(U(T), U_{\text{target}}) = \frac{1}{d^2} \left| \text{Tr}\Big(U_{\text{target}}^\dagger U(T) \Big) \right|^2 . \tag{2.11}$$

This quantity satisfies

$$0 \leq F_{\text{gate}} \leq 1, \tag{2.12}$$

and reaches its maximal value $F_{\text{gate}} = 1$ if and only if

$$U(T) = e^{i\phi} U_{\text{target}}, \qquad \phi \in \mathbb{R}, \tag{2.13}$$

that is, when the realized operation agrees with the target up to an irrelevant global phase. The associated *gate infidelity* is then defined as

$$\varepsilon_{\text{gate}} := 1 - F_{\text{gate}}. \tag{2.14}$$

This is the quantity most frequently minimized in practical control optimization, since it vanishes precisely at perfect implementation and is cheap to evaluate numerically. This thesis adopts exactly this form of gate infidelity as the first contribution to its control cost function, emphasizing that it provides a computationally convenient measure of control inaccuracy [**?**].

In realistic devices, however, the implemented operation is not exactly unitary on the logical subspace, since calibration errors, stochastic fluctuations, decoherence, and leakage make the effective map better described by a quantum channel $\mathcal{E}$.

TODO: decidere quale usare (spostare nel caso la seconda frase da un altra parte)

A *quantum channel* is a linear map

$$\mathcal{E} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}), \tag{2.15}$$

which describes the most general physically admissible evolution of a quantum state and is required to be *completely positive* and *trace preserving* (CPTP)[1]. In finite dimension, every

[1] Complete positivity means that, for any ancillary Hilbert space $\mathcal{H}_{\text{anc}}$, the extended map $\mathcal{E} \otimes \mathbb{I}_{\text{anc}}$ maps positive operators to positive operators. This requirement is stronger than ordinary positivity and is necessary because a physical system may be entangled with an external ancilla; the evolution must therefore remain well defined even when applied only to part of a larger composite system. Trace preservation means that $\text{Tr}\big[\mathcal{E}(\rho)\big] = \text{Tr}(\rho)$ for every density operator ρ, so that total probability is conserved. The reason one uses a quantum channel rather than a unitary operator in the present context is that the effective action of a realistic gate on the computational subspace is, in general, not exactly unitary. Even when the full dynamics of the enlarged physical system is unitary, the reduced evolution seen on the logical degrees of freedom may become non-unitary because of leakage outside the computational subspace, coupling to uncontrolled degrees of freedom, stochastic control fluctuations, model mismatch, or decoherence. More formally, if the system interacts with an environment initially in state ρ_E, then the effective evolution on the system alone takes the form $\mathcal{E}(\rho) = \text{Tr}_E\left[U(\rho \otimes \rho_E)U^\dagger\right]$, which is generally a CPTP map rather than a unitary conjugation. For this reason, the language of quantum channels is the natural one when discussing noisy gates, average gate fidelity, and robustness beyond the ideal closed-system approximation.

quantum channel admits a Kraus representation,

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger, \qquad \sum_k E_k^\dagger E_k = \mathbb{I}, \tag{2.16}$$

where the operators $\{E_k\}$ are the Kraus operators associated with the channel. The unitary case is recovered as the special situation in which there is a single Kraus operator $E_1 = U$, so that

$$\mathcal{E}(\rho) = U \rho U^\dagger. \tag{2.17}$$

TODO: decidere quale footnote usare

In that more general setting, an experimentally meaningful notion of quality is the *average gate fidelity*, defined by averaging over all pure input states with respect to the Haar measure [2]

$$\overline{F}(\mathcal{E}, U_{\text{target}}) = \int d\psi \; \langle\psi| U_{\text{target}}^\dagger \, \mathcal{E}(|\psi\rangle\langle\psi|) \; U_{\text{target}} |\psi\rangle. \tag{2.18}$$

Equation (2.18) measures the average probability that an arbitrary input state is transformed as intended. It is therefore more operational than a purely state-specific fidelity, and it is particularly useful when one wishes to characterize a gate rather than a single state-transfer task. As discussed in the literature, average fidelity is also the natural quantity for assessing robustness under stochastic control perturbations, since one may compare how the performance of a control protocol degrades under repeated noisy realizations [**?**]. A related quantity is the *average gate infidelity*,

$$\overline{\varepsilon}_{\text{gate}} := 1 - \overline{F}(\mathcal{E}, U_{\text{target}}) \,. \tag{2.19}$$

For trace-preserving quantum channels, the average fidelity is closely related to the entanglement fidelity and process fidelity. In particular, if F_{e} denotes the entanglement fidelity with respect to the ideal target unitary, then one has the standard relation

$$\overline{F} = \frac{d\, F_{\text{e}} + 1}{d+1}. \tag{2.20}$$

In the special case where the implemented evolution is itself unitary, Eq. (2.18) reduces to

$$\overline{F}(U(T), U_{\text{target}}) = \frac{\left|\text{Tr}\left(U_{\text{target}}^\dagger U(T)\right)\right|^2 + d}{d(d+1)}. \tag{2.21}$$

Hence, Eqs. (2.11) and (2.21) encode the same mismatch information up to a dimension-dependent affine transformation. This is one reason why overlap-based infidelity is so widely used in numerical optimal control: it is simpler to evaluate, while still remaining closely tied to experimentally relevant benchmarking quantities [**?**, **?**]. In the present thesis, the distinction between these notions will remain important: the optimization problem is naturally formulated in terms of gate infidelity, whereas robustness analyses are more naturally reported in terms of average fidelity under noisy implementations, exactly as done in the reference paper [**?**].

[2] In Eq. (2.18), the symbol $d\psi$ denotes integration with respect to the Haar-induced probability measure on the space of pure states. More precisely, let $S(\mathcal{H}) = \{|\psi\rangle \in \mathcal{H} : \|\psi\| = 1\}$ be the unit sphere of the Hilbert space, and let the space of pure states be identified with the complex projective space $\mathbb{P}(\mathcal{H}) = S(\mathcal{H})/\sim, \quad |\psi\rangle \sim e^{i\theta}|\psi\rangle, \quad \theta \in \mathbb{R}$, that is, unit vectors modulo physically irrelevant global phase. We denote by $\mathcal{B}\mathbb{P}(\mathcal{H})$ the Borel σ-algebra generated by the natural topology on $\mathbb{P}(\mathcal{H})$. The symbol μ then denotes the unique normalized Borel probability measure $\mu : \mathcal{B}\mathbb{P}(\mathcal{H}) \longrightarrow [0,1]$ that is invariant under the action of the unitary group, namely $\mu(UA) = \mu(A)$ for every $A \in \mathcal{B}\mathbb{P}(\mathcal{H})$ and every unitary U. In this sense, μ is the natural notion of a uniform probability distribution over pure quantum states: it assigns measure 1 to the whole pure-state space and treats all directions in Hilbert space on an equal footing. [**?**] Equivalently, one may define μ as the pushforward of the normalized Haar measure on the compact unitary group $U(d)$, where $d = \dim \mathcal{H}$. Fixing any reference unit vector $|\psi_0\rangle \in S(\mathcal{H})$, one obtains $\mu(A) = \nu\{V \in U(d) : [V|\psi_0\rangle] \in A\}$, where ν is the normalized Haar measure on $U(d)$ and $[V|\psi_0\rangle]$ denotes the corresponding ray in $\mathbb{P}(\mathcal{H})$. This construction makes explicit that averaging with respect to $d\psi$ is basis-independent and unitarily invariant [**?**].

Accuracy alone, however, is not sufficient. A gate that is extremely faithful but requires too long a control time may still be of limited practical use, because quantum systems are inevitably exposed to decoherence, relaxation, dephasing, and additional control noise throughout the entire duration of the pulse. For this reason, the total runtime T of a control protocol must be treated as a second fundamental metric. At the most basic level, runtime is simply the final integration time appearing in the propagator

$$U(T,0) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^T H(t)\,dt\right], \tag{2.22}$$

but physically it quantifies the temporal exposure of the quantum state to all imperfections that accumulate during the gate execution. This is why shorter gate runtimes extend the achievable circuit depth by attempting to complete the intended operation before uncorrectable errors induced by noise and dissipation become dominant [**?**].

A simple phenomenological way to express this point is to note that, in the presence of a coherence timescale T_{coh}, one generically expects the quality of a gate to deteriorate as its duration becomes a non-negligible fraction of that timescale. At the level of a rough heuristic model, one may write

$$F_{\text{coh}}(T) \sim e^{-T/T_{\text{coh}}}, \tag{2.23}$$

or, equivalently,

$$\varepsilon_{\text{coh}}(T) \sim 1 - e^{-T/T_{\text{coh}}}. \tag{2.24}$$

These expressions should not be interpreted as a complete device model, since the actual decay law depends on the physical noise channels and on the precise gate under consideration. Their role is instead conceptual: they make explicit why gate duration is not a neutral parameter. All else being equal, faster gates are generally preferable because they reduce the time window over which incoherent errors can act. This is precisely why runtime minimization is a standard objective in quantum optimal control, and why the UFO framework includes an explicit runtime penalty term in its cost function [**?**, **?**] as it is stated in section 3.2.3.

The role of runtime is even clearer when one considers that quantum algorithms are implemented as long sequences of elementary or composite operations. If a circuit contains N_{gates} successive gates with durations $\{T_j\}_{j=1}^{N_{\text{gates}}}$, the total physical runtime is approximately

$$T_{\text{circ}} = \sum_{j=1}^{N_{\text{gates}}} T_j \tag{2.25}$$

up to idle times and synchronization overhead. Thus, improving a single gate by shortening its duration can have a multiplicative effect at the circuit level. In this sense, gate runtime is not merely a low-level engineering quantity; it is a parameter that constrains the accessible algorithmic depth before decoherence overwhelms the computation. The reference paper exploits exactly this logic when comparing learned analog controls with a discrete gate-synthesis baseline: one of its main claims is that direct pulse-level optimization can realize certain target unitaries significantly faster than their decomposition into a standard gate sequence [**?**].

The central conceptual point is therefore that quantum control is intrinsically a *multi-objective* problem. Maximizing fidelity alone may lead to excessively long or experimentally fragile pulses, while minimizing runtime alone may produce aggressive controls that suffer from leakage, control non-adiabaticity, or large sensitivity to calibration errors. In practice one seeks a compromise among several competing desiderata, with fidelity and runtime representing the two most basic ones. This tradeoff can be stated abstractly by introducing a generic control objective of the form

$$\mathcal{J}[\boldsymbol{u}] = w_F\,\varepsilon_{\text{gate}}[\boldsymbol{u}] + w_T\,T[\boldsymbol{u}] + \cdots, \tag{2.26}$$

Table 2.1: Core performance metrics used in gate-level quantum control.

Quantity	Definition / interpretation
Gate fidelity	$F_{\text{gate}} = \frac{1}{d^2}\left\lvert \text{Tr}(U_{\text{target}}^\dagger U)\right\rvert^2$
Gate infidelity	$\varepsilon_{\text{gate}} = 1 - F_{\text{gate}}$
Average gate fidelity	$\overline{F}(\mathcal{E}, U_{\text{target}}) = \int d\psi \, \langle\psi\lvert U_{\text{target}}^\dagger \mathcal{E}(\lvert\psi\rangle\langle\psi\rvert) U_{\text{target}}\rvert\psi\rangle$
Average gate infidelity	$\overline{\varepsilon}_{\text{gate}} = 1 - \overline{F}$
Runtime	Final control duration T entering the propagator $U(T, 0)$

where $\boldsymbol{u}$ denotes the control protocol and the omitted terms may encode additional physical penalties such as leakage, bandwidth constraints, boundary conditions, or energy costs. Equation (2.26) is intentionally schematic, but it captures the philosophy that underlies much of modern quantum control. In the present thesis, this general viewpoint will become concrete in the next chapter, where the UFO cost function will be introduced as a specific realization of this multi-objective principle, with explicit terms for infidelity, leakage, boundary constraints, and runtime [**?**].

From the viewpoint of the thesis structure, the discussion above motivates the distinction that will be used throughout the rest of the work. When the implemented evolution can be treated as approximately unitary, gate infidelity as in Eq. (2.14) provides a convenient optimization target. When robustness against stochastic errors is studied, average fidelity as in Eq. (2.18) becomes the more informative diagnostic. Runtime, on the other hand, remains a fundamental metric in both settings, because it directly mediates the tradeoff between fast control and exposure to imperfections. Together, these quantities provide the minimal language needed to formulate the control problem addressed in this thesis.

2.1.3 Universal gates and direct analog control

A central notion in quantum computation is that of *universality*. At the abstract circuit level, one usually says that a collection of elementary gates $\mathcal{G}$ is universal if arbitrary unitary operations on a finite number of qubits can be generated exactly or to arbitrary accuracy by composing gates from $\mathcal{G}$ [**?**, **?**]. For an n-qubit register with logical Hilbert space

$$\mathcal{H}_{\text{comp}} \cong (\mathbb{C}^2)^{\otimes n}, \tag{2.27}$$

the ideal set of logical operations is the unitary group $U(2^n)$, or equivalently $SU(2^n)$ up to a global phase. A finite gate set $\mathcal{G}$ is therefore called universal when the subgroup it generates is dense in $SU(2^n)$:

$$\overline{\langle \mathcal{G} \rangle} = SU(2^n). \tag{2.28}$$

Equation (2.28) formalizes the standard circuit-model view: even if the hardware natively exposes only a limited menu of primitive operations, it is still computationally sufficient provided those primitives can approximate any desired logical transformation to arbitrary precision.

In this framework, a target gate $U_{\text{target}} \in SU(2^n)$ is not implemented directly, but rather compiled into a finite sequence of elementary operations,

$$U_{\text{syn}} = G_L G_{L-1} \cdots G_1, \qquad G_\ell \in \mathcal{G}, \tag{2.29}$$

such that

$$\left\| U_{\text{target}} - e^{i\phi} U_{\text{syn}} \right\| \leq \varepsilon, \qquad \phi \in \mathbb{R}, \tag{2.30}$$

for a prescribed synthesis accuracy $\varepsilon > 0$. The existence of such approximations for universal sets is guaranteed in general, and the Solovay–Kitaev theorem further shows that efficient

approximations can be constructed with sequence length growing only polylogarithmically in $1/\varepsilon$ for standard assumptions on the gate set [**?**, **?**]. From the perspective of fault-tolerant quantum computing, this discrete picture is fundamental: logical algorithms are compiled into a standard finite library of protected primitives, and the abstraction is both mathematically elegant and operationally robust.

However, the circuit-synthesis viewpoint is not the only physically relevant one. In an actual device, each elementary gate G_ℓ must itself be realized by controlled time evolution. Thus, the synthesized circuit in Eq. (2.29) ultimately corresponds to a concatenation of pulse segments, each with its own control fields, calibration overhead, and execution time. As stated in 2.25, if τ_ℓ denotes the runtime of the ℓ-th elementary gate, the total physical duration of the synthesized implementation is approximately

$$T_{\text{syn}} = \sum_{\ell=1}^{L} \tau_\ell, \tag{2.31}$$

while the logical depth is the number of time layers required to schedule these operations,

$$D_{\text{syn}} \leq L. \tag{2.32}$$

Equation (2.31) makes explicit a simple but important fact: even when the number of compiled gates is information-theoretically optimal, the resulting implementation may still be physically expensive, because every additional gate consumes coherence time and introduces further opportunities for control error and leakage. This observation becomes especially relevant in pre-fault-tolerant and noisy intermediate-scale quantum devices, where the experimentally accessible circuit depth is strongly limited by decoherence and imperfect control [**?**].

For this reason, near-term hardware often benefits from a more direct viewpoint, in which one seeks to realize the full target unitary through a single analog control protocol rather than through a discrete decomposition. In the notation introduced earlier, one instead searches for control fields $\boldsymbol{u}(t)$ such that

$$U(T,0) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^T H(t;\boldsymbol{u})\, dt\right] \approx e^{i\phi} U_{\text{target}}, \tag{2.33}$$

with T as small as possible and with all relevant physical constraints satisfied. The contrast with Eq. (2.29) is conceptual as much as technical. In circuit synthesis, the target unitary is approximated through a universal symbolic alphabet of gates. In direct analog control, by contrast, one exploits the continuous dynamics naturally generated by the underlying Hamiltonian in order to realize the target in one coherent control episode. The referenced paper is explicitly motivated by this distinction: for near-term devices, quantum operations need not be restricted to a finite universal gate set, and that implementing a desired unitary with one pulse sequence can reduce resource overhead relative to a longer synthesized circuit [**?**].

This distinction is closely tied to the notion of *controllability*. Let the controlled Hamiltonian be written as

$$H(t) = H_{\text{d}} + \sum_{k=1}^{m} u_k(t) H_k, \tag{2.34}$$

where H_{d} is the drift term and the H_k are control generators. A standard sufficient condition for full operator controllability on a d-dimensional Hilbert space is that the dynamical Lie algebra generated by the skew-Hermitian operators $\{iH_{\text{d}}, iH_1, \ldots, iH_m\}$ spans the whole algebra $\mathfrak{su}(d)$:

$$\text{Lie}\{iH_{\text{d}}, iH_1, \ldots, iH_m\} = \mathfrak{su}(d). \tag{2.35}$$

Equation (2.35) expresses the ideal mathematical limit of analog universality: if the available drift and control Hamiltonians generate all of $\mathfrak{su}(d)$, then any unitary in $SU(d)$ is reachable in principle through suitable time-dependent controls [**?**]. In practice, of course, reachability alone

is not enough. One must still contend with finite amplitudes, bandwidth limits, leakage outside the computational subspace, calibration uncertainty, and noise. These constraints make the *effective* implementation of arbitrary target gates highly nontrivial even when controllability holds in the abstract.

The practical advantage of direct analog control is most pronounced when the target unitary is naturally aligned with the native interactions of the hardware. If a device Hamiltonian contains, for example, an entangling term of the form

$$H_{\text{nat}} = J\left(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y\right), \tag{2.36}$$

then free or weakly modulated evolution under this interaction generates the gate family

$$U_{\text{nat}}(t) = \exp\left[-\frac{i}{\hbar}tH_{\text{nat}}\right] = \exp[-i\theta(t)\left(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y\right)], \tag{2.37}$$

with $\theta(t) = Jt/\hbar$. In such a situation, certain entangling operations may be realized almost directly by the device physics itself, possibly supplemented by a small number of local rotations, instead of being compiled into a longer generic circuit. This is precisely the type of hardware-structure exploitation that makes analog control attractive: the control procedure can adapt to the geometry and couplings of a given platform rather than forcing every task through a fixed universal template [**?**, **?**].

From a computational perspective, one may therefore distinguish two different notions of efficiency. The first is *gate-set efficiency*, meaning how economically a target unitary can be decomposed into a prescribed universal library. The second is *hardware efficiency*, meaning how economically that same unitary can be realized on a specific physical platform once its native interactions are taken into account. These notions need not coincide. A gate may admit an optimal decomposition within a discrete universal set and yet still be slower than a direct analog realization that exploits the natural Hamiltonian of the device. This tension is one of the central motivations of the present thesis, and it is made explicit in the corrisponding literature, which compares direct analog controls with optimal gate-synthesis counterparts for a family of two-qubit gates and reports significant runtime reductions for some targets [**?**]. In the following chapters, this distinction will become concrete in the superconducting-gmon setting, where a family of physically relevant two-qubit gates can be analyzed both from the perspective of optimal discrete synthesis and from that of direct pulse-level realization.

It is important, however, not to overstate the case. Direct analog control does not replace the circuit model as a fundamental description of quantum computation, nor does it eliminate the need for universal gate sets in fault-tolerant architectures. Rather, it provides a complementary layer of optimization that is particularly valuable in near-term devices and in hardware-aware quantum algorithms. The relevant question is not whether one should abandon universal discrete gates, but whether a given target operation should be implemented through a compiled sequence or through a direct continuous control trajectory. The answer depends on the interaction between algorithmic structure, hardware native couplings, noise, and control constraints. This is exactly the regime in which modern quantum-control methods, including reinforcement-learning-based approaches, become especially relevant [**?**, **?**].

2.2 Physical setting: superconducting qubits and leakage

2.2.1 Superconducting qubits as weakly anharmonic multilevel systems

Among the leading physical platforms for gate-based quantum information processing, superconducting circuits occupy a particularly important position because they combine lithographic scalability, strong microwave control, and compatibility with fast entangling interactions [**?**, **?**]. From the theoretical point of view, however, a superconducting qubit is not fundamentally a

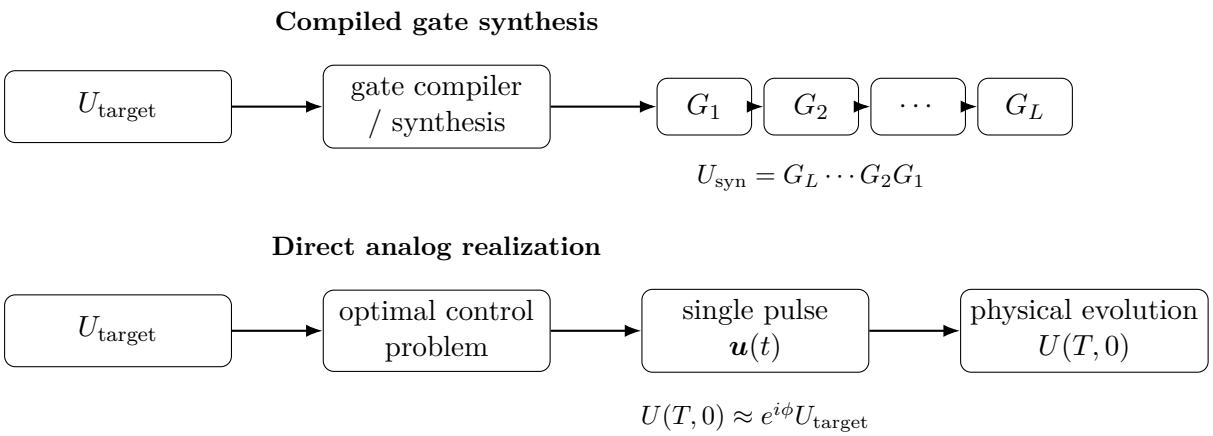

Figure 2.2: Two conceptually distinct routes to implementing a target unitary. In the upper route, the gate is compiled into a sequence of elements from a finite universal set. In the lower route, one directly optimizes a continuous control protocol so that the full physical evolution approximates the target in a single coherent pulse sequence.

two-level system. Rather, it is a quantized nonlinear electrical circuit built from capacitive, inductive, and Josephson elements. The Josephson junction provides the essential nonlinearity that breaks the equal spacing of a harmonic oscillator and thereby allows the two lowest energy eigenstates to be selectively addressed as a logical qubit [**?**, **?**]. This observation is central for the present thesis, because the control problem studied later is formulated on a hardware platform of this kind, and the possibility of populating levels outside the logical subspace is precisely what gives rise to leakage errors.

A standard starting point is the circuit Hamiltonian of a Cooper-pair-box or transmon-type qubit,

$$\hat{H}_{\text{tr}} = 4E_C(\hat{n} - n_g)^2 - E_J \cos\hat{\varphi}, \tag{2.38}$$

where E_C is the charging energy, E_J the Josephson energy, $\hat{n}$ the number operator for transferred Cooper pairs, $\hat{\varphi}$ the superconducting phase difference across the junction, and n_g an offset charge [**?**]. The canonical variables satisfy

$$[\hat{\varphi}, \hat{n}] = i, \tag{2.39}$$

so that Eq. (2.38) describes a nonlinear quantum oscillator. In the transmon regime,

$$\frac{E_J}{E_C} \gg 1, \tag{2.40}$$

charge dispersion is exponentially suppressed, which strongly reduces sensitivity to charge noise, while the residual nonlinearity remains sufficient for qubit operation [**?**]. This is one of the main reasons why transmon-like devices, including xmon- and gmon-based architectures, have become so prominent in superconducting quantum computing [**?**, **?**, **?**].

The physics of weak anharmonicity becomes transparent by expanding the cosine potential around one of its minima,

$$-E_J \cos\hat{\varphi} = -E_J + \frac{E_J}{2}\hat{\varphi}^2 - \frac{E_J}{24}\hat{\varphi}^4 + O(\hat{\varphi}^6). \tag{2.41}$$

Substituting Eq. (2.41) into Eq. (2.38), one obtains to lowest nontrivial order

$$\hat{H}_{\text{tr}} \approx 4E_C\hat{n}^2 + \frac{E_J}{2}\hat{\varphi}^2 - \frac{E_J}{24}\hat{\varphi}^4 - E_J. \tag{2.42}$$

The first two terms define a harmonic oscillator, while the quartic correction introduces a weak nonlinearity. It is therefore natural to interpret the device as an oscillator whose level spacing is almost, but not exactly, uniform. In other words, superconducting transmon-like qubits are *weakly anharmonic oscillators* rather than exact spin-1/2 systems [**?**, **?**]. This distinction is crucial: without anharmonicity, a resonant drive tuned to the $|0\rangle \leftrightarrow |1\rangle$ transition would also excite higher transitions, making selective qubit control impossible.

Introducing bosonic ladder operators in the usual way,

$$\hat{\varphi} = \varphi_{\text{zpf}}(\hat{a} + \hat{a}^\dagger), \qquad \hat{n} = i n_{\text{zpf}}(\hat{a}^\dagger - \hat{a}), \tag{2.43}$$

with zero-point fluctuation amplitudes chosen so as to diagonalize the quadratic part, one may rewrite the Hamiltonian approximately as a weakly nonlinear oscillator,

$$\hat{H}_{\text{tr}} \approx \omega_p \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) + \frac{\alpha}{2} \hat{a}^\dagger \hat{a}^\dagger \hat{a}\hat{a} + \cdots , \tag{2.44}$$

where

$$\omega_p \approx \sqrt{8E_J E_C}, \tag{2.45}$$

in units with $\hbar = 1$, and where α denotes the anharmonicity. More precisely, perturbation theory yields the well-known large-E_J/E_C approximation for the energy levels

$$E_m \approx -E_J + \sqrt{8E_J E_C}\left(m + \frac{1}{2}\right) - \frac{E_C}{12}\left(6m^2 + 6m + 3\right), \qquad m = 0, 1, 2, \ldots, \tag{2.46}$$

from which the transition frequencies follow as

$$\omega_{m,m+1} = E_{m+1} - E_m \approx \sqrt{8E_J E_C} - E_C(m+1). \tag{2.47}$$

In particular,

$$\omega_{01} \approx \sqrt{8E_J E_C} - E_C, \qquad \omega_{12} \approx \sqrt{8E_J E_C} - 2E_C, \tag{2.48}$$

so that the anharmonicity is approximately

$$\alpha := \omega_{12} - \omega_{01} \approx -E_C. \tag{2.49}$$

Equation (2.49) shows explicitly why the transmon is only weakly anharmonic: the nonuniformity of the spectrum is governed by E_C, while the overall transition frequency is of order $\sqrt{8E_J E_C}$. In the transmon regime of Eq. (2.40), one therefore has

$$|\alpha| \ll \omega_{01}, \tag{2.50}$$

which is the mathematical expression of the weakly anharmonic limit [**?**, **?**].

The logical qubit is then encoded in the two-dimensional computational subspace

$$\mathcal{H}_{\text{comp}} = \text{span}\left\{|0\rangle, |1\rangle\right\}, \tag{2.51}$$

where $|0\rangle$ and $|1\rangle$ denote the ground and first excited eigenstates of the circuit Hamiltonian. The full device Hilbert space, however, is larger:

$$\mathcal{H}_{\text{phys}} = \text{span}\left\{|0\rangle, |1\rangle, |2\rangle, |3\rangle, \ldots\right\}. \tag{2.52}$$

Thus, the qubit approximation is not fundamental but effective: it consists in restricting the dynamics to Eq. (2.51) and neglecting population transfer into higher excited states. This approximation is excellent when control pulses are sufficiently weak, spectrally selective, and adiabatic with respect to unwanted transitions, but it is never exact. In particular, short

or strong pulses can admix the states $|2\rangle, |3\rangle, \ldots,$ making the multilevel structure physically relevant even for nominally two-level gate operations [**?**, **?**].

For this reason, it is often useful to describe a driven superconducting qubit directly in bosonic language. A generic driven weakly anharmonic mode may be modeled as

$$\hat{H}(t) = \omega_q \hat{a}^\dagger \hat{a} + \frac{\alpha}{2} \hat{a}^\dagger \hat{a}^\dagger \hat{a}\hat{a} + \Omega(t) \left(\hat{a} + \hat{a}^\dagger \right), \tag{2.53}$$

where ω_q is the fundamental transition frequency, $\alpha < 0$ the weak anharmonicity, and $\Omega(t)$ an externally applied microwave drive. Equation (2.53) makes two points transparent. First, the creation and annihilation operators act on the full ladder of states and therefore naturally include higher-level dynamics. Second, the same control field that is intended to drive the $|0\rangle \leftrightarrow |1\rangle$ transition also couples, in principle, to $|1\rangle \leftrightarrow |2\rangle$ and higher transitions, with selectivity arising only because the spectrum is not exactly harmonic. The control task is therefore intrinsically multilevel.

TODO: Riguardare sto paragrafo e come tratta il reference paper, in particolare la eventuale trattazione della appendice

This multilevel perspective is not merely a pedagogical refinement; it is the viewpoint adopted in the reference paper analyzed in this thesis. In its Appendix A, the gmon architecture is modeled by quantizing the circuit degrees of freedom in terms of bosonic modes and then deriving an effective rotating-wave Hamiltonian with explicit anharmonicity, tunable coupling, detuning controls, and microwave drives [**?**]. In that formulation, each superconducting qubit is treated as a weakly anharmonic oscillator before projection onto the qubit basis, exactly in line with the general discussion above. This is the correct framework for studying high-performance control, because it preserves the higher excited levels that are responsible for leakage and because it allows one to assess whether a pulse remains confined to the intended logical sector.

When two such devices are coupled, the same principle persists at the multi-qubit level. If each qubit is truncated to d_{loc} levels, the full two-body Hilbert space has dimension

$$\dim \mathcal{H}^{(2)}_{\text{phys}} = d_{\text{loc}}^2, \tag{2.54}$$

whereas the logical computational sector retains dimension

$$\dim \mathcal{H}^{(2)}_{\text{comp}} = 4. \tag{2.55}$$

Already with $d_{\text{loc}} = 3$, one obtains a 9-dimensional physical space for a nominally two-qubit gate. This illustrates why the multilevel description rapidly becomes unavoidable in realistic control simulations: even if the target gate acts only on the logical subspace, the actual Schrödinger evolution takes place in a larger Hilbert space that contains leakage channels.

In summary, superconducting qubits should be viewed as engineered nonlinear oscillators whose weak anharmonicity enables qubit selectivity but does not eliminate the presence of higher excited states. The logical qubit is encoded in the two lowest levels, yet the full control problem is inherently multilevel. This fact underlies the entire structure of modern superconducting quantum control: it explains why one uses bosonic or multilevel Hamiltonians at the modeling stage, why fast control can induce unwanted excitations, and why leakage must later be treated as an explicit optimization objective. The next paragraph will build directly on this point by introducing the computational subspace more carefully and defining leakage as population transfer outside it.

2.2.2 Computational subspace and leakage

The effective qubit description of a superconducting circuit is obtained by selecting a low-dimensional sector of the full Hilbert space and declaring it to be the logical space used for

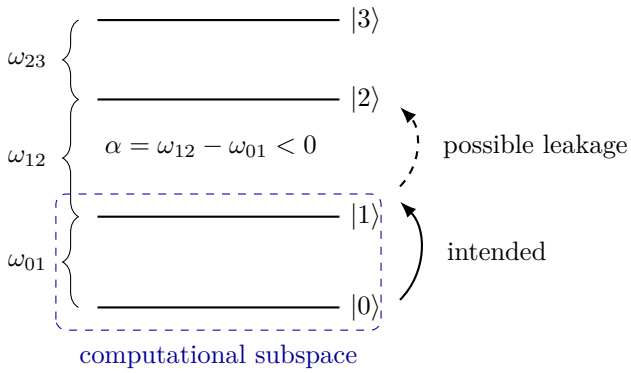

Figure 2.3: Schematic spectrum of a weakly anharmonic superconducting qubit. The two lowest levels define the computational subspace, while the higher levels remain physically present. Because the transition spacings are unequal but not widely separated, a drive resonant with the logical transition can still weakly couple to higher transitions, especially under strong or short pulses.

computation. For a single weakly anharmonic mode, this logical sector is the span of the two lowest eigenstates,

$$\mathcal{H}^{(1)}_{\text{comp}} = \text{span}\{|0\rangle, |1\rangle\}, \tag{2.56}$$

while the full physical Hilbert space remains

$$\mathcal{H}^{(1)}_{\text{phys}} = \text{span}\{|0\rangle, |1\rangle, |2\rangle, |3\rangle, \dots\}. \tag{2.57}$$

For a register of n such devices, the logical computational subspace is the tensor product of the local qubit sectors,

$$\mathcal{H}_{\text{comp}} = \left(\mathcal{H}^{(1)}_{\text{comp}}\right)^{\otimes n} \cong (\mathbb{C}^2)^{\otimes n}, \tag{2.58}$$

with dimension

$$d_{\text{comp}} = 2^n. \tag{2.59}$$

The full physical space is larger,

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_{\text{comp}} \oplus \mathcal{H}_{\text{leak}}, \tag{2.60}$$

where $\mathcal{H}_{\text{leak}}$ collects all states outside the intended qubit sector. The decomposition in Eq. (2.60) is conceptually central: a quantum gate is specified only on $\mathcal{H}_{\text{comp}}$, but the actual time evolution takes place on $\mathcal{H}_{\text{phys}}$ [**?**, **?**, **?**].

It is therefore natural to introduce the orthogonal projectors onto the computational and leakage sectors,

$$\Pi_{\text{comp}} = \sum_{\mu=1}^{d_{\text{comp}}} |\mu\rangle\langle\mu|, \qquad \Pi_{\text{leak}} = \mathbb{I} - \Pi_{\text{comp}}, \tag{2.61}$$

where $\{|\mu\rangle\}$ is any orthonormal basis of $\mathcal{H}_{\text{comp}}$. For a state $|\psi(t)\rangle \in \mathcal{H}_{\text{phys}}$, initialized inside the computational subspace,

$$\Pi_{\text{comp}} |\psi(0)\rangle = |\psi(0)\rangle, \tag{2.62}$$

the instantaneous leakage probability is defined as

$$L_\psi(t) := \langle\psi(t)|\Pi_{\text{leak}}|\psi(t)\rangle = 1 - \langle\psi(t)|\Pi_{\text{comp}}|\psi(t)\rangle. \tag{2.63}$$

Equation (2.63) gives the most direct meaning of leakage: it is simply the population that has escaped the logical sector. Since Π_{leak} is a projector, one has

$$0 \le L_\psi(t) \le 1. \tag{2.64}$$

If $L_\psi(t) = 0$, the state remains fully inside $\mathcal{H}_{\text{comp}}$; if $L_\psi(t) > 0$, then part of the wavefunction has reached states that are outside the intended qubit encoding.

At the gate level one is interested not in a single initial state, but in how well the full control protocol preserves the computational subspace on average. Let $U(t, 0)$ denote the physical unitary evolution generated by the driven multilevel Hamiltonian. For an arbitrary pure input state $|\psi\rangle \in \mathcal{H}_{\text{comp}}$, the survival probability in the computational subspace after time t is

$$P_{\text{stay}}(\psi, t) = \langle\psi|U^\dagger(t, 0)\, \Pi_{\text{comp}}\, U(t, 0)|\psi\rangle, \tag{2.65}$$

and the corresponding leakage is

$$L(\psi, t) = 1 - P_{\text{stay}}(\psi, t). \tag{2.66}$$

A natural averaged quantity is obtained by integrating over all pure states in the computational subspace with respect to the Haar measure,

$$\overline{L}(t) = 1 - \int_{\mathcal{H}_{\text{comp}}} d\psi\ \langle\psi|U^\dagger(t, 0)\, \Pi_{\text{comp}}\, U(t, 0)|\psi\rangle. \tag{2.67}$$

Using the identity

$$\int_{\mathcal{H}_{\text{comp}}} d\psi\ \langle\psi|A|\psi\rangle = \frac{1}{d_{\text{comp}}} \text{Tr}_{\text{comp}}(A), \tag{2.68}$$

valid for any operator A acting on $\mathcal{H}_{\text{comp}}$, Eq. (2.67) becomes

$$\overline{L}(t) = 1 - \frac{1}{d_{\text{comp}}} \text{Tr}\Big(\Pi_{\text{comp}} U^\dagger(t, 0) \Pi_{\text{comp}} U(t, 0) \Pi_{\text{comp}}\Big). \tag{2.69}$$

Equation (2.69) is especially useful in numerical simulations, because it directly quantifies the average fraction of logical population that fails to remain in the computational sector.

To understand the origin of leakage, it is useful to decompose the full Hamiltonian with respect to the direct sum in Eq. (2.60). Writing

$$H(t) = \Pi_{\text{comp}} H(t) \Pi_{\text{comp}} + \Pi_{\text{leak}} H(t) \Pi_{\text{leak}} + \Pi_{\text{comp}} H(t) \Pi_{\text{leak}} + \Pi_{\text{leak}} H(t) \Pi_{\text{comp}}, \tag{2.70}$$

one may define the block-diagonal part

$$H_{\text{diag}}(t) := \Pi_{\text{comp}} H(t) \Pi_{\text{comp}} + \Pi_{\text{leak}} H(t) \Pi_{\text{leak}}, \tag{2.71}$$

and the block-off-diagonal part

$$H_{\text{off}}(t) := \Pi_{\text{comp}} H(t) \Pi_{\text{leak}} + \Pi_{\text{leak}} H(t) \Pi_{\text{comp}}. \tag{2.72}$$

The interpretation is immediate: $H_{\text{diag}}(t)$ generates dynamics *within* each sector, whereas $H_{\text{off}}(t)$ couples the computational subspace to the leakage sector and is therefore the fundamental microscopic source of leakage. In this sense, leakage is not an external add-on to the control problem; it is already encoded in the multilevel structure of the Hamiltonian itself [**?**, **?**, **?**].

This block structure is particularly transparent in the two-qubit gmon model studied later in this thesis. There, the computational subspace is

$$\Omega_0 = \text{span}\{|00\rangle, |10\rangle, |01\rangle, |11\rangle\}, \tag{2.73}$$

while higher-excitation sectors, such as

$$\Omega_1 = \text{span}\{|20\rangle, |21\rangle, |12\rangle, |02\rangle\}, \tag{2.74}$$

belong to the leakage space. In the notation of the reference paper, the block-off-diagonal Hamiltonian term is precisely the component that couples different energy subspaces and is

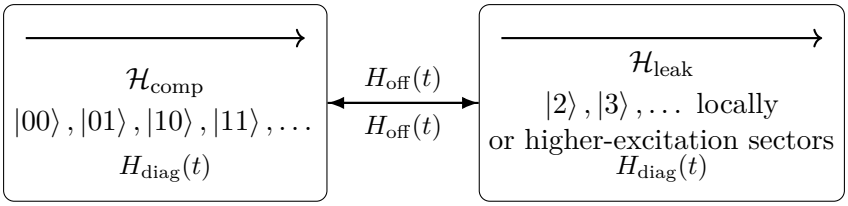

Figure 2.4: Schematic decomposition of the physical Hilbert space into computational and leakage sectors. The block-diagonal Hamiltonian generates evolution within each sector, while the block-off-diagonal part $H_{\text{off}}(t)$ couples the two sectors and is the microscopic source of leakage.

identified as the culprit behind leakage errors [**?**]. The same paper further emphasizes that the relevant energy-scale separation is between intra-subspace couplings and the gap Δ between Ω_0 and higher sectors, which is why leakage becomes especially severe when controls are too strong or too rapidly modulated [**?**].

A further distinction, important both conceptually and practically, is that between *coherent* and *incoherent* leakage. Coherent leakage refers to a reversible unitary transfer of amplitude between $\mathcal{H}_{\text{comp}}$ and $\mathcal{H}_{\text{leak}}$, generated by the Hamiltonian couplings in Eq. (2.72). Since the full evolution remains unitary, the leaked amplitude can in principle return to the computational sector at a later time. In contrast, incoherent leakage denotes effectively irreversible loss from the computational subspace, for example through decay to environmental modes, photon loss, or other open-system processes. In the language of the reference literature, even non-adiabatic excursions induced during time-dependent modulation may be treated as effectively incoherent in practice when the backflow to the computational subspace does not occur on the relevant control timescale [**?**].

This distinction matters because coherent and incoherent leakage affect gate design differently. Coherent leakage may partially cancel through interference, and therefore depends sensitively on pulse shape, phase accumulation, and gate duration. Incoherent leakage, by contrast, accumulates more like a genuine loss mechanism and directly degrades the logical channel implemented on the qubit subspace. A control pulse can therefore have high apparent fidelity on a restricted logical model and yet still perform poorly on the real device if it transiently or permanently populates higher levels. This is one of the main reasons why modern pulse-engineering methods for superconducting qubits explicitly account for leakage rather than optimizing only a projected two-level model [**?**, **?**, **?**].

It is also useful to distinguish leakage from ordinary logical errors that remain inside the computational subspace. If a pulse maps $|0\rangle$ and $|1\rangle$ to the wrong superpositions of $|0\rangle$ and $|1\rangle$, then the error is purely logical and can still be described by an incorrect unitary acting on $\mathcal{H}_{\text{comp}}$. Leakage is different: once population reaches $\mathcal{H}_{\text{leak}}$, the resulting process is no longer representable as a unitary gate on the logical Hilbert space alone. This means that gate infidelity and leakage quantify genuinely different failure modes, and both must be monitored separately in any realistic multilevel control problem.

For this reason, the preservation of the computational subspace should be regarded as a structural requirement of quantum control, not merely as a secondary diagnostic. The goal is not only to approximate a target logical unitary but to do so while keeping the evolution effectively confined to $\mathcal{H}_{\text{comp}}$. In the remainder of the thesis, this idea will reappear in a more quantitative form through explicit leakage penalties and bounds. In particular, the UFO framework studied later augments the gate-infidelity objective by a dedicated leakage term constructed from the block-off-diagonal dynamics of the full multilevel Hamiltonian [**?**].

2.2.3 The speed–leakage tradeoff

A central difficulty of quantum control in weakly anharmonic superconducting devices is that the desire for short gate times competes directly with the need to suppress leakage outside the computational subspace. In principle, faster gates are desirable because they reduce the exposure of the system to decoherence, relaxation, dephasing, and low-frequency calibration drift. In practice, however, the implementation of a shorter gate typically requires either stronger control amplitudes, more abrupt temporal modulation, or both. In multilevel systems, these same features tend to increase the coupling to unwanted transitions and therefore raise the probability of populating non-computational states [**?**, **?**, **?**]. This tension is one of the defining constraints of pulse engineering in superconducting qubits and is a key reason why realistic control optimization must treat runtime and leakage simultaneously rather than as independent objectives [**?**].

TODO: Demonstrate this The origin of this tradeoff can already be seen in the simplest driven weakly anharmonic model. Consider a three-level truncation of a superconducting mode, with logical states $|0\rangle$ and $|1\rangle$ and the first leakage level $|2\rangle$. In a frame rotating resonantly with the $|0\rangle \leftrightarrow |1\rangle$ transition, a standard effective Hamiltonian takes the form

$$\hat{H}_{3\mathrm{lvl}}(t) = \frac{\Omega(t)}{2}\left(|0\rangle\langle 1| + |1\rangle\langle 0| + \lambda\,|1\rangle\langle 2| + \lambda\,|2\rangle\langle 1|\right) + \alpha\,|2\rangle\langle 2|\,, \tag{2.75}$$

where $\Omega(t)$ is the drive envelope, $\alpha = \omega_{12} - \omega_{01} < 0$ is the anharmonicity, and λ is the relative matrix element for the $|1\rangle \leftrightarrow |2\rangle$ transition. Equation (2.75) makes the problem explicit: the same drive that implements logical rotations also couples to the leakage transition, which is only detuned by the finite anharmonicity α. Thus, unlike in an ideal two-level system, faster control does not merely accelerate the intended dynamics; it also enhances the unwanted coupling to higher levels [**?**, **?**, **?**].

In the perturbative off-resonant regime,

$$|\Omega(t)| \ll |\alpha|, \tag{2.76}$$

the unwanted excitation of the leakage level can be estimated heuristically from standard off-resonant driving arguments. For a drive amplitude Ω that is approximately constant over the relevant interval, the leakage amplitude scales as

$$|c_2(t)| \sim \frac{\lambda\,\Omega}{2|\alpha|}, \tag{2.77}$$

so that the corresponding leakage probability obeys the rough scaling law

$$P_{\mathrm{leak}} \sim \left(\frac{\lambda\,\Omega}{2|\alpha|}\right)^2. \tag{2.78}$$

Although Eqs. (2.77)–(2.78) are only heuristic estimates, they capture an essential fact: increasing the control amplitude in order to shorten the gate time tends to increase leakage quadratically relative to the small parameter $\Omega/|\alpha|$. This is precisely why the short-time regime is particularly challenging in weakly nonlinear oscillators, and why the leading error mechanism for fast gates is often leakage rather than an ordinary logical over-rotation [**?**, **?**].

A complementary perspective comes from the time-frequency structure of the pulse. A control signal with characteristic duration T_{g} necessarily has a spectral width of order

$$\Delta\omega_{\mathrm{pulse}} \sim \frac{1}{T_{\mathrm{g}}}, \tag{2.79}$$

so that shorter pulses possess broader frequency content. Spectral selectivity of the logical transition requires that the pulse bandwidth remain small compared with the detuning to the nearest unwanted transition. In weakly anharmonic qubits, this means roughly

$$\Delta\omega_{\mathrm{pulse}} \ll |\alpha|, \tag{2.80}$$

or equivalently,

$$T_{\mathrm{g}} \gg \frac{1}{|\alpha|}. \tag{2.81}$$

Equation (2.81) should not be interpreted as a rigorous lower bound on the gate time, since shaped pulses and interference effects can do substantially better than naive square-pulse estimates. Its conceptual role is instead to show that fast gating is constrained by spectral crowding: once the pulse becomes too short, its Fourier components inevitably overlap the leakage transition and the two-level approximation begins to break down [**?**, **?**].

The same conclusion can be reached from an adiabaticity viewpoint. As discussed in the previous subsection, the full Hamiltonian may be decomposed into block-diagonal and block-off-diagonal components with respect to the splitting $\mathcal{H}_{\mathrm{phys}} = \mathcal{H}_{\mathrm{comp}} \oplus \mathcal{H}_{\mathrm{leak}}$. The block-off-diagonal term $H_{\mathrm{off}}(t)$ couples the computational and leakage sectors, while rapid temporal modulation enhances non-adiabatic transfer between them. **TODO:** Write the complete derivation of this expression However it is possible to derive analitically a leakage bound through the utilization of the time-dependent Schrieffer–Wolff framework, where the accumulated coherent leakage is controlled not only by the size of the block-off-diagonal Hamiltonian but also by the rate at which it varies in time. In schematic form, their bound reads

$$L_{\mathrm{tot}} \lesssim \frac{\|\hat{H}_{\mathrm{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\mathrm{od}}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\mathrm{od}}(t)}{dt^2} \right\| dt, \tag{2.82}$$

where $\Delta(t)$ denotes the relevant energy gap and $\hat{H}_{\mathrm{od}}(t)$ the effective block-off-diagonal Hamiltonian in the transformed basis [**?**]. Equation (2.82) is highly instructive for the present discussion: it shows that leakage is aggravated both by large off-diagonal couplings and by rapid pulse modulation, since faster pulses typically make the derivative term larger. Thus, trying to shorten the gate time too aggressively can increase leakage even if the control amplitudes themselves remain moderate.

The speed–leakage tradeoff is therefore not merely a matter of stronger fields; it is also a matter of pulse smoothness and spectral engineering. This is the physical motivation behind derivative-based correction schemes such as DRAG, where an additional quadrature proportional to the time derivative of the main pulse is introduced to cancel leakage-inducing terms order by order [**?**, **?**]. More generally, modern optimal-control methods shape the pulse so that fast logical evolution is achieved while destructive interference suppresses net population transfer to higher levels. Experimental work on superconducting transmons confirms that this balance is nontrivial: very short gates can indeed be realized, but only with carefully optimized pulse shapes that explicitly control leakage [**?**].

At the level of control design, the existence of this tradeoff implies that gate optimization is intrinsically multi-objective. Let $\boldsymbol{u}(t)$ denote the control protocol and $T[\boldsymbol{u}]$ its final runtime. Then an abstract cost functional consistent with the preceding discussion is

$$\mathcal{J}[\boldsymbol{u}] = w_F\, \varepsilon_{\mathrm{gate}}[\boldsymbol{u}] + w_L\, L[\boldsymbol{u}] + w_T\, T[\boldsymbol{u}] + \cdots, \tag{2.83}$$

where $\varepsilon_{\mathrm{gate}}$ measures logical inaccuracy, $L[\boldsymbol{u}]$ is a leakage functional, and the omitted terms may encode boundary constraints, bandwidth penalties, fluence costs, or robustness objectives. Equation (2.83) summarizes the central message of this subsection: making a gate faster is not automatically beneficial unless the resulting increase in leakage remains under control. This perspective is directly aligned with the philosophy of the UFO framework, which later combines gate infidelity, leakage, control constraints, and total runtime into a single optimization objective [**?**].

From the viewpoint of superconducting-qubit physics, one may summarize the tradeoff as follows. Long pulses are usually easier to make spectrally selective and therefore tend to suppress leakage, but they suffer from greater exposure to decoherence and slow noise. Very short pulses

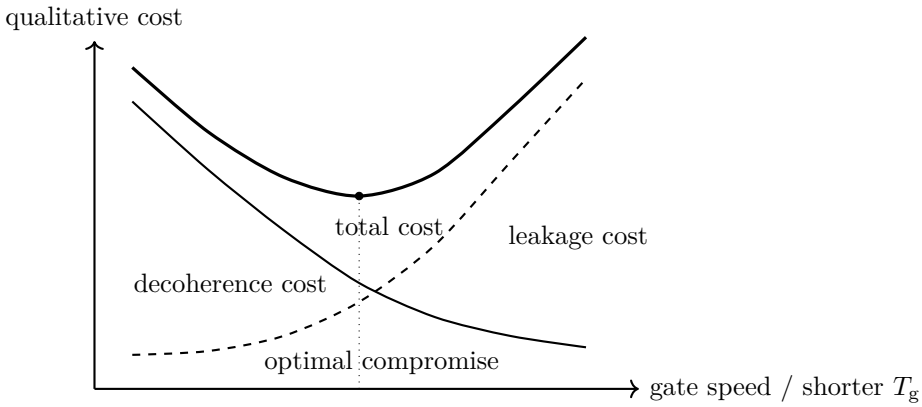

Figure 2.5: Qualitative illustration of the speed–leakage tradeoff in weakly anharmonic superconducting qubits. Slower gates suffer more from decoherence and low-frequency noise, whereas faster gates typically increase leakage through stronger driving, broader spectral content, and larger non-adiabatic effects. The physically relevant control problem is therefore a compromise between these competing tendencies.

reduce the total interaction time with the environment, but they broaden the pulse spectrum, enhance non-adiabaticity, and increase the risk of population transfer into higher levels. The practically relevant regime is therefore neither the slowest nor the fastest one in an absolute sense, but the regime in which runtime has been reduced as much as possible while leakage and control-induced errors remain acceptably small. This balance is one of the central organizing principles of high-fidelity gate design in weakly anharmonic superconducting systems [**?**, **?**, **?**].

2.3 Quantum control theory background

2.3.1 Open-loop and closed-loop quantum control

A basic conceptual distinction in control theory is that between *open-loop* and *closed-loop* strategies. In an open-loop protocol, the control fields are prescribed in advance and then applied to the system without using measurement outcomes obtained during the same realization of the dynamics. In a closed-loop protocol, by contrast, the controls are adapted using information returned by the system itself, either continuously or iteratively, through some form of measurement and feedback [**?**, **?**, **?**]. This distinction is especially important in quantum systems, because measurement is not a passive readout operation: it can introduce statistical uncertainty, backaction, and experimental latency, all of which influence the practicality of feedback-based control. For this reason, the open-loop/closed-loop dichotomy in quantum engineering is both conceptually rich and experimentally consequential.

In the model-based open-loop setting, one begins from a dynamical model

$$\dot{\rho}(t) = \mathcal{L}_{\boldsymbol{u}(t)}[\rho(t)], \tag{2.84}$$

where $\rho(t)$ is the quantum state (or density matrix), $\mathcal{L}_{\boldsymbol{u}(t)}$ is the generator of the dynamics, and $\boldsymbol{u}(t)$ denotes the externally chosen controls. For a closed quantum system this reduces to the Schrödinger or von Neumann equation,

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H(t;\boldsymbol{u}), \rho(t)], \tag{2.85}$$

while in the presence of dissipation one may instead work with a Lindblad generator. The optimization problem is then posed entirely at the level of the model: one selects a parameterized

family of controls,

$$\boldsymbol{u}(t) = \boldsymbol{u}(t; \boldsymbol{\theta}), \tag{2.86}$$

and solves

$$\boldsymbol{\theta}^\star = \arg\min_{\boldsymbol{\theta}} \ \mathcal{J}_{\text{model}}(\boldsymbol{\theta}), \tag{2.87}$$

where $\mathcal{J}_{\text{model}}$ is a model-based objective such as gate infidelity, leakage, runtime, or some weighted combination thereof. Once $\boldsymbol{\theta}^\star$ has been found numerically, the resulting pulse is implemented experimentally without any intra-shot adaptation. This is the standard framework of much of quantum optimal control, including GRAPE-, Krotov-, GOAT-, and many reinforcement-learning-based approaches when training is performed in simulation [**?**].

The main advantage of open-loop control is that it can exploit a detailed physical model and powerful numerical optimization tools before any experiment is run. It is therefore well suited to platforms for which the Hamiltonian and noise mechanisms are sufficiently well characterized. Moreover, since no measurements are required during the gate execution itself, open-loop control avoids the complications of quantum backaction and avoids the hardware overhead associated with real-time feedback. However, its main weakness is equally clear: the optimized pulse is only as good as the model on which it is based. If the actual device differs significantly from the assumed Hamiltonian, if transfer functions distort the waveform, or if noise sources are incompletely modeled, then a pulse that appears optimal in simulation may perform suboptimally on hardware [**?**, **?**]. This model mismatch is one of the main reasons why purely open-loop design can become unreliable in realistic experiments.

Closed-loop control is introduced precisely to address this gap between model and hardware. In its most literal form, it means that the control applied at time t_k depends on measurement information gathered at earlier times,

$$\boldsymbol{u}(t_k) = \pi_k(y_0, y_1, \dots, y_{k-1}) \,, \tag{2.88}$$

where $\{y_j\}$ denotes the measurement record and π_k is a control law or feedback policy. In this picture, the system is repeatedly observed and the control is updated in real time according to the observed outcomes. Because quantum measurements disturb the state, such feedback must be described together with the measurement backaction. If $\mathcal{M}_y$ denotes the measurement operation associated with outcome y, then the post-measurement state is

$$\rho \ \longmapsto \ \frac{\mathcal{M}_y(\rho)}{\mathrm{Tr}[\mathcal{M}_y(\rho)]}, \tag{2.89}$$

with outcome probability

$$p(y) = \mathrm{Tr}[\mathcal{M}_y(\rho)] \,. \tag{2.90}$$

Thus, unlike in classical control, the act of acquiring information modifies the system whose dynamics one is trying to steer. This makes genuine feedback control more delicate in the quantum regime, especially for fast gate operations where measurement latency and finite measurement fidelity can become serious bottlenecks [**?**].

TODO: Recheck if I want this in my thesis since In quantum-information experiments, however, the phrase *closed-loop optimization* is often used in a broader and more practical sense. Rather than updating the control during a single gate execution, one performs repeated experimental shots with a fixed pulse $\boldsymbol{u}(t; \boldsymbol{\theta}_r)$, estimates a performance metric from the measurement outcomes, and then updates the pulse parameters between shots according to some classical optimization rule. If $\widehat{\mathcal{J}}_{\text{exp}}(\boldsymbol{\theta}_r)$ denotes the experimentally estimated cost at iteration r, then the iterative update can be written schematically as

$$\boldsymbol{\theta}_{r+1} = \mathcal{A}\Big(\boldsymbol{\theta}_r, \widehat{\mathcal{J}}_{\text{exp}}(\boldsymbol{\theta}_r), \widehat{\nabla \mathcal{J}}_{\text{exp}}(\boldsymbol{\theta}_r), \dots\Big) \,, \tag{2.91}$$

where $\mathcal{A}$ denotes an optimization rule such as gradient descent, finite-difference search, Bayesian optimization, CRAB/dCRAB-style updates, or a reinforcement-learning policy improvement step. In this experimentally relevant sense, the loop is "closed" not because the Hamiltonian is modified in real time during the gate, but because the control design is iteratively informed by measured performance on the actual device [**?**, **?**]. This distinction is worth stating explicitly, because the words *closed-loop control* may refer either to continuous measurement-based feedback or to shot-to-shot in situ calibration.

The practical strengths of closed-loop approaches are clear. Since the performance metric is measured directly on the physical device, systematic model errors, waveform distortions, and unknown hardware nonlinearities are automatically incorporated into the optimization. This can make closed-loop strategies more robust than purely model-based open-loop methods when the system is imperfectly characterized. Experimental studies indeed show that closed-loop quantum control can outperform open-loop control in settings where model inaccuracies or hardware distortions dominate the error budget [**?**, **?**]. In addition, closed-loop schemes can target figures of merit that are directly operational, such as randomized-benchmarking-based performance estimates or experimentally reconstructed gate fidelities, rather than relying exclusively on idealized simulation objectives.

The price paid for this robustness is experimental cost. Closed-loop methods generally require many repeated measurements, and therefore many experimental iterations, in order to estimate objective functions or gradients with acceptable statistical precision. In fast gate optimization this can become a serious limitation, since readout is often much slower than the gate itself and can perturb subsequent operations. This concern is emphasized in modern literature: the authors note that many efforts to improve robustness against control errors have focused on closed-loop feedback optimization, but that frequent measurements remain relatively slow and can degrade later fidelities, making such approaches less practical for near-term devices [**?**]. In contrast, their approach remains fundamentally open-loop during gate execution, while incorporating robustness and leakage suppression directly into an analytic cost function that can be optimized numerically.

This distinction leads naturally to a hybrid viewpoint. Purely open-loop control is efficient when the model is accurate enough; purely closed-loop control is attractive when the device itself must reveal the relevant imperfections; and many modern workflows combine the two. A common strategy is to first compute a high-quality pulse from a detailed simulation and then refine or calibrate it experimentally through an iterative in situ procedure. Schematically, one may write

$$\boldsymbol{\theta}_0 = \arg\min_{\boldsymbol{\theta}} \mathcal{J}_{\text{model}}(\boldsymbol{\theta}), \qquad \boldsymbol{\theta}^\star = \text{Refine}\Big(\boldsymbol{\theta}_0, \widehat{\mathcal{J}}_{\text{exp}}\Big)\,, \tag{2.92}$$

where the first step is model-based and the second experimental. This hybrid logic has become increasingly important in quantum control software and experimental practice, precisely because it attempts to combine the numerical efficiency of open-loop design with the hardware realism of closed-loop refinement [**?**, **?**].

For the purposes of this thesis, the most important point is the following. The UFO framework analyzed later is an *open-loop* control framework in the sense that the control pulse is optimized from a Hamiltonian model and then applied without measurement-based feedback during the gate. At the same time, its motivation is shaped by the practical limitations of closed-loop techniques: because repeated measurements are slow and because the exact error model is often unavailable, the framework seeks to encode robustness and leakage suppression directly into the cost function rather than outsourcing them to experimental feedback [**?**]. The distinction developed in this subsection is therefore not merely terminological; it explains why the thesis focuses on analytic objectives and model-based control design, while still recognizing that future hardware-level implementations may benefit from hybrid or partially closed-loop refinements.

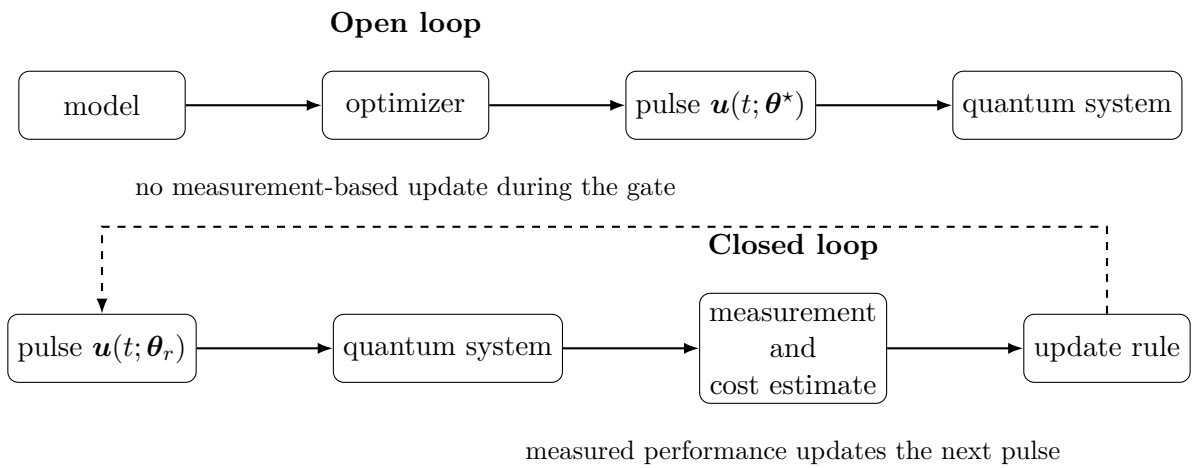

Figure 2.6: Schematic distinction between open-loop and closed-loop quantum control. In open-loop control, pulse design is performed from a model and the resulting pulse is applied without feedback during the gate. In closed-loop control, experimental information is fed back to improve the next control decision, either in real time or iteratively between repeated runs.

2.3.2 Classical optimal-control methods for quantum systems

Before introducing reinforcement-learning-based control, it is useful to summarize the main families of *classical* optimal-control methods that have shaped modern quantum pulse engineering. In all of them, the basic problem is the same: one specifies a dynamical model, chooses a parameterization of the control fields, defines a cost functional that encodes the target task, and then numerically optimizes the control so as to minimize that cost [?, ?, ?, ?, ?]. These methods are typically *open-loop* and *model-based*: they rely on repeated simulation of the controlled Schrödinger or Lindblad dynamics, and they assume that the model captures the relevant physics with sufficient fidelity.

A general quantum optimal-control problem can be written as follows. Let the system dynamics be governed by a controlled Hamiltonian

$$H(t;\boldsymbol{u}) = H_{\mathrm{d}} + \sum_{k=1}^{m} u_k(t)\, H_k, \tag{2.93}$$

with propagator $U(t)$ satisfying

$$i\hbar \frac{d}{dt} U(t) = H(t;\boldsymbol{u})\, U(t), \qquad U(0) = \mathbb{I}. \tag{2.94}$$

The optimization objective is encoded in a cost functional of the generic form

$$\mathcal{J}[\boldsymbol{u}] = \Phi\big(U(T)\big) + \int_0^T g\big(\boldsymbol{u}(t), t\big)\, dt, \tag{2.95}$$

where Φ is a terminal cost, for example a gate-infidelity or state-transfer error, and g is a running cost encoding penalties such as pulse energy, smoothness, boundary conditions, or robustness terms. In practice the control is often discretized on a time grid

$$u_k(t) = u_{k,\ell}, \qquad t \in [t_\ell, t_{\ell+1}), \qquad \ell = 0, \ldots, N-1, \tag{2.96}$$

which reduces the infinite-dimensional search problem to a finite-dimensional optimization over the amplitudes $\{u_{k,\ell}\}$.

One of the most influential methods in this setting is *GRAPE* (Gradient Ascent Pulse Engineering), originally developed in the NMR community and later widely adopted in quantum

information processing [?, ?]. The central idea of GRAPE is to compute the gradient of the cost functional with respect to each control amplitude $u_{k,\ell}$ efficiently by combining forward propagation of the state or propagator with backward propagation of an adjoint variable. In its simplest gradient-descent or gradient-ascent form, the update reads

$$u_{k,\ell}^{(r+1)} = u_{k,\ell}^{(r)} - \eta \frac{\partial \mathcal{J}}{\partial u_{k,\ell}}, \tag{2.97}$$

where η is a step size and r indexes the optimization iteration. The practical strength of GRAPE is that all control amplitudes can be optimized simultaneously with gradients that scale efficiently once the forward and backward trajectories have been computed. This makes it especially powerful for high-dimensional pulse optimization with fine time discretization.

For state-transfer objectives, the structure of the GRAPE gradient can be illustrated in a standard bilinear setting. Let $|\psi_\ell\rangle$ be the forward-propagated state at time step ℓ and let $|\chi_{\ell+1}\rangle$ be the corresponding backward-propagated adjoint state. Then the gradient contribution of the control amplitude $u_{k,\ell}$ has the generic form

$$\frac{\partial \mathcal{J}}{\partial u_{k,\ell}} \propto -\operatorname{Im} \langle \chi_{\ell+1} | H_k | \psi_\ell \rangle \,, \tag{2.98}$$

up to convention-dependent prefactors and discretization factors. Equation (2.98) makes clear why GRAPE is computationally attractive: all gradients are obtained from quantities already available during propagation. Later refinements, such as second-order GRAPE and quasi-Newton acceleration, improve convergence speed and numerical robustness without changing this core adjoint-based structure [?].

A second major family is based on *Krotov-type* updates. In the quantum-control literature, Krotov's method is valued for its variational foundation and for the possibility of obtaining *monotonic convergence*, meaning that the objective improves at every iteration under appropriate assumptions [?, ?]. Instead of viewing the optimization merely as a local gradient step in a large Euclidean parameter space, Krotov's method constructs an update directly from the stationarity conditions of the augmented control functional. In a common first-order setting, one obtains an update of the form

$$u_k^{(r+1)}(t) = u_k^{(r)}(t) + \frac{S_k(t)}{\lambda_k} \operatorname{Im} \left\langle \chi^{(r)}(t) \middle| \frac{\partial H}{\partial u_k} \middle| \psi^{(r+1)}(t) \right\rangle, \tag{2.99}$$

where $S_k(t)$ is a shape function enforcing boundary smoothness or turn-on/turn-off constraints, λ_k is a step-size-like penalty parameter, and $|\chi^{(r)}(t)\rangle$ and $|\psi^{(r+1)}(t)\rangle$ are the adjoint and forward states associated with consecutive iterations. The characteristic feature of Eq. (2.99) is that the new control depends on the newly propagated state $|\psi^{(r+1)}(t)\rangle$, which is one reason why Krotov-type schemes differ structurally from straightforward GRAPE updates.

While GRAPE and Krotov are both gradient-based, they differ in philosophy and practical use. GRAPE is often especially convenient when the controls are piecewise constant and when one wishes to use standard unconstrained optimization tools such as BFGS or L-BFGS in a high-dimensional parameter space. Krotov's method is often preferred when monotonic convergence, time-dependent constraints, or specific functional forms of the objective are important. Both, however, belong to the same broad class: they optimize a cost functional by exploiting gradient information computed from the controlled dynamics [?, ?].

A rather different strategy is embodied by *CRAB* (Chopped Random Basis), which reduces the search space by expanding the control in a truncated basis and optimizing only the corresponding coefficients [?]. A typical CRAB parameterization reads

$$u_k(t) = u_k^{(0)}(t) \left[1 + \sum_{m=1}^{M} \left(A_{k,m} \sin \omega_{k,m} t + B_{k,m} \cos \omega_{k,m} t \right) \right], \tag{2.100}$$

where $u_k^{(0)}(t)$ is a reference pulse, the frequencies $\omega_{k,m}$ are chosen or randomized within a suitable range, and the coefficients $A_{k,m}, B_{k,m}$ become the optimization variables. The conceptual advantage is clear: instead of optimizing hundreds or thousands of piecewise-constant amplitudes, one works in a much lower-dimensional space of coefficients that can naturally encode smoothness, bandwidth restrictions, and experimental simplicity. Because of this reduced parametrization, CRAB is particularly useful when gradients are expensive or unavailable, and it is often combined with gradient-free optimizers such as Nelder–Mead or simplex-type search strategies [**?**]. This basis-restricted perspective is also attractive in experimental closed-loop refinements, where each additional optimization variable may carry a substantial measurement cost.

TODO: Deciding if I want to keep this stuff in, since crab and krotov are not the main focus of my thesis

A more recent method that addresses some of the limitations of both piecewise-constant gradient methods and purely gradient-free basis search is *GOAT* (Gradient Optimization of Analytic conTrols) [**?**]. The guiding idea of GOAT is to keep the controls in a low-dimensional but analytic parameterization,

$$u_k(t) = u_k(t; \boldsymbol{\theta}), \tag{2.101}$$

and to compute exact gradients with respect to the parameters $\boldsymbol{\theta}$ by propagating not only the system evolution but also its derivatives. If $U(t; \boldsymbol{\theta})$ denotes the propagator, then differentiation of Eq. (2.94) with respect to a parameter θ_j yields the coupled evolution equation

$$i\hbar \frac{d}{dt}\left(\frac{\partial U}{\partial \theta_j}\right) = H(t; \boldsymbol{\theta})\frac{\partial U}{\partial \theta_j} + \frac{\partial H(t; \boldsymbol{\theta})}{\partial \theta_j} U(t; \boldsymbol{\theta}), \tag{2.102}$$

with initial condition

$$\frac{\partial U(0; \boldsymbol{\theta})}{\partial \theta_j} = 0. \tag{2.103}$$

Thus, once the augmented system formed by U and its derivatives is propagated, one obtains the gradient of the final cost $\partial \mathcal{J}/\partial \theta_j$ directly. The principal appeal of GOAT is that it combines gradient efficiency with flexible, low-dimensional, hardware-friendly pulse parametrizations, making it particularly attractive for superconducting qubits where pulse simplicity and subsequent experimental calibration matter greatly [**?**].

From a broader perspective, GRAPE, Krotov, CRAB, and GOAT should not be viewed as mutually exclusive competitors so much as different design philosophies for the same underlying problem. GRAPE and Krotov prioritize gradient information and direct optimization in control space; CRAB prioritizes dimensional reduction and basis flexibility; GOAT attempts to retain exact gradients while working in a compact analytic parameter space. The best method depends on the physics, the size of the system, the availability of gradients, the presence of constraints, and whether the pulse will later need to be calibrated on hardware [**?**, **?**, **?**]. In superconducting circuits, for example, piecewise-constant pulses may be numerically convenient but experimentally awkward, while smooth low-parameter pulses may be easier to calibrate and less sensitive to transfer-function distortions.

For the purposes of this thesis, these methods serve two roles. First, they provide the standard background language of open-loop model-based quantum control: objective functional, parameterization, gradient computation, and iterative pulse improvement. Second, they define the natural baseline against which reinforcement-learning-based methods should be interpreted. Reinforcement learning does not replace the optimal-control problem itself; rather, it offers an alternative way to search the space of control laws, especially in settings where the cost landscape is highly nonconvex, robustness matters strongly, or sequential structure can be exploited. The value of the approach studied in the reference paper is therefore best understood not in isolation, but against the mature backdrop of these classical optimal-control techniques.

It is useful to summarize the common structure of all these methods schematically as

$$\text{model} + \text{control ansatz} + \text{cost functional} + \text{numerical optimizer} \longrightarrow \text{optimized pulse}. \tag{2.104}$$

What distinguishes GRAPE, Krotov, CRAB, and GOAT is not the overall control objective, but the specific choice of control ansatz, gradient computation, and update rule used to navigate the optimization landscape.

2.3.3 Why robustness is difficult in quantum control

In an idealized optimal-control problem one assumes that the system Hamiltonian, the control transfer function, and the relevant noise model are all known accurately. Under these assumptions, one can optimize a pulse for a *nominal* model and obtain extremely high fidelities in simulation. In practice, however, quantum control is almost never deployed under such perfectly known conditions. Device parameters drift, couplings are incompletely calibrated, control electronics distort the intended waveform, environmental noise is only partially characterized, and higher-level dynamics may not be fully captured by the chosen effective model. As a result, a pulse that is optimal for a nominal model may perform poorly once implemented on the actual device [**?**,**?**,**?**]. This gap between nominal optimality and physical reliability is the basic reason robustness is difficult.

A convenient way to formalize this point is to write the dynamics as depending not only on the control $\boldsymbol{u}(t)$ but also on uncertain parameters $\boldsymbol{\delta}$:

$$\dot{\rho}(t) = \mathcal{L}_{\boldsymbol{u}(t),\,\boldsymbol{\delta}}[\rho(t)]. \tag{2.105}$$

Here $\boldsymbol{\delta}$ may represent static detuning errors, drift in anharmonicity, imprecise couplings, control-amplitude distortions, fluctuating calibration offsets, or environmental parameters. If one defines a nominal performance functional $\mathcal{J}(\boldsymbol{u};\boldsymbol{\delta})$, then the standard open-loop optimization problem corresponds to minimizing

$$\mathcal{J}_{\text{nom}}(\boldsymbol{u}) := \mathcal{J}(\boldsymbol{u};\boldsymbol{\delta}_0), \tag{2.106}$$

for a single reference value $\boldsymbol{\delta}_0$. A robust-control problem, by contrast, must account for a whole *set* or *distribution* of possible uncertainties. This immediately makes the problem more difficult, because the optimizer is no longer asked to perform well at one point in parameter space, but over a family of models.

There are several inequivalent ways to formulate robustness. In an *average-case* formulation one assumes that the uncertainty is drawn from a probability distribution $p(\boldsymbol{\delta})$ and optimizes the expected cost

$$\mathcal{J}_{\text{avg}}(\boldsymbol{u}) = \mathbb{E}_{\boldsymbol{\delta}\sim p}\left[\mathcal{J}(\boldsymbol{u};\boldsymbol{\delta})\right] = \int \mathcal{J}(\boldsymbol{u};\boldsymbol{\delta})\, p(\boldsymbol{\delta})\, d\boldsymbol{\delta}. \tag{2.107}$$

In a *worst-case* or minimax formulation one instead seeks to control the largest possible degradation over an uncertainty set Δ:

$$\mathcal{J}_{\text{wc}}(\boldsymbol{u}) = \sup_{\boldsymbol{\delta}\in\Delta} \mathcal{J}(\boldsymbol{u};\boldsymbol{\delta}). \tag{2.108}$$

A third common viewpoint is threshold-based, where one requires the fidelity or success probability to remain above a prescribed level with high probability:

$$\mathbb{P}_{\boldsymbol{\delta}\sim p}(F(\boldsymbol{u};\boldsymbol{\delta}) \geq F_{\min}) \geq 1-\varepsilon. \tag{2.109}$$

These three criteria encode genuinely different priorities. A control that is excellent on average can still fail badly on rare but important perturbations, while a minimax-optimal control may be too conservative for average experimental operation [**?**, **?**]. Thus, robustness is not a single scalar property but a design choice that depends on what failure modes one wishes to suppress.

A second source of difficulty is that the uncertainty set itself is often only partially known. In many experiments one does not know the exact noise law $p(\boldsymbol{\delta})$, and even if some approximate characterization is available, it may drift in time. This issue is explicitly emphasized in the literature, where the authors note that the exact model of control errors is often unavailable and may vary over time, so that one should ideally seek control strategies that remain reliable not merely for one precisely specified perturbation model, but across a broader family of experimentally relevant error processes [**?**]. Formally, this means that one often trains or optimizes under one distribution $p_{\text{train}}(\boldsymbol{\delta})$ while the actual deployment conditions follow another distribution $p_{\text{deploy}}(\boldsymbol{\delta})$, with

$$p_{\text{train}}(\boldsymbol{\delta}) \neq p_{\text{deploy}}(\boldsymbol{\delta}). \tag{2.110}$$

Equation (2.110) is the control-theoretic version of distribution shift: a pulse may appear robust in simulation but degrade under out-of-distribution perturbations that were absent or underrepresented during optimization.

A third difficulty is computational. To evaluate the nominal objective in Eq. (2.106), one usually needs only a single propagation of the system dynamics per iteration. By contrast, evaluating a robust objective typically requires many propagations over different uncertainty realizations or uncertainty-grid points. If one approximates the average-case objective by Monte Carlo sampling with M samples,

$$\mathcal{J}_{\text{avg}}(\boldsymbol{u}) \approx \frac{1}{M}\sum_{j=1}^{M} \mathcal{J}(\boldsymbol{u}; \boldsymbol{\delta}^{(j)}), \tag{2.111}$$

then every optimization step becomes roughly M times more expensive than the nominal one. This sampling burden can become severe in multi-qubit systems, especially when the control space is already high-dimensional. The difficulty is compounded when one wishes to include realistic open-system evolution, since each propagation may then require solving a Lindblad master equation rather than a closed-system Schrödinger equation [**?**, **?**, **?**]. Robustness is therefore not hard only because of physics; it is hard because it is computationally expensive to certify.

A fourth source of difficulty is geometric. The control landscape of a nominal problem is already high-dimensional and nonconvex. Introducing robustness means that the optimizer must now navigate not one landscape, but an *aggregate* landscape produced by averaging or extremizing over many perturbations. Even if each individual cost $\mathcal{J}(\boldsymbol{u}; \boldsymbol{\delta})$ were relatively benign, their superposition can produce a far more intricate structure with flatter regions, competing descent directions, and stronger tradeoffs between objectives. In local sensitivity analysis one may expand the cost around a nominal control $\boldsymbol{u}_0$ as

$$\mathcal{J}(\boldsymbol{u}_0 + \Delta\boldsymbol{u}) \approx \mathcal{J}(\boldsymbol{u}_0) + \nabla\mathcal{J}(\boldsymbol{u}_0)^\top \Delta\boldsymbol{u} + \frac{1}{2}\Delta\boldsymbol{u}^\top H_{\mathcal{J}}(\boldsymbol{u}_0)\Delta\boldsymbol{u}, \tag{2.112}$$

where $H_{\mathcal{J}}$ is the Hessian of the control objective. Robustness analysis based on such curvature information can be very informative, because it quantifies sensitivity to small perturbations in the controls. But it is also expensive in large systems: as the literature points out, canonical Hessian-based robustness analysis becomes computationally prohibitive as the number of control degrees of freedom grows, which is one reason the authors adopt a more scalable practical robustness definition [**?**]. Thus, even diagnosing whether a pulse is robust can already be difficult, before one attempts to optimize robustness itself.

Another complication is that different sources of robustness may conflict with one another. A control that is robust against amplitude noise may be fragile against detuning error; a control that is robust against quasistatic drift may be more sensitive to bandwidth truncation; a pulse that suppresses leakage may require longer duration and thus suffer more strongly from decoherence.

In this sense, robustness is not usually a single-axis improvement but a multidimensional tradeoff problem. A more realistic objective is therefore often of the form

$$\mathcal{J}_{\text{rob}}[\boldsymbol{u}] = w_F\, \varepsilon_{\text{gate}} + w_L\, L + w_T\, T + w_R\, \mathcal{R}[\boldsymbol{u}], \tag{2.113}$$

where $\mathcal{R}[\boldsymbol{u}]$ denotes a chosen robustness functional and the weights $\{w_F, w_L, w_T, w_R\}$ encode the tradeoff between fidelity, leakage, runtime, and robustness. There is generally no reason for these objectives to be simultaneously minimized by the same pulse. A robust-control problem is therefore often a Pareto problem rather than a single-objective one.

In superconducting qubits, the robustness problem is made sharper by the multilevel character of the hardware. Parameter uncertainty does not merely perturb the logical gate within the computational subspace; it can alter detunings, anharmonicities, and couplings in a way that changes the leakage channels themselves. Thus, robustness against model uncertainty and robustness against leakage are intertwined rather than separable. This is one reason why robust pulse design in weakly anharmonic systems often requires the full multilevel Hamiltonian rather than a simplified two-level approximation [**?**, **?**]. The same observation is directly relevant for the reference paper: its robust RL optimization is performed in a stochastic environment where perturbations are injected into physically important control parameters, and the resulting performance is then judged by averaged noisy gate fidelity and fidelity variance rather than by nominal accuracy alone [**?**].

It is therefore useful to distinguish two questions:

1. *Robustness evaluation:* given a control pulse, how sensitive is it to perturbations?

2. *Robustness optimization:* how should the pulse be modified so as to reduce that sensitivity?

The first is already nontrivial because it requires specifying the relevant uncertainty model and computing the corresponding sensitivity metric. The second is harder still because robustness must be optimized jointly with all the other physical criteria of the control problem. In that sense, robustness is difficult not because it is a secondary correction to nominal control, but because it fundamentally changes the optimization problem itself.

For the purposes of this thesis, the most important takeaway is that robustness cannot be treated as an afterthought. A pulse optimized only for nominal gate fidelity may be scientifically interesting, but it is not yet a practically convincing control solution unless one also understands how it behaves under uncertainty, noise, and imperfect modeling. This is precisely why the literature moves beyond a purely nominal objective and evaluates control performance under a family of stochastic perturbations, using average fidelity and fidelity variance as practical diagnostics of robustness [**?**]. The extension work proposed later in this thesis builds directly on this point by asking whether robustness learned under one uncertainty model generalizes to others.

2.4 Machine learning and RL for control

2.4.1 Why reinforcement learning is relevant for quantum control

Reinforcement learning (RL) is relevant to quantum control because many pulse-design problems are naturally sequential, stochastic, high-dimensional, and lack labeled training data. In contrast to supervised learning, where one learns from example input-output pairs, RL learns by trial and error through repeated interaction with an environment and by maximizing a scalar reward signal [**?**, **?**]. This makes it appealing for quantum-control settings in which one can evaluate the quality of a control protocol through simulation or experiment, but cannot provide a database of known optimal pulses in advance. The reference paper studied in this thesis emphasizes precisely this point: in the presence of leakage, control noise, and poorly understood high-dimensional

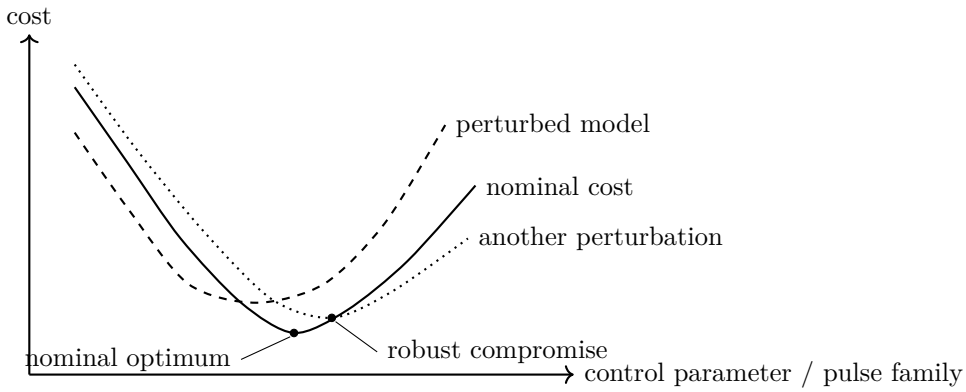

Figure 2.7: Schematic illustration of why robustness is difficult. A pulse that is optimal for a nominal model need not remain optimal when the Hamiltonian, noise parameters, or control transfer functions are perturbed. Robust optimization must account for an ensemble of such landscapes rather than for a single nominal one.

control landscapes, RL offers a flexible framework for learning good control trajectories directly from the optimization objective rather than from labeled data [**?**].

At a formal level, RL models the control task as a Markov decision process (MDP), specified by a state space $\mathcal{S}$, an action space $\mathcal{A}$, a transition kernel $P(s'|s,a)$, and a reward function $r(s,a,s')$. At discrete time step t, the agent observes a state $s_t \in \mathcal{S}$, selects an action $a_t \in \mathcal{A}$ according to a policy $\pi_{\boldsymbol{\theta}}(a|s)$ parameterized by $\boldsymbol{\theta}$, receives a reward r_t, and transitions to a new state s_{t+1}. The cumulative performance of the policy is quantified through the return

$$G_t = \sum_{k=t}^{T} \gamma^{k-t} r_k, \tag{2.114}$$

where $\gamma \in [0,1]$ is the discount factor and T is the horizon of an episode. The learning objective is then

$$J(\pi_{\boldsymbol{\theta}}) = \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}, P}\left[G_0\right], \tag{2.115}$$

where the expectation is taken over trajectories

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots, s_T). \tag{2.116}$$

In words, the agent seeks a policy that maximizes expected cumulative reward over entire control episodes [**?**].

This formalism maps naturally onto quantum control. The *state* may encode the current time step, the propagated wavefunction or density matrix, effective features derived from the system state, or a partial measurement record in a feedback setting. The *action* corresponds to the control values applied over the next time interval, for example a vector of microwave amplitudes, phases, or detunings. The *environment* is the controlled quantum system itself, either simulated numerically or accessed experimentally. The *reward* is derived from the control objective, such as gate fidelity, state-transfer fidelity, leakage suppression, runtime penalties, or some weighted combination thereof. Thus, quantum pulse design can be interpreted as a sequential decision problem in which the agent gradually constructs a control protocol while observing its consequences through the system dynamics [**?**, **?**, **?**].

A particularly useful feature of the RL perspective is that it unifies open-loop and closed-loop control under the same conceptual language. If the state contains only the time index,

$$s_t = t, \tag{2.117}$$

then the learned policy

$$a_t = \pi_{\boldsymbol{\theta}}(t) \tag{2.118}$$

defines an effectively open-loop pulse schedule: the action at time t is predetermined once the policy parameters are fixed. If, on the other hand, the state includes a measurement history or experimentally observed features,

$$s_t = (t, y_0, y_1, \ldots, y_{t-1}), \tag{2.119}$$

then the same policy formalism becomes a closed-loop feedback controller. This flexibility is one reason RL has become attractive in quantum technologies: it can be used both to learn open-loop pulse sequences and, at least in principle, to learn adaptive feedback strategies within a single mathematical framework [**?**, **?**].

Another reason RL is relevant is that many quantum-control objectives are naturally *delayed*. A control action applied early in the pulse sequence may only reveal its true value at the final time, when fidelity, leakage, or some other terminal metric is evaluated. In such cases, it is often convenient to use a reward structure of the form

$$r_t = \begin{cases} 0, & t < T, \\ -\mathcal{J}_{\text{final}}, & t = T, \end{cases} \tag{2.120}$$

or, equivalently, a positive terminal reward based on fidelity,

$$r_T = F_{\text{final}}. \tag{2.121}$$

This sparse-reward structure is awkward for naive local optimization, but it is entirely natural in RL, where the goal is to learn sequences of actions whose value may only become apparent at the end of the episode. In this sense, RL is well suited to control problems in which the quality of a full pulse sequence matters more than the immediate effect of each individual control action [**?**, **?**].

RL is also attractive when stochasticity is intrinsic to the problem. If the environment contains random perturbations, sampled parameter drifts, or control noise, then the transition law becomes stochastic,

$$s_{t+1} \sim P(\,\cdot \mid s_t, a_t, \xi_t), \tag{2.122}$$

where ξ_t represents a random disturbance. The policy objective in Eq. (2.115) then automatically becomes an expectation over noisy trajectories. This is a particularly natural way to formulate robust control: instead of first optimizing a nominal pulse and then testing it under perturbations, one can train the agent directly in a noisy environment and thereby encourage the policy to discover actions that perform well *on average* across many stochastic realizations [**?**]. This thesis uses exactly this principle by training RL agents in environments with sampled control noise so as to optimize not only fidelity and runtime, but also robustness against stochastic control errors.

A further advantage of RL is that it can exploit structure beyond local differential information. Classical gradient-based optimal control methods often update a pulse by using local derivatives of a cost functional with respect to control parameters. RL, by contrast, can treat the control problem as a long-horizon sequential optimization problem and may therefore exploit nonlocal regularities of the trajectory space. This perspective has been particularly fruitful in problems where the control landscape is highly nonconvex or where the action sequence has strong temporal dependencies. The literature stresses this point explicitly, arguing that deep RL can harness nonlocal regularities of noisy control trajectories and facilitate transfer learning between related control tasks [**?**]. This is especially relevant for families of quantum gates that share common structure, because a learned policy or value representation may generalize across nearby targets better than a pulse optimized from scratch each time.

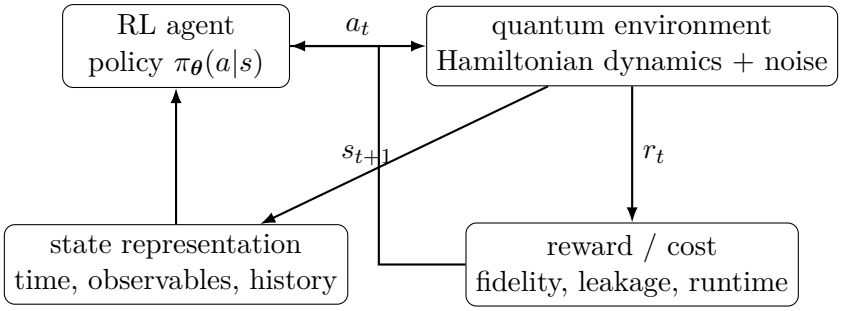

Figure 2.8: Agent–environment view of quantum control in reinforcement learning. The agent chooses control actions, the quantum system evolves under those controls and possible noise, and the resulting state and reward are fed back to improve the policy. Depending on the choice of state representation, the same formalism can describe either open-loop pulse generation or closed-loop feedback control.

Of course, RL is not automatically superior to classical optimal control. It introduces its own challenges: large sample requirements, sensitivity to reward shaping, instability during training, and the need to choose a suitable state representation and action parameterization. In addition, many RL algorithms were originally developed for generic machine-learning tasks rather than for the special structure of quantum dynamics. For this reason, a productive viewpoint is not to see RL as replacing optimal control, but rather as providing an alternative search strategy within the same control-theoretic framework [**?**, **?**]. In deterministic model-based settings, RL may be interpreted as learning to solve the same optimization problem attacked by GRAPE or Krotov, but by optimizing a policy over sequential decisions rather than directly differentiating a cost with respect to a fixed pulse vector.

For the purposes of this thesis, the relevance of RL is therefore twofold. First, it offers a mathematically natural framework for sequential pulse construction in noisy and uncertain environments. Second, it provides a route to incorporating practical objectives such as leakage suppression, robustness to sampled perturbations, and gate-time reduction into a single reward function. This is precisely the role RL plays in the UFO framework analyzed later: the control problem is cast as a sequential decision process, the total objective is encoded into a reward/cost signal, and the agent learns pulse trajectories that balance fidelity, leakage, runtime, and robustness under stochastic control errors as it will be covered in section 3.2.3. In this sense, RL is relevant not because quantum control is a machine-learning problem in disguise, but because the structure of many realistic quantum-control tasks already has the architecture of a sequential decision problem.

2.4.2 Policy optimization for continuous-action quantum control

Once a quantum-control task has been formulated as a sequential decision problem, the next issue is how the policy should be represented and optimized. This question is especially important in pulse engineering, because the controls applied to a quantum device are typically *continuous* variables: microwave amplitudes, phases, detunings, and coupler biases take values in subsets of $\mathbb{R}^m$ rather than in a small discrete action set. For this reason, many quantum-control problems are more naturally approached with *policy optimization* methods than with purely value-based reinforcement-learning algorithms originally designed for discrete actions [**?**, **?**, **?**]. The reference paper studied in this thesis fits exactly into this setting: at each time step, the agent chooses continuous control values, and the resulting pulse trajectory is optimized through a policy-gradient method, specifically trust-region policy optimization (TRPO) [**?**].

To make this precise, let the control horizon be divided into discrete time steps $t = 0, 1, \ldots, T-$

1, and let the action at time t be a continuous vector

$$a_t \in \mathcal{A} \subseteq \mathbb{R}^m. \tag{2.123}$$

In the context of quantum pulse design, one may identify

$$a_t \equiv \boldsymbol{u}_t, \tag{2.124}$$

where $\boldsymbol{u}_t$ denotes the set of control amplitudes applied over the next time slice. A parameterized stochastic policy then specifies a probability density over actions conditioned on the current state:

$$a_t \sim \pi_{\boldsymbol{\theta}}(\cdot \mid s_t), \tag{2.125}$$

with parameters $\boldsymbol{\theta}$. In continuous-control applications, a common choice is a Gaussian policy,

$$\pi_{\boldsymbol{\theta}}(a \mid s) = \mathcal{N}(a;\, \boldsymbol{\mu}_{\boldsymbol{\theta}}(s),\, \Sigma_{\boldsymbol{\theta}}(s)), \tag{2.126}$$

where $\boldsymbol{\mu}_{\boldsymbol{\theta}}(s)$ is the policy mean and $\Sigma_{\boldsymbol{\theta}}(s)$ is the covariance matrix. Equation (2.126) is particularly natural for pulse optimization, because it allows the policy to output a mean control action together with an exploration scale, thereby balancing exploitation of already promising pulse shapes with continued sampling of nearby alternatives [**?**, **?**].

The fundamental reason policy methods are attractive in this setting is that value-based action maximization becomes awkward in continuous spaces. In discrete-action reinforcement learning, a greedy policy can be obtained by choosing

$$a^\star(s) = \arg\max_{a \in \mathcal{A}} Q(s, a), \tag{2.127}$$

since the maximization involves only finitely many candidate actions. In continuous spaces, however, the analogous operation

$$a^\star(s) = \arg\max_{a \in \mathbb{R}^m} Q(s, a) \tag{2.128}$$

is itself a nontrivial optimization problem that must, in principle, be solved at every time step. This greatly complicates the use of purely value-based methods in continuous-control domains. Policy optimization avoids this issue by directly parameterizing the control law $a = \pi_{\boldsymbol{\theta}}(s)$ or the action density $\pi_{\boldsymbol{\theta}}(a \mid s)$ and updating its parameters to improve the expected return [**?**, **?**]. For quantum control, where the action is already the pulse amplitude vector, this direct parameterization is especially natural.

The objective remains the expected discounted return,

$$J(\boldsymbol{\theta}) = \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}} \left[\sum_{t=0}^{T} \gamma^t r_t \right], \tag{2.129}$$

where τ denotes a trajectory sampled under the policy and the system dynamics. Policy-gradient methods optimize Eq. (2.129) by differentiating it with respect to the policy parameters. In one of its standard forms, the policy-gradient theorem states that

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}} \left[\sum_{t=0}^{T} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t \mid s_t)\, Q^{\pi_{\boldsymbol{\theta}}}(s_t, a_t) \right], \tag{2.130}$$

where $Q^{\pi_{\boldsymbol{\theta}}}(s, a)$ is the action-value function under the current policy [**?**, **?**]. Equation (2.130) is the central mathematical reason policy optimization is feasible in continuous spaces: one does not need to solve Eq. (2.128); instead, one estimates a gradient that directly tells how to change the policy parameters so as to increase expected return.

In practice, the variance of the estimator in Eq. (2.130) can be large. A standard improvement is to subtract a baseline $b(s_t)$ without changing the expected gradient, yielding

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{\tau\sim\pi_{\boldsymbol{\theta}}}\left[\sum_{t=0}^{T} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t \mid s_t)\left(Q^{\pi_{\boldsymbol{\theta}}}(s_t, a_t) - b(s_t)\right)\right]. \tag{2.131}$$

Choosing the baseline to be the state-value function,

$$b(s) = V^{\pi_{\boldsymbol{\theta}}}(s), \tag{2.132}$$

one obtains the *advantage function*

$$A^{\pi_{\boldsymbol{\theta}}}(s, a) = Q^{\pi_{\boldsymbol{\theta}}}(s, a) - V^{\pi_{\boldsymbol{\theta}}}(s), \tag{2.133}$$

and the gradient becomes

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{\tau\sim\pi_{\boldsymbol{\theta}}}\left[\sum_{t=0}^{T} \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a_t \mid s_t)\, A^{\pi_{\boldsymbol{\theta}}}(s_t, a_t)\right]. \tag{2.134}$$

This form is especially important because most modern policy-gradient algorithms, including TRPO, are built around advantage estimates rather than raw returns [**?**, **?**].

For completeness, it is worth noting that continuous-control RL also admits *deterministic* policy gradients. If one uses a deterministic control law

$$a = \mu_{\boldsymbol{\theta}}(s), \tag{2.135}$$

then, under suitable assumptions, the objective gradient can be written as

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}_{s\sim\rho^{\mu_{\boldsymbol{\theta}}}}\left[\nabla_{\boldsymbol{\theta}}\mu_{\boldsymbol{\theta}}(s)\, \nabla_a Q^{\mu_{\boldsymbol{\theta}}}(s, a)\big|_{a=\mu_{\boldsymbol{\theta}}(s)}\right], \tag{2.136}$$

where $\rho^{\mu_{\boldsymbol{\theta}}}$ is the discounted state-visitation distribution under the deterministic policy [**?**]. Deterministic methods can be highly effective in continuous control, especially when the action dimension is large. However, this thesis employs a stochastic policy optimized by TRPO, which is particularly attractive when one wants stable learning under noise and controlled exploration in a stochastic environment [**?**].

This brings us to the main algorithmic idea relevant for the thesis: *trust-region policy optimization*. A naive policy-gradient update of the form

$$\boldsymbol{\theta}_{r+1} = \boldsymbol{\theta}_r + \eta\, \widehat{\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}_r)} \tag{2.137}$$

can be unstable, especially when the policy is represented by a large neural network and the return estimates are noisy. In such cases, a single large step may change the action distribution so drastically that the new policy performs much worse than the old one, even if the local gradient estimate pointed uphill. TRPO addresses this problem by replacing the unconstrained update with a constrained optimization in policy space, built around a local surrogate objective and a Kullback–Leibler (KL) trust-region constraint [**?**].

Let $\boldsymbol{\theta}_{\text{old}}$ denote the current policy parameters. TRPO optimizes the surrogate objective

$$\mathcal{L}_{\boldsymbol{\theta}_{\text{old}}}(\boldsymbol{\theta}) = \mathbb{E}_{(s,a)\sim\pi_{\boldsymbol{\theta}_{\text{old}}}}\left[\frac{\pi_{\boldsymbol{\theta}}(a \mid s)}{\pi_{\boldsymbol{\theta}_{\text{old}}}(a \mid s)}\, \hat{A}_{\boldsymbol{\theta}_{\text{old}}}(s, a)\right], \tag{2.138}$$

subject to the trust-region constraint

$$\mathbb{E}_{s\sim\pi_{\boldsymbol{\theta}_{\text{old}}}}\left[D_{\text{KL}}(\pi_{\boldsymbol{\theta}_{\text{old}}}(\cdot \mid s) \parallel \pi_{\boldsymbol{\theta}}(\cdot \mid s))\right] \le \delta, \tag{2.139}$$

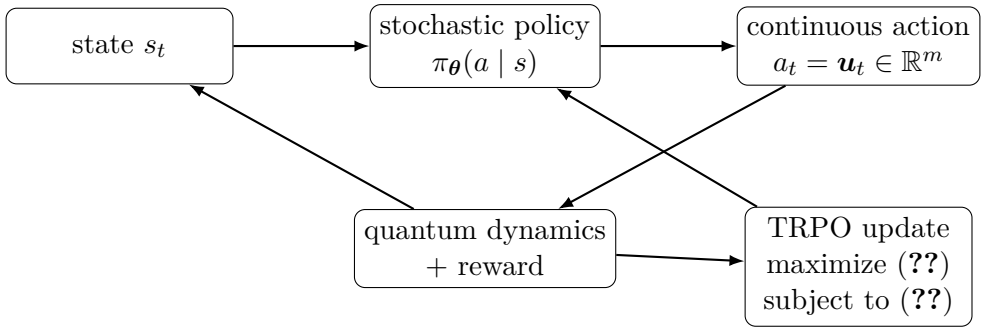

Figure 2.9: Continuous-action policy optimization in quantum control. The policy maps the current state to a distribution over continuous control amplitudes, the quantum system returns the resulting trajectory information and reward, and TRPO updates the policy through a surrogate objective constrained by a KL trust region.

where $\hat{A}_{\boldsymbol{\theta}_{\text{old}}}(s, a)$ is an estimate of the advantage function and $\delta > 0$ is a prescribed step-size radius in distribution space [**?**]. Equation (2.138) measures the expected first-order improvement of the new policy relative to the old one, while Eq. (2.139) ensures that the update does not move too far from the current action distribution. In this sense, TRPO does not merely limit parameter changes; it limits changes in the *behavior* of the policy.

The KL constraint is particularly meaningful in continuous-action control. For a Gaussian policy as in Eq. (2.126), a large parameter step can correspond to a substantial change in both the mean control signal and its exploration variance. In quantum pulse design this can be dangerous: a sudden policy shift may produce control sequences that are wildly different from those already found to be promising, causing catastrophic drops in fidelity or erratic training dynamics. By constraining the expected KL divergence, TRPO enforces a notion of *stable policy improvement*, which is one reason it has historically been effective for nonlinear continuous-control tasks [**?**]. This is directly relevant to the task studied in this thesis, where the agent must learn pulse trajectories in a stochastic environment and where training instability would make robust gate optimization far more difficult [**?**].

There is also a conceptual reason why policy optimization is a good match for quantum control. In pulse-design problems, the action at each time step is not an isolated move but part of a temporally correlated control trajectory whose final value is only revealed at the end of the episode. The quality of a pulse depends on interference, accumulated phase, transient leakage, and the interaction between early and late parts of the control sequence. Policy-based methods are well suited to this because they optimize the entire action-generating rule rather than treating each action selection as a stand-alone maximization problem. This makes them particularly attractive when the action space is continuous, the rewards are delayed, and the control landscape is highly nonconvex-precisely the regime encountered in robust quantum pulse design [**?**, **?**].

For the purposes of this thesis, the key message is the following. Once the control amplitudes are treated as continuous actions, reinforcement learning naturally leads to policy-gradient methods rather than to purely value-based discrete-action algorithms. Among policy-gradient methods, TRPO is especially relevant because it combines direct optimization of continuous stochastic policies with a principled mechanism for stabilizing updates through a trust-region constraint. This is why it forms the algorithmic core of the reference paper and why it provides the natural bridge between the general RL background developed so far and the specific deep-RL control framework analyzed in the following chapter.

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2.4.3 Reward design as the bridge between physics and reinforcement learning

In reinforcement learning, the reward function is not a secondary implementation detail: it is the mathematical object that specifies *what* the agent is being asked to achieve. For this reason, reward design is the crucial bridge between the physical control problem and the learning algorithm. In quantum control, one does not ultimately care about abstract return maximization for its own sake; one cares about realizing a target unitary, suppressing leakage, respecting hardware constraints, and doing so within a practically useful runtime. All of these physical desiderata must therefore be translated into a scalar reward or cost signal that the agent can optimize [**?**, **?**, **?**]. In this sense, reward design is the point at which domain knowledge enters reinforcement learning most directly.

Let $\tau = (s_0, a_0, r_0, \ldots, s_T)$ denote a control trajectory generated by a policy $\pi_{\boldsymbol{\theta}}$. The standard discounted return is

$$G_0 = \sum_{t=0}^{T} \gamma^t r_t, \tag{2.140}$$

and the policy objective is

$$J(\boldsymbol{\theta}) = \mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}}[G_0]. \tag{2.141}$$

If the physical task is naturally expressed as the minimization of a cost functional $\mathcal{C}[\tau]$, then the RL problem can be made equivalent by choosing the reward to be the negative cost. In the simplest episodic formulation,

$$r_t = \begin{cases} 0, & t < T, \\ -\mathcal{C}[\tau], & t = T, \end{cases} \tag{2.142}$$

so that maximizing expected return is equivalent to minimizing expected control cost:

$$J(\boldsymbol{\theta}) = -\mathbb{E}_{\tau \sim \pi_{\boldsymbol{\theta}}} \left[\mathcal{C}[\tau] \right]. \tag{2.143}$$

Equations (2.142)–(2.143) make explicit that RL is not replacing the control objective; it is providing a different optimization machinery for the same physical target.

In quantum pulse design, however, a purely terminal reward may be too sparse to learn from efficiently. A pulse is built action by action, but the quality of the full control sequence is often only known at the end of the episode, once fidelity, leakage, or runtime can be evaluated. This creates a credit-assignment problem: early control decisions may strongly influence the final outcome, yet receive no immediate feedback. A common remedy is *reward shaping*, where one augments the reward with intermediate signals that guide learning without changing the intended task [**?**, **?**]. In its most principled form, potential-based shaping adds a term

$$F_\Phi(s_t, s_{t+1}) = \gamma \Phi(s_{t+1}) - \Phi(s_t), \tag{2.144}$$

so that the shaped reward becomes

$$r'_t = r_t + F_\Phi(s_t, s_{t+1}), \tag{2.145}$$

where Φ is a scalar potential function on the state space. A standard result is that this transformation preserves the optimal policy under suitable assumptions [**?**]. The importance of Eq. (2.145) in the present context is conceptual: it shows that one can provide denser learning signals while still aiming at the same physical control objective.

For quantum control, the reward should encode the same performance criteria that would appear in a classical optimal-control cost functional. At the most generic level, one may write a physically meaningful episodic cost as

$$\mathcal{C}[\tau] = w_F\, \varepsilon_{\text{gate}}[\tau] + w_L\, L[\tau] + w_T\, T[\tau] + w_B\, B[\tau] + w_S\, S[\tau] + \cdots, \tag{2.146}$$

where $\varepsilon_{\text{gate}}$ is the gate infidelity, L is a leakage functional, T is the total runtime, B encodes boundary-condition violations, and S may penalize fluence, nonsmoothness, or bandwidth-demanding control features. The reward is then typically chosen as

$$r_T = -\mathcal{C}[\tau], \tag{2.147}$$

or, alternatively, distributed over time through a running-cost decomposition

$$\mathcal{C}[\tau] = \Phi(s_T) + \sum_{t=0}^{T-1} c(s_t, a_t, s_{t+1}), \tag{2.148}$$

which leads to shaped rewards of the form

$$r_t = -\,c(s_t, a_t, s_{t+1}), \qquad r_T = -\,\Phi(s_T). \tag{2.149}$$

This decomposition is especially convenient in sequential pulse design, because it allows the control algorithm to receive intermediate feedback about properties such as action magnitude, smoothness, or deviation from boundary conditions before the final fidelity is known.

A key lesson is that reward design determines the *behavioral bias* of the learned policy. If the reward contains only terminal fidelity, then the agent is encouraged to maximize accuracy but has no direct incentive to avoid long runtimes, excessive pulse amplitudes, strong boundary discontinuities, or transient excursions into leakage states. Conversely, if leakage or runtime is weighted too strongly relative to fidelity, the agent may discover controls that are conservative but fail to realize the target gate accurately. Thus, the coefficients in Eq. (2.146) are not merely numerical hyperparameters: they encode the scientific priorities of the control problem. In a realistic quantum-control application, these tradeoff weights determine whether the learned policy prefers fast but aggressive gates, smooth but slower gates, or highly robust but somewhat less time-optimal ones [**?**, **?**].

This point becomes even more important in stochastic environments. Suppose the dynamics depend on random disturbances ξ_t, representing amplitude fluctuations, parameter drift, or noise in the control electronics. Then the trajectory cost becomes random,

$$\mathcal{C}[\tau; \xi], \tag{2.150}$$

and the RL objective becomes

$$J(\boldsymbol{\theta}) = -\mathbb{E}_{\tau,\xi}\left[\mathcal{C}[\tau; \xi]\right]. \tag{2.151}$$

Equation (2.151) shows how robustness enters reward design naturally: by training in a stochastic environment and rewarding low expected cost, one asks the agent not merely to find a good pulse for a nominal model, but to find a control strategy that performs well on average across noisy realizations. This is one of the strongest conceptual advantages of the RL formulation for quantum control, since robustness can be built directly into the training objective instead of being added only as a post hoc evaluation criterion [**?**].

At the same time, reward design in physics-based RL is delicate. A badly chosen reward may lead to *reward hacking*, meaning that the agent exploits loopholes in the scalar objective rather than learning the intended physical behavior. In quantum control this can happen, for example, if a reward based only on projected logical fidelity allows large transient leakage that later returns to the computational subspace, or if a poorly scaled runtime penalty encourages premature episode termination rather than genuine fast gate synthesis. It can also happen if the reward mixes terms of very different numerical scales, making optimization dominated by whichever term has the largest magnitude rather than by the physically most important criterion. For this reason, professional reward design requires more than writing down a sum of penalties: it requires checking that the resulting scalar objective faithfully reflects the experimental and theoretical meaning of "good control."

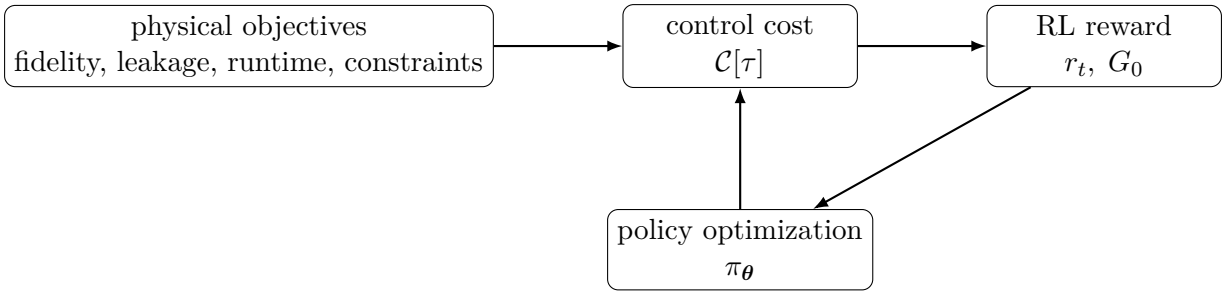

Figure 2.10: Reward design as the interface between physics and learning. The relevant physical control objectives are first encoded into a scalar cost functional, which is then converted into the reward signal used by the RL algorithm to optimize the policy.

A useful principle is therefore to start from the *physics* and only then translate it into reward form. One first asks: what quantities genuinely determine whether a control pulse is acceptable? For the present thesis, the answer includes at least target-gate accuracy, suppression of leakage, runtime, and compatibility with control constraints. Only after these quantities have been identified should one define the scalar reward. In that sense, RL does not remove the need for careful control theory; it makes the formulation of the control objective even more important, because the reward is the sole channel through which physical priorities are communicated to the learning algorithm [**?**, **?**].

This is precisely the conceptual role played by the framework analyzed in the next chapter. The literature explicitly argues that an effective control cost function is essential for efficient optimization and for meaningful quantum control, and it constructs the UFO objective by combining gate infidelity, leakage penalties, total runtime, and boundary-condition penalties into a single scalar cost [**?**]. That cost is then used as the reward signal for a continuous-variable policy-gradient agent trained with TRPO [**?**]. In other words, the paper does not first design an RL agent and then ask what to optimize; it begins from the physics of the control problem and uses the reward as the mechanism for injecting that physics into the learning dynamics. This is exactly why reward design is the final and most important conceptual bridge between the generic RL background of this chapter and the paper-specific framework introduced next.

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2.4.4 Neural parameterization of policy and value functions

The reinforcement-learning formulation introduced in the previous subsections defines the control problem at the level of states, actions, rewards, and policy updates. In order to make this formulation operational in a continuous-action and high-dimensional setting, one must still specify how the policy and the value function are represented. In simple low-dimensional problems, one could in principle store these objects explicitly in tabular form. In the present control setting, however, such an approach is not feasible. The state space is continuous, the action space is continuous, and the relevant control law must be able to capture highly nonlinear relations between the current dynamical configuration of the system and the next control decision. For this reason, both the policy and the value function are represented by trainable parametric function approximators.

In the framework considered in this thesis, these approximators are implemented as feed-forward neural networks. This choice should be understood in a pragmatic rather than purely algorithmic sense. The goal is not to introduce a separate layer of physical modeling based on machine learning, but to provide a flexible finite-dimensional representation for the two central objects of actor–critic reinforcement learning: the policy

$$\pi_\theta(a \mid s), \tag{2.152}$$

which determines how control actions are sampled from the current state, and the value function

$$V_\omega(s), \tag{2.153}$$

which estimates the expected discounted future reward associated with that state. In this way, the role of the neural networks is not to redefine the control objective, but to make its optimization tractable in a setting where explicit representations would be impossible.

The use of feed-forward architectures is particularly natural in the present context because the control problem is formulated in discrete time and each decision step is based on a fixed-dimensional state representation. The network is therefore asked to learn a nonlinear map from a vector encoding the current status of the gate-construction process to either a distribution over the next control action or a scalar estimate of future performance. No recurrence is required at this level, since the relevant dynamical history is already encoded in the state variable itself. For this reason, fully connected feed-forward networks provide a sufficiently expressive and conceptually simple parameterization for both the actor and the critic.

Policy function

In continuous-action reinforcement learning, the policy is most naturally represented as a probability distribution over actions conditioned on the current state. A standard choice, and the one relevant for the present framework, is a Gaussian policy of the form

$$\pi_\theta(\tilde{a}_t \mid s_t) = \mathcal{N}\big(\tilde{a}_t;\, \mu_\theta(s_t), \Sigma_\theta(s_t)\big), \tag{2.154}$$

where $\mu_\theta(s_t)$ is the state-dependent mean produced by the policy network and $\Sigma_\theta(s_t)$ is the covariance matrix controlling exploration. In practice, one often works with an unconstrained auxiliary action variable $\tilde{a}_t$, sampled from the Gaussian distribution in Eq. (2.154), and then maps it to the admissible action domain through a bounded nonlinear transformation, schematically written as

$$a_t = \varphi(\tilde{a}_t), \tag{2.155}$$

where φ is typically chosen so as to enforce finite action bounds. This construction is especially convenient in control problems, since the physically allowed amplitudes and phases of the control channels are bounded by hardware constraints.

The conceptual role of the policy network is therefore clear: it does not output a complete pulse sequence in one shot, but rather parameterizes a sequential control law that generates the control trajectory step by step. At each decision time, the network receives the current state s_t and produces the parameters of the distribution from which the next action is sampled. The resulting sequence of actions defines the full control protocol. In this sense, the policy network is the object through which reinforcement learning translates the abstract optimization objective into an adaptive rule for constructing pulse trajectories.

Value function

The second neural component is the value network, which represents the state-value function

$$V^\pi(s_t) = \mathbb{E}_\pi\left[\sum_{k=t}^{T} \gamma^{k-t} r_k \,\middle|\, s_t\right]. \tag{2.156}$$

Equation (2.156) measures the expected discounted future reward obtainable from the current state under the policy π. In actor–critic methods, this quantity plays a central auxiliary role: it does not define the physical objective itself, but provides an estimate of how promising a given state is in terms of future return.

This estimate is especially important for policy-gradient optimization. Rather than updating the policy directly from raw returns, one typically uses the value function to construct an advantage estimate, schematically of the form

$$A^{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - V^{\pi}(s_t), \tag{2.157}$$

so that the policy update depends on whether the chosen action performs better or worse than what is expected on average from that state. The practical consequence is a substantial reduction in gradient variance and a more stable training process. In this sense, the value network acts as a critic: it evaluates the long-term quality of the states visited by the policy and thereby supports the optimization of the actor.

From the viewpoint of the control problem, the distinction between the two networks should be kept conceptually sharp. The policy network is the object that generates candidate control actions. The value network is an auxiliary estimator that improves optimization efficiency by approximating the expected future return. Neither replaces the underlying physical model, the Hamiltonian evolution, or the control objective. Rather, both are embedded within the reinforcement-learning loop as trainable parametric representations that make actor–critic optimization possible in a continuous and nonlinear control landscape.

In the present thesis, this neural parameterization should therefore be viewed as the algorithmic core of the learning agent rather than as the main scientific object of study. The scientific content remains the same as in the previous subsections: the relevant questions concern fidelity, leakage, runtime, robustness, and their combination in the UFO objective. The role of the neural networks is to provide a sufficiently expressive representation of the policy and value functions so that these physically motivated objectives can be optimized effectively. The concrete realization of this parameterization (including the precise choice of inputs, outputs, architecture, and action mapping used in the experiments) will be specified in the implementation chapter.

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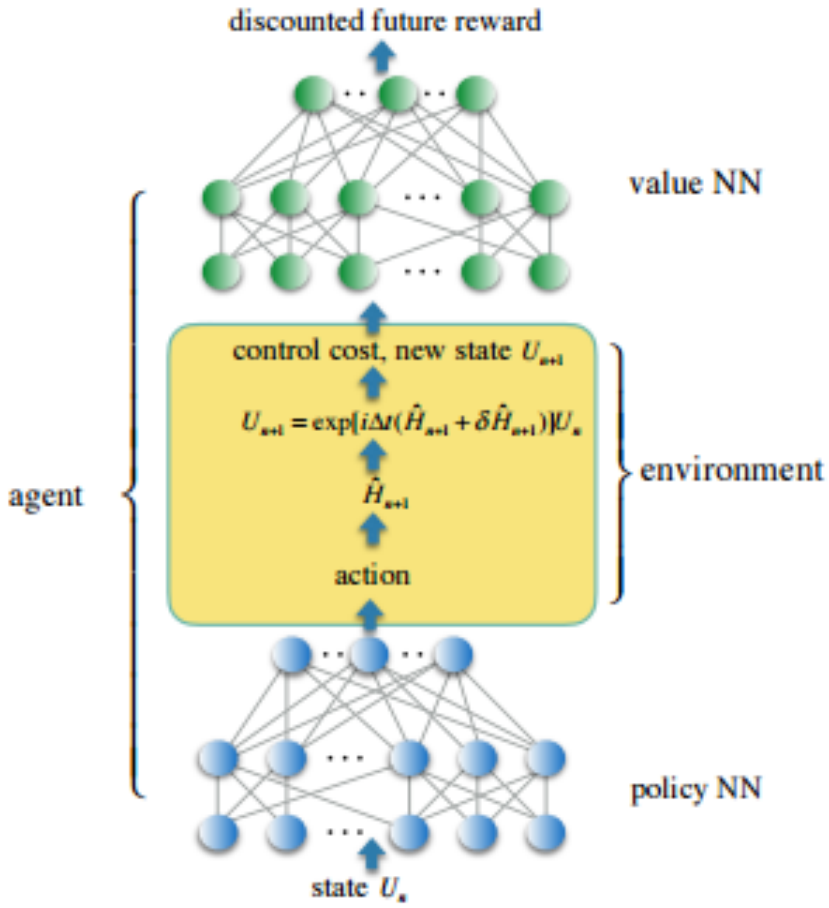


Figure 2.11: Actor–critic view of continuous-action quantum control. The policy network maps the current state to a distribution over the next control action, while the value network estimates the expected discounted future reward associated with that state. The environment propagates the controlled quantum dynamics and returns the next state and reward.

Chapter 3

Implementation

3.1 Physical model and control setting

3.1.1 The gmon architecture as a platform for robust analog quantum control

The control framework studied in this thesis is developed on a superconducting two-qubit platform based on the *gmon* architecture. This choice is physically well motivated. The gmon combines two key features that are particularly desirable for high-performance quantum control: on the one hand, it inherits from transmon- and xmon-like superconducting qubits the advantages of relatively high coherence and convenient microwave addressability; on the other hand, it introduces a *tunable inter-qubit coupling*, which can be varied dynamically during the gate operation [**?**, **?**, **?**]. This makes the platform especially suitable for studying direct analog realization of two-qubit operations, since both local control and entangling interaction strength can be shaped in time rather than being treated as fixed background resources.

At a conceptual level, the architecture consists of two weakly anharmonic superconducting qubits, each locally driven and frequency-controlled, together with a flux-tunable coupler that mediates their interaction. If one denotes the two qubits by Q_1 and Q_2, then the controlled dynamics may be viewed schematically as arising from a Hamiltonian of the form

$$H(t) = H_{\text{loc}}^{(1)}(t) + H_{\text{loc}}^{(2)}(t) + H_{\text{int}}(t), \tag{3.1}$$

where the local terms describe the single-qubit dynamics and the interaction term encodes the tunable coupling between the devices. In a simplified bosonic picture, the coupling contribution may be thought of schematically as

$$H_{\text{int}}(t) \sim g(t) \left(a_1^\dagger a_2 + a_1 a_2^\dagger \right), \tag{3.2}$$

with $g(t)$ a time-dependent coupling strength and $a_j, a_j^\dagger$ the annihilation and creation operators associated with qubit Q_j. Equation (3.2) is not yet the full model used in the paper, but it already captures the essential architectural feature: the interaction itself is a control resource, rather than a static parameter.

This tunability is one of the main reasons the gmon is an attractive testbed for analog gate design. In architectures with fixed coupling, two-qubit gate synthesis is constrained by the native interaction strength and by the limited set of ways in which that interaction can be modulated indirectly through local controls. In the gmon, by contrast, the coupling can be turned on, suppressed, or reshaped during the control sequence, thereby enlarging the family of dynamically accessible pulse protocols [**?**, **?**]. From the viewpoint of optimal control, this additional flexibility is valuable because it may reduce the runtime of entangling operations and improve hardware efficiency relative to purely synthesized gate decompositions. It also makes the architecture particularly relevant for the problem studied in the reference paper, whose aim is precisely to

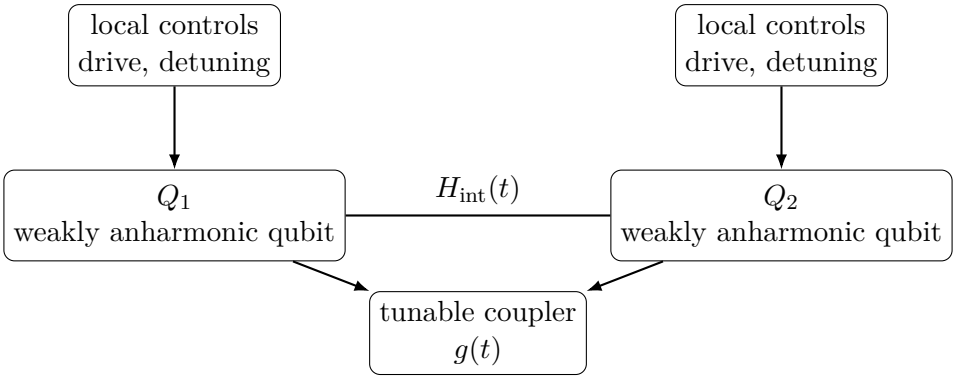


Figure 3.1: Conceptual schematic of the gmon architecture. Two weakly anharmonic superconducting qubits are subject to local controls and coupled through a tunable interaction channel. This combination of local addressability and time-dependent coupling makes the platform particularly suitable for pulse-level optimization of two-qubit gates.

optimize quantum gates directly at the pulse level rather than only at the circuit-compilation level [**?**].

At the same time, the gmon remains a *weakly anharmonic multilevel* superconducting system. Therefore, the extra controllability provided by the tunable coupler does not eliminate the main physical difficulties of fast gate design. On the contrary, the platform naturally exhibits the same speed–leakage tension discussed in the background chapter: stronger and faster controls can shorten the gate time and exploit the tunable interaction more aggressively, but they also increase the risk of population transfer outside the computational subspace [**?**, **?**]. For this reason, the gmon is not merely a convenient platform with many knobs; it is a realistic environment in which the competing goals of fidelity, speed, and leakage suppression all appear simultaneously and must be balanced in a quantitatively meaningful way.

This balance is exactly what makes the architecture particularly suitable for the control problem addressed in this thesis. A platform with only fixed interactions would make it harder to study the benefits of direct analog gate optimization, while an idealized two-level system would hide the physically important leakage mechanisms associated with weak anharmonicity. The gmon lies precisely in the interesting middle ground: it is experimentally motivated, it offers enough flexibility to make nontrivial pulse-level optimization worthwhile, and it remains sufficiently realistic that multilevel effects and robustness issues cannot be ignored. In this sense, the platform is not incidental to the paper's contribution; it is an essential part of why the proposed framework is physically meaningful [**?**, **?**].

For the purposes of the thesis, this subsection therefore plays a mainly motivational role. It identifies the gmon architecture as the physical stage on which the control problem is posed and explains why it is a natural setting for studying robust analog quantum control. The next paragraph will move from this architectural overview to the explicit mathematical model used in the paper, introducing the effective Hamiltonian and the corresponding control channels in more detail.

3.1.2 Effective Hamiltonian and control channels

Having identified the gmon as the physical platform of interest, the next step is to specify the effective dynamical model used in the control problem. In the reference literature, the starting point is a phenomenological circuit quantization of two coupled nonlinear Josephson L–C oscillators, followed by a transformation to the interaction picture and a rotating-wave approximation (RWA) valid in the regime where the fast oscillating terms average out over the control timescale [**?**]. The resulting description is particularly convenient because it retains

the multilevel bosonic structure needed to study leakage, while at the same time making the experimentally relevant control channels explicit.

At the bosonic level, the effective two-mode Hamiltonian used takes the form

$$\hat{H}_{\mathrm{RWA}}(t) = \frac{\eta}{2}\sum_{j=1}^{2}\hat{n}_j(\hat{n}_j-1) + g(t)(\hat{a}_2^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_2) + \sum_{j=1}^{2}\delta_j(t)\hat{n}_j + \sum_{j=1}^{2} i f_j(t)\left(\hat{a}_j e^{i\phi_j(t)} - \hat{a}_j^\dagger e^{-i\phi_j(t)}\right), \tag{3.3}$$

where $\hat{a}_j$ and $\hat{a}_j^\dagger$ are the annihilation and creation operators of qubit mode j, and

$$\hat{n}_j = \hat{a}_j^\dagger\hat{a}_j \tag{3.4}$$

is the corresponding number operator. Equation (3.3) is the central effective Hamiltonian of the control model, and it already makes transparent the physical meaning of the available control knobs.

- The first term in Eq. (3.3),

$$\frac{\eta}{2}\sum_{j=1}^{2}\hat{n}_j(\hat{n}_j-1), \tag{3.5}$$

  encodes the weak anharmonicity of the superconducting modes. This is the term that breaks the exact harmonic spacing of the oscillator spectrum and thereby allows the lowest two levels of each mode to function as an effective qubit. At the same time, because η is finite rather than infinitely large, the system remains intrinsically multilevel, so the higher excited states cannot be ignored in fast control protocols. For this reason, Eq. (3.5) is not just a detail of the model: it is the term that makes leakage a meaningful physical issue in the first place.

- The second term,

$$g(t)(\hat{a}_2^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_2), \tag{3.6}$$

  represents the tunable inter-qubit coupling. It has the structure of an exchange interaction between the two bosonic modes and is the main source of entangling dynamics in the model. The time dependence of $g(t)$ is particularly important: unlike in fixed-coupling architectures, the interaction strength itself becomes a control parameter that can be modulated during the gate. This is one of the central architectural advantages of the gmon and one of the reasons why it is a natural platform for direct analog gate optimization [**?**,**?**].

- The third term,

$$\sum_{j=1}^{2}\delta_j(t)\hat{n}_j, \tag{3.7}$$

  describes local detunings of the qubit frequencies. These terms act diagonally in the occupation-number basis and allow one to shift the effective local energies during the pulse. Physically, they correspond to frequency control of the individual qubits and provide an additional mechanism for shaping the relative phase accumulation and resonance conditions throughout the evolution. Together with the coupling pulse $g(t)$, the detunings $\delta_j(t)$ enlarge the set of dynamically accessible unitary transformations.

- Finally, the last term in Eq. (3.3),

$$\sum_{j=1}^{2}\left(\hat{a}_j e^{i\phi_j(t)} - \hat{a}_j^\dagger e^{-i\phi_j(t)}\right), \tag{3.8}$$

  models the externally applied microwave drives. Here $f_j(t)$ is the drive amplitude and $\phi_j(t)$ its phase.

These controls generate local coherent rotations of each qubit mode and, because of the phase dependence, allow one to address different quadratures of the local dynamics. Thus, the full set of control channels in the effective model consists of:

$$\{g(t),\ \delta_1(t), \delta_2(t),\ f_1(t), f_2(t),\ \phi_1(t), \phi_2(t)\}. \tag{3.9}$$

This explicit decomposition is very useful in the thesis, because it makes clear which physical quantities the learning agent is allowed to shape.

Although Eq. (3.3) is the correct model for studying multilevel dynamics and leakage, it is also helpful to project it onto the logical qubit basis in order to interpret the corresponding two-qubit control structure. After truncation to the qubit subspace, we can re-write the Hamiltonian as

$$\hat{H}_{\mathrm{qb}}(t) = \frac{g(t)}{2}\left(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y\right) + \sum_{j=1}^{2}\left[\frac{\delta_j(t)}{2}\sigma_j^z - f_j(t)\Big(\sin\phi_j(t)\,\sigma_j^x + \cos\phi_j(t)\,\sigma_j^y\Big)\right], \tag{3.10}$$

where $\sigma_j^x, \sigma_j^y, \sigma_j^z$ are the Pauli operators acting on qubit j [**?**]. Equation (3.10) gives a clear logical interpretation of the control resources:

- the coupling pulse $g(t)$ generates an XY-type entangling interaction,
- the detuning pulses $\delta_j(t)$ generate local Z-type control,
- the microwave amplitudes and phases generate local rotations in the XY plane.

From the logical point of view, Eq. (3.10) already suggests why the architecture is sufficiently expressive for universal two-qubit control: one has access to both entangling interactions and arbitrary local single-qubit rotations. However, the projected model alone is not sufficient for the purposes of this thesis, because the control problem is not only to realize a target logical unitary, but to do so while suppressing population transfer outside the computational subspace. For this reason, the bosonic Hamiltonian in Eq. (3.3) remains the fundamental model, while the projected Hamiltonian in Eq. (3.10) serves mainly as an interpretive tool.

A further important aspect of the paper is that the control problem is posed under realistic amplitude constraints and experimental limitations. In particular, the magnitudes of the coupling pulse, detuning controls, and microwave amplitudes are chosen to remain substantially smaller than the anharmonicity scale, so as to reduce leakage-inducing transitions, and the modulation bandwidth is restricted to remain compatible with experimental hardware limitations [**?**]. This is conceptually important: the Hamiltonian is not introduced as a purely formal theoretical model, but as a realistic effective control model in which both physical accessibility and multilevel errors are taken seriously.

For the structure of the thesis, the main role of the present subsection is therefore to identify the dynamical variables of the problem. Equation (3.3) gives the effective multilevel control model on which the optimization is performed, while Eq. (3.10) clarifies the corresponding logical meaning of the available controls. The next paragraph will build on this by specializing the computational and leakage subspaces in the notation of the paper, thereby preparing the ground for the analytical leakage treatment that follows.

3.1.3 Computational and leakage subspaces in the paper's notation

The effective bosonic Hamiltonian introduced in the previous subsection acts on a multilevel Hilbert space spanned by occupation-number states of the two weakly anharmonic modes. It is therefore natural to use the tensor-product Fock basis

$$|n_1 n_2\rangle = |n_1\rangle_1 \otimes |n_2\rangle_2\,, \tag{3.11}$$

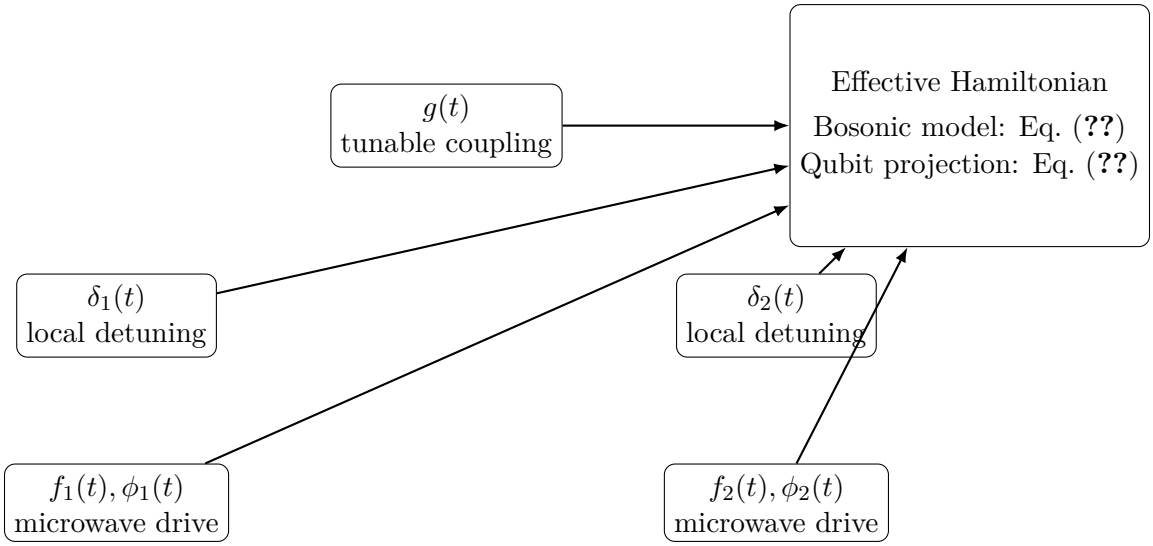


Figure 3.2: Control channels entering the effective gmon Hamiltonian. The tunable coupling $g(t)$, local detunings $\delta_j(t)$, and microwave amplitudes/phases $(f_j(t), \phi_j(t))$ together define the time-dependent control resources available for quantum gate optimization.

where $n_j \in \{0, 1, 2, \dots\}$ denotes the excitation number of qubit mode j. In this notation, the logical two-qubit sector corresponds to the four states for which each mode is restricted to its lowest two levels. Let's denote this computational sector by

$$\Omega_0 = \text{span}\left\{|00\rangle, |10\rangle, |01\rangle, |11\rangle\right\}. \tag{3.12}$$

Equation (3.12) is the paper-specific form of the computational subspace introduced more abstractly in the previous chapter. All target logical gates are defined on this four-dimensional space.

Because the underlying superconducting modes are weakly anharmonic rather than ideal two-level systems, the full physical Hilbert space is larger than Ω_0. We can therefore write the physical Hilbert space as a direct sum of the computational subspace and the higher-energy subspaces

$$\mathcal{H}_{\text{phys}} = \bigoplus_{\alpha=0}^{\infty} \Omega_\alpha, \tag{3.13}$$

where Ω_0 is the qubit subspace and the subspaces Ω_α with $\alpha \geq 1$ collect states outside the computational sector. In particular, the first leakage subspace is

$$\Omega_1 = \text{span}\left\{|20\rangle, |21\rangle, |12\rangle, |02\rangle\right\}, \tag{3.14}$$

which is the nearest higher-energy sector most directly coupled to the computational dynamics under the available controls [**?**]. In other words, Ω_1 plays a distinguished role because it contains the leading leakage channels generated by fast driving and tunable coupling.

This subspace structure is not merely a matter of bookkeeping. It reflects the hierarchy of energy scales in the effective Hamiltonian. The large anharmonic term

$$\hat{H}_0 = \sum_{\alpha} \sum_{m \in \Omega_\alpha} E_\alpha |m\rangle \langle m|, \tag{3.15}$$

assigns different characteristic energies to the different subspaces and thereby creates the gap between the computational sector and the higher excited sectors. The minimum energy separation between the qubit subspace and any leakage subspace is denoted in the paper by

$$\Delta = \min_{\alpha \neq 0} |E_\alpha - E_0|. \tag{3.16}$$

This gap is one of the central control parameters of the theory, because it is the large scale against which the strengths and modulation rates of the control terms are compared.

To formulate leakage systematically, the full Hamiltonian can be decomposed into three parts,

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) + \hat{H}_2(t), \tag{3.17}$$

where $\hat{H}_0$ is the time-independent diagonal part introduced in Eq. (3.15). The second term,

$$\hat{H}_1(t) = \sum_{\alpha} \sum_{m,m' \in \Omega_\alpha} \langle m| \hat{H}_1^{\alpha}(t) |m'\rangle |m\rangle \langle m'| , \tag{3.18}$$

contains the couplings *within* each subspace and is therefore block-diagonal with respect to the decomposition in Eq. (3.13). By contrast, the third term,

$$\hat{H}_2(t) = \sum_{\alpha \neq \alpha'} \sum_{\substack{m \in \Omega_\alpha \\ m' \in \Omega_{\alpha'}}} \langle m| \hat{H}_2^{\alpha,\alpha'}(t) |m'\rangle |m\rangle \langle m'| , \tag{3.19}$$

contains the couplings *between* different subspaces and is therefore block-off-diagonal. This is the term responsible for direct coherent leakage from the computational sector into higher-energy sectors.

This decomposition has a clear physical meaning in the present gmon model. The qubit operations one wants to realize logically are encoded in the dynamics of $\hat{H}_1(t)$ restricted to Ω_0, since this term preserves the subspace structure. However, the actual experimental controls that produce the desired logical rotations and entangling interactions also generate the block-off-diagonal contribution $\hat{H}_2(t)$. Therefore, one cannot simply turn off the leakage-generating couplings without also removing the very controls that make universal gate implementation possible. This is precisely why leakage suppression is a nontrivial control-theoretic problem rather than a trivial projection argument [**?**].

More concretely, $\hat{H}_1(t)$ contains not only the desired couplings within Ω_0, but also the couplings within Ω_1. Thus, the first leakage sector is not just a passive reservoir of unwanted states; it is itself dynamically structured. This point is important for two reasons. First, it explains why the dominant leakage cannot be understood simply as isolated transitions into a single stray level. Second, it motivates the later use of a transformed basis, since the full time-dependent dynamics involves a nontrivial interplay between intra-subspace and inter-subspace couplings.

The perturbative regime assumed by the paper is expressed through a separation of scales:

$$\epsilon \sim \|\hat{H}_1(t)\| \sim \|\hat{H}_2(t)\| \ll \Delta, \tag{3.20}$$

or equivalently

$$\frac{\epsilon}{\Delta} \ll 1. \tag{3.21}$$

Here ϵ denotes the characteristic magnitude of the control-induced couplings, while Δ denotes the large energy gap separating the qubit sector from higher sectors. Equations (3.20)–(3.21) are fundamental because they justify the perturbative treatment that follows in the next subsection. Intuitively, the control fields are assumed to be strong enough to generate nontrivial logical gates, but still weak enough relative to the inter-subspace energy gap that leakage can be treated systematically rather than completely overwhelming the dynamics.

From the viewpoint of the thesis, the main importance of the present subsection is that it introduces the exact language in which this thesis formulates leakage. The computational space is not referred to generically, but specifically as the lowest-energy subspace Ω_0; leakage is not described only as population outside the qubit basis, but as coupling from Ω_0 to the higher subspaces $\Omega_1, \Omega_2, \ldots$ through the block-off-diagonal Hamiltonian $\hat{H}_2(t)$. This notation is essential for the following subsection, where the time-dependent Schrieffer–Wolff transformation is introduced precisely to suppress and bound the effect of these inter-subspace couplings.

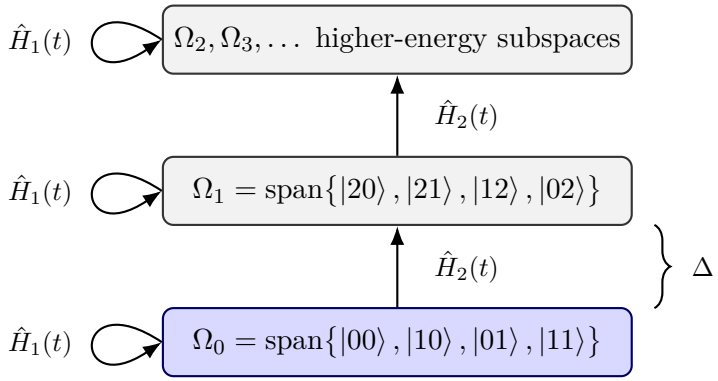


Figure 3.3: Subspace structure used in the paper. The computational sector is the lowest-energy subspace Ω_0, while $\Omega_1, \Omega_2, \ldots$ are higher-energy leakage subspaces. The block-diagonal term $\hat{H}_1(t)$ acts within each subspace, whereas the block-off-diagonal term $\hat{H}_2(t)$ couples different subspaces and induces leakage.

3.2 Leakage treatment and control objective

3.2.1 Why leakage is treated analytically: the role of the time-dependent Schrieffer–Wolff transformation

Once the control problem has been formulated on the full multilevel gmon Hilbert space, a natural question arises: why not simply optimize the exact time evolution numerically and measure leakage directly, without introducing any additional analytic machinery? The reason is that, although direct simulation remains essential, it does not by itself provide a sufficiently structured description of the *mechanism* of leakage. A projected qubit model is too crude because it completely removes the higher excited levels responsible for leakage, while a brute-force multilevel simulation, though accurate, does not automatically yield a compact control-oriented quantity that identifies which features of the Hamiltonian are most responsible for leakage generation. The reference paper therefore seeks an intermediate description: one that keeps the multilevel physics, but reorganizes it into a form better suited for analysis and optimization [**?**].

The key observation is that the Hamiltonian decomposition introduced in the previous subsection,

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) + \hat{H}_2(t), \tag{3.22}$$

already separates the dynamics into qualitatively different parts. The block-diagonal term $\hat{H}_1(t)$ generates evolution within each subspace Ω_α, while the block-off-diagonal term $\hat{H}_2(t)$ directly couples different subspaces and is therefore the immediate source of coherent leakage. Since the control regime of interest satisfies

$$\frac{\epsilon}{\Delta} \ll 1, \tag{3.23}$$

with ϵ the characteristic strength of the control-induced couplings and Δ the gap separating the computational sector from the leakage sectors, it becomes natural to ask whether one can perturbatively transform the Hamiltonian into a basis in which these direct inter-subspace couplings are suppressed to higher order [**?**]. This is precisely the purpose of the Schrieffer–Wolff transformation.

In its standard time-independent form, the Schrieffer–Wolff transformation is a perturbative change of basis designed to eliminate block-off-diagonal couplings order by order in a small parameter. The idea is to replace the original state $|\psi\rangle$ by a rotated state

$$|\tilde{\psi}(t)\rangle = e^{-\hat{S}(t)} |\psi(t)\rangle, \tag{3.24}$$

where $\hat{S}(t)$ is an anti-Hermitian generator,

$$\hat{S}^\dagger(t) = -\hat{S}(t), \tag{3.25}$$

chosen so that the transformed Hamiltonian becomes as block-diagonal as possible. In the present setting this means finding a rotation that decouples, to the desired perturbative order, the computational subspace Ω_0 from the higher subspaces $\Omega_1, \Omega_2, \ldots$. Conceptually, the transformation does not remove leakage states from the theory; rather, it changes the basis so that the dominant direct leakage couplings are absorbed into an effective block-diagonal description.

Because the controls are explicitly time dependent, the transformation itself must also be time dependent. This is why the paper uses the *time-dependent* Schrieffer–Wolff transformation (TSWT) rather than the standard static version. If one inserts Eq. (3.24) into the Schrödinger equation, the transformed state obeys

$$i\frac{d}{dt}\left|\tilde{\psi}(t)\right\rangle = \hat{\mathbb{H}}(t)\left|\tilde{\psi}(t)\right\rangle, \tag{3.26}$$

with effective Hamiltonian

$$\hat{\mathbb{H}}(t) = -i\,\frac{de^{-\hat{S}(t)}}{dt}e^{\hat{S}(t)} + e^{-\hat{S}(t)}\hat{H}(t)e^{\hat{S}(t)}. \tag{3.27}$$

Equation (3.27) shows immediately why time dependence matters: even if the static similarity transformation could remove block-off-diagonal terms coming from $\hat{H}(t)$, the derivative term generated by the moving basis would itself reintroduce additional couplings. Thus, in a driven system, leakage is not caused only by direct Hamiltonian couplings between subspaces, but also by the non-adiabatic motion of the rotated basis itself [**?**].

This is the main conceptual reason the paper treats leakage analytically rather than only numerically. The TSWT separates the problem into two physically distinct contributions. First, there is the *direct coupling induced leakage*, originating from the block-off-diagonal part of the original Hamiltonian. Second, there is the *non-adiabatic leakage* generated by the time dependence of the transformation. In the language of the paper, one first chooses the perturbative generator $\hat{S}(t)$ so that the block-off-diagonal part of the effective Hamiltonian is suppressed to higher order, and only then estimates the residual non-adiabatic leakage that remains due to the time dependence of the control [**?**]. This decomposition is far more informative than a raw leakage probability computed at the end of the pulse, because it identifies which structural aspects of the control protocol are responsible for leakage.

A second major motivation for TSWT is that it produces a leakage estimate that is much better aligned with optimization. A direct numerical computation of leakage requires full propagation of the multilevel dynamics for every candidate control pulse. This is certainly possible, but it does not immediately reveal how leakage scales with control amplitudes, modulation rates, or energy gaps. By contrast, the TSWT framework expresses leakage in terms of the size of block-off-diagonal couplings and the rate of temporal variation of the Hamiltonian. In this way, it translates a difficult dynamical error mechanism into a more transparent perturbative quantity that can be inserted directly into the control objective. This is exactly the strategy of the paper: rather than treating leakage as a purely empirical output of the simulation, it derives an analytic bound that can serve as a control penalty [**?**].

The perturbative structure of the method is encoded by expanding the generator as

$$\hat{S}(t) = \epsilon\hat{S}_1(t) + \epsilon^2\hat{S}_2(t) + \cdots + \epsilon^n\hat{S}_n(t), \tag{3.28}$$

with the aim of choosing the coefficients $\hat{S}_1, \hat{S}_2, \ldots$ so that the block-off-diagonal terms of the effective Hamiltonian are cancelled order by order. The paper states this goal explicitly: for an

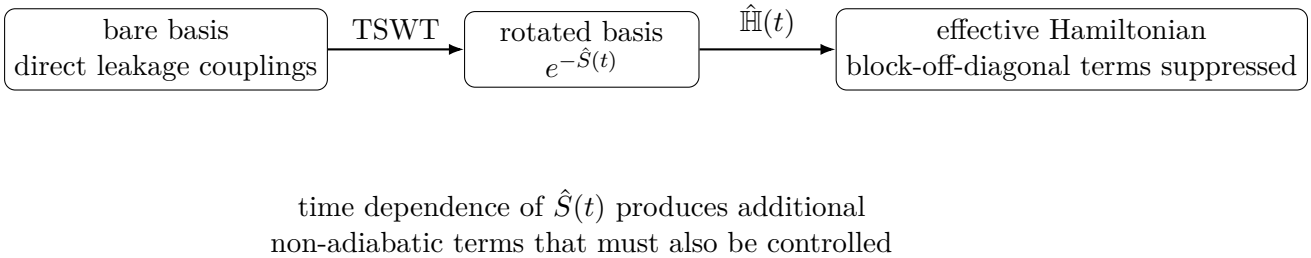


Figure 3.4: Conceptual role of the time-dependent Schrieffer–Wolff transformation. The original multilevel Hamiltonian is transformed to a rotated basis in which direct inter-subspace couplings are perturbatively suppressed. Because the transformation is time dependent, one must also account for non-adiabatic contributions generated by the motion of the basis itself.

nth-order perturbative solution, the block-off-diagonal part of the transformed Hamiltonian is suppressed to order

$$O\left(\frac{\epsilon^{n+1}}{\Delta^{n}}\right). \tag{3.29}$$

Equation (3.29) is highly revealing. It shows that the transformation does not merely re-label the basis; it systematically weakens direct inter-subspace couplings in powers of the small ratio ϵ/Δ. This is precisely what makes it useful for leakage analysis in the weak-coupling, weakly non-adiabatic regime relevant to the paper.

It is worth stressing that the purpose of TSWT here is not to solve the dynamics exactly. Rather, it is to construct a rotated computational basis in which the physically relevant leakage channels are perturbatively exposed and organized. This rotated basis is more natural for control analysis because it distinguishes between errors that are intrinsic to the desired block-diagonal effective dynamics and errors that arise from residual inter-subspace coupling. In other words, TSWT provides the analytical language needed to say not just that leakage occurs, but *why* it occurs and at which perturbative order it enters.

For the structure of the thesis, this subsection therefore plays a purely motivational and conceptual role. It explains why the paper introduces the time-dependent Schrieffer–Wolff transformation before defining its leakage penalty. The next subsection will build on this framework more concretely by presenting the leakage-related bound itself and explaining how it enters the control objective as a penalty term.

3.2.2 The leakage bound and its role as a control penalty

The conceptual role of the time-dependent Schrieffer–Wolff transformation becomes concrete once one asks how leakage should actually be quantified during control optimization. In the reference paper, the key idea is that after applying the second-order TSWT, the leading leakage mechanisms can be organized into two physically distinct contributions: direct coupling leakage caused by the residual block-off-diagonal Hamiltonian, and non-adiabatic leakage caused by the time dependence of the transformed basis itself [**?**]. Rather than treating these effects only through full numerical propagation or experimental benchmarking, the paper derives an explicit analytic bound that captures the dominant coherent leakage accumulated during the gate.

The residual inter-subspace coupling after the second-order TSWT is described by an effective block-off-diagonal Hamiltonian $\hat{H}_{\mathrm{od}}(t)$, whose magnitude scales as

$$\hat{H}_{\mathrm{od}}(t) = O\left(\frac{\epsilon^{3}}{\Delta^{2}}\right), \tag{3.30}$$

where ϵ is the characteristic scale of the inter- and intra-subspace couplings and Δ is the minimum energy gap separating the computational subspace from the higher subspaces. Equation (3.30)

is crucial: it expresses the fact that the direct leakage-inducing couplings have already been perturbatively suppressed to higher order by the TSWT. Leakage is therefore not controlled by the original raw off-diagonal Hamiltonian, but by the smaller residual coupling that remains after the rotated basis has been chosen appropriately.

Starting from this residual Hamiltonian, the paper derives in Appendix B a practical upper bound for the accumulated coherent leakage,

$$L_{\mathrm{tot}}(T) \lesssim \frac{\|\hat{H}_{\mathrm{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\mathrm{od}}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\mathrm{od}}(t)}{dt^2} \right\| dt. \tag{3.31}$$

This is the central leakage-related quantity used in the main text of the paper [**?**]. Even though Eq. (3.31) is compact, it carries a rich physical interpretation and is particularly well suited to control design.

The first two terms in Eq. (3.31),

$$\frac{\|\hat{H}_{\mathrm{od}}(0)\|}{\Delta(0)} \qquad \text{and} \qquad \frac{\|\hat{H}_{\mathrm{od}}(T)\|}{\Delta(T)}, \tag{3.32}$$

are boundary contributions. They penalize the presence of residual block-off-diagonal couplings at the initial and final times of the control sequence. Their physical meaning is intuitive: if the control pulse does not start and end in a configuration where the computational basis aligns well with the physical Fock basis, then the gate may begin or terminate with unwanted admixture of leakage components. These terms therefore strongly motivate boundary conditions that force the relevant control pulses to vanish at the beginning and at the end of the gate.

The integral term,

$$\int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\mathrm{od}}(t)}{dt^2} \right\| dt, \tag{3.33}$$

has a different interpretation. It penalizes rapid temporal variation of the residual block-off-diagonal Hamiltonian and is therefore closely connected to non-adiabaticity. In qualitative terms, this contribution says that leakage is not only controlled by how large the unwanted inter-subspace coupling is, but also by how sharply it changes in time. A pulse that is too abruptly modulated may remain acceptable from the point of view of instantaneous amplitude, yet still induce leakage because the time dependence itself excites transitions between subspaces. Equation (3.33) is thus the mathematical expression of the speed–leakage tradeoff discussed earlier in the background chapter.

One of the most attractive features of Eq. (3.31) is that it unifies the dominant leakage behavior in both off-resonant and on-resonant regimes. The paper argues that, after comparing the different terms arising from the direct-coupling and generalized adiabatic-theorem analyses, the bound in Eq. (3.31) captures the dominant leakage contribution in both regimes [**?**]. More specifically, in slowly modulated off-resonant situations the boundary terms may dominate, whereas in faster on-resonant modulation regimes the time-derivative contribution becomes more important. This makes the bound especially valuable for optimization, since it does not apply only to one narrow operational regime.

From a control-theoretic perspective, the importance of Eq. (3.31) lies in the fact that it is *optimization-friendly.* Unlike a leakage estimate based purely on final-state population obtained after a full time evolution, the bound is written directly in terms of the effective Hamiltonian and its derivatives. This means that it can be evaluated as a physically meaningful soft penalty during optimization, and it also gives immediate qualitative guidance:

- reduce the magnitude of the residual block-off-diagonal coupling,
- enforce smooth temporal modulation,

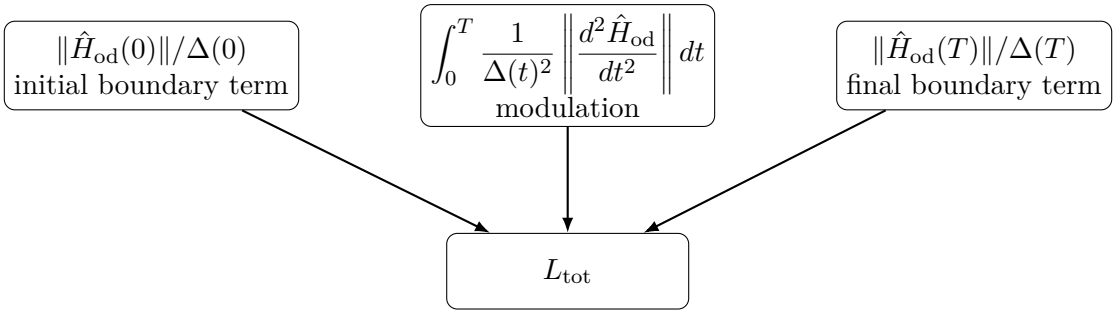


Figure 3.5: Structure of the leakage bound used in the paper. The total leakage penalty L_{tot} combines two boundary contributions, associated with residual off-diagonal couplings at the initial and final times, with an integral term that penalizes rapid modulation of the effective block-off-diagonal Hamiltonian.

- satisfy appropriate boundary conditions,
- keep the relevant energy gap from becoming too small.

In this sense, the bound is not only a diagnostic; it is also an analytic design principle.

This is precisely why the paper elevates leakage from a post hoc error metric to an explicit optimization target. Instead of relying on randomized benchmarking or brute-force leakage estimation at the end of training, it introduces the bound L_{tot} as a penalty term in the control objective. The significance of this move should not be understated. It means that the reinforcement-learning agent is not merely rewarded for producing a good final logical unitary, but is also discouraged, throughout optimization, from exploring pulse trajectories that are predicted to be leakage-prone by the perturbative physics of the full multilevel Hamiltonian [**?**].

For the structure of the thesis, Eq. (3.31) is the main mathematical outcome of the paper's leakage analysis. It takes the abstract discussion of TSWT and turns it into a concrete scalar quantity that can be inserted into the cost function. The next subsection will build directly on this result by introducing the full UFO objective, where L_{tot} appears alongside gate infidelity, boundary-condition penalties, and total runtime.

3.2.3 The Universal cost Function Optimization objective

The central contribution of the reference paper is the introduction of a control objective that combines, within a single scalar cost function, the main physical requirements of realistic quantum gate design: logical accuracy, suppression of leakage, compatibility with control boundary conditions, and short runtime [**?**]. This objective is called *Universal cost Function control Optimization* (UFO). The adjective "universal" is important: the aim is not merely to optimize a specific state-transfer task or a particular entangling gate under highly specialized constraints, but to define a general control objective that remains meaningful for arbitrary target unitaries while preserving the controllability of the underlying quantum system.

The first term of the UFO objective is the gate infidelity. If the target unitary is denoted by U_{target} and the physical control pulse generates the final propagator

$$U(T) = \mathcal{T} \exp\left(-i \int_0^T \hat{H}_{\mathrm{RWA}}(t)\, dt \right), \tag{3.34}$$

then the paper measures control inaccuracy through

$$1 - F[U(T)] = 1 - \frac{1}{16} \left| \mathrm{Tr}\Big(U^\dagger(T)\, U_{\mathrm{target}} \Big) \right|^2, \tag{3.35}$$

where the factor 1/16 reflects the dimension $d = 4$ of the two-qubit computational subspace. Equation (3.35) vanishes exactly when the implemented gate agrees with the target gate up to a global phase. The paper chooses this quantity as the first part of the cost function because it is computationally inexpensive to evaluate and directly linked to the quality of the controlled logical operation [**?**].

The second term is the leakage penalty discussed in the previous subsection. Rather than measuring leakage only after optimization, the paper inserts the analytic leakage bound L_{tot} directly into the objective. In this way, leakage suppression becomes a genuine optimization target rather than a posterior diagnostic. Conceptually, this is one of the strongest features of the UFO framework: the control agent is not only rewarded for implementing the correct logical unitary, but also discouraged from exploiting pulse trajectories that are predicted to generate substantial leakage in the full multilevel dynamics.

The third ingredient of the UFO cost function is a boundary-condition penalty. The paper imposes that the microwave control pulses and the tunable coupling pulse vanish at the initial and final times, so that the computational basis and the Fock basis coincide at the boundaries of the gate. This is enforced through the penalty

$$P_{\text{bdry}} = \sum_{t\in\{0,T\}} \left[g(t)^2 + f(t)^2\right], \tag{3.36}$$

where, for compactness, one may write

$$f(t)^2 := f_1(t)^2 + f_2(t)^2. \tag{3.37}$$

The physical role of Eq. (3.36) is twofold. First, it facilitates convenient gate concatenation by ensuring that the pulse begins and ends in a clean logical basis. Second, as emphasized in the paper, it helps reduce errors associated with deviations from the rotating-wave approximation, since non-RWA oscillatory contributions are less problematic when the control envelopes vanish smoothly at the boundaries [**?**].

The fourth and last ingredient is the runtime penalty. If T denotes the total duration of the control pulse, then a linear penalty in T is added to the cost function in order to encourage shorter gates:

$$P_{\text{time}} = T. \tag{3.38}$$

This term encodes the physically important idea that long gates are undesirable not only because they are operationally slower, but also because they increase the exposure of the quantum system to decoherence, control drift, and stochastic control errors. In this sense, the runtime penalty is not an arbitrary numerical regularizer; it is a direct expression of the hardware-level desirability of fast control.

Combining these four ingredients, the paper defines the full UFO cost function as

$$C(\chi,\beta,\mu,\kappa) = \chi\big[1 - F(U(T))\big] + \beta L_{\text{tot}} + \mu \sum_{t\in\{0,T\}} \left[g(t)^2 + f(t)^2\right] + \kappa T, \tag{3.39}$$

where $\chi, \beta, \mu, \kappa > 0$ are adjustable hyperparameters. Equation (3.39) is the defining equation of the framework. It is the scalar objective minimized throughout the control optimization, and each coefficient controls the relative priority assigned to a different physical desideratum:

$$\chi: \text{ gate accuracy,} \tag{3.40}$$
$$\beta: \text{ leakage suppression,} \tag{3.41}$$
$$\mu: \text{ boundary-condition enforcement,} \tag{3.42}$$
$$\kappa: \text{ runtime reduction.} \tag{3.43}$$

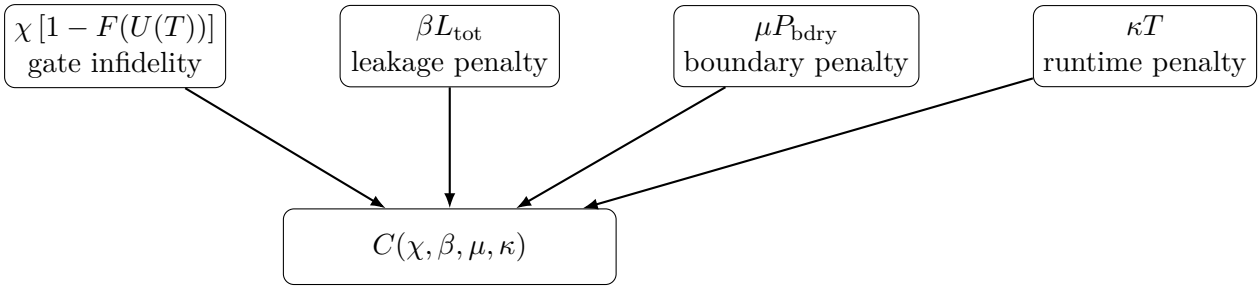


Figure 3.6: Structure of the UFO objective. The full control cost is obtained by combining four physically motivated penalty terms: gate infidelity, accumulated leakage, violation of boundary conditions, and total runtime.

The conceptual importance of Eq. (3.39) is not just that it contains several terms, but that it encodes them as *soft constraints* rather than hard restrictions. This is a major design choice of the paper. Much of the previous literature had reduced leakage by turning off certain control channels entirely or by imposing strong structural restrictions on the admissible controls. While such hard constraints may simplify the problem or suppress certain error mechanisms, they can also impair the full controllability of the system. The UFO framework takes the opposite approach: it preserves access to all relevant independent control channels and instead penalizes physically undesirable behavior through tunable cost terms [**?**]. This offers the optimizer more flexibility while still discouraging leakage, runtime inflation, and poor boundary behavior.

This soft-constraint philosophy is one of the reasons the objective deserves to be called universal. The framework does not depend on a single special-purpose gate ansatz, nor does it assume that the best strategy is obtained by imposing rigid analytical restrictions on the control. Instead, it defines a physically motivated cost landscape over the full admissible control space. The optimizer is then free to explore that landscape and discover control solutions that balance the competing objectives according to the chosen weights. From a control-theoretic perspective, Eq. (3.39) is therefore best viewed as a multi-objective scalarization of the realistic gate-design problem.

Another important feature is that the UFO objective is modular. The paper explicitly notes that, when applying the framework to a different quantum-computing platform, the individual terms of the cost function can be modified so as to reflect the relevant underlying physics [**?**]. For example, if a different hardware platform required alternative boundary conditions, bandwidth penalties, or noise-aware robustness measures, these could be incorporated by replacing or augmenting the corresponding terms. Thus, universality here should not be interpreted as meaning that one fixed formula solves every platform unchanged; rather, it means that the framework provides a reusable blueprint for constructing a physically meaningful optimization objective.

From the perspective of reinforcement learning, Eq. (3.39) is the crucial bridge between physics and algorithm. Once this scalar objective has been defined, it can be used directly as the quantity to be minimized, or equivalently as the negative reward to be maximized by the RL agent. This is why the UFO cost function occupies such a central position in the paper: it is the place where all the physics of the control problem is distilled into a form that is both analytically meaningful and numerically optimizable.

For the structure of the thesis, Eq. (3.39) should be treated as the centerpiece of the paper-specific framework chapter. The earlier subsections prepare its ingredients: the gmon Hamiltonian defines the control model, the subspace decomposition identifies the leakage channels, and the TSWT analysis produces the leakage bound. The later subsections then explain how this cost function is used inside the reinforcement-learning formulation, and how the resulting controls are evaluated on the target gate family studied in the paper.

3.3 Reinforcement-learning formulation of the control problem

3.3.1 Casting the control problem as an RL problem

Once the UFO objective has been defined, the control problem can be recast as a sequential decision problem and solved with reinforcement learning. In the reference paper, the role of the agent is to construct a complete time-dependent control trajectory by proposing control actions step by step, while the environment simulates the resulting quantum evolution and returns the associated control cost [**?**]. In this formulation, reinforcement learning is not used to replace the underlying Schrödinger dynamics; rather, it provides an optimization strategy for navigating the high-dimensional space of admissible pulse sequences.

The time axis is discretized into control intervals of duration Δt, and the quantum gate is built iteratively over these time steps. Let U_n denote the simulated unitary propagator after the n-th step of the control trajectory. Then the state used by the RL agent is taken to contain the information of the current simulated gate,

$$s_n \sim U_n, \tag{3.44}$$

that is, the policy network receives as input a representation of the quantum evolution accumulated up to time step n. This is a natural choice from the viewpoint of control: the current gate U_n summarizes the dynamical history relevant for deciding which control action should be applied next.

At each step, the agent proposes a continuous control action corresponding to the Hamiltonian parameters for the next interval. In the notation of the effective gmon Hamiltonian, one may write the action as

$$a_n = \big(g_n,\ \delta_{1,n}, \delta_{2,n},\ f_{1,n}, f_{2,n},\ \phi_{1,n}, \phi_{2,n}\big), \tag{3.45}$$

where each component specifies one of the control channels available during the next time slice. Thus, one RL episode is nothing but the sequential construction of a full analog pulse schedule

$$\boldsymbol{a} = (a_0, a_1, \ldots, a_{N-1}), \tag{3.46}$$

from which the complete time-dependent Hamiltonian trajectory is assembled.

Given the proposed action a_n, the training environment propagates the quantum dynamics for one time step. In schematic form, the paper represents this update as

$$U_{n+1} = \exp\Big[-i\Delta t\,\big(\hat{H}_{n+1} + \delta\hat{H}_{n+1}\big)\Big]\,U_n, \tag{3.47}$$

where $\hat{H}_{n+1}$ is the Hamiltonian specified by the control action and $\delta\hat{H}_{n+1}$ denotes a stochastic perturbation used during noisy training [**?**]. Equation (3.47) captures the core RL-environment interaction of the paper: the agent proposes a control, the environment applies the corresponding noisy Hamiltonian for one interval, and the propagated gate U_{n+1} is returned as the next state.

The reward is derived directly from the UFO cost. Since reinforcement learning is naturally formulated as a reward-maximization problem, while the control objective is written as a cost to be minimized, the two are related simply by a sign change:

$$r_n = -\,c_n, \tag{3.48}$$

where c_n denotes the control cost evaluated by the environment at step n, or more generally the increment of the trajectory cost associated with the newly applied control. At the level of a full episode, the return is therefore

$$G_0 = \sum_{n=0}^{N-1} \gamma^n r_n, \tag{3.49}$$

and maximizing expected return is equivalent to minimizing the expected UFO control cost over sampled pulse trajectories. In this way, the entire physics of the problem enters the RL agent only through the environment update in Eq. (3.47) and the reward signal in Eq. (3.48).

The paper further interprets optimization as a sequence of episodes, each of which contains all the time steps of one complete control trajectory. The episode length is determined by the smaller of:

$$T_{\text{episode}} = \min(T_{\max},\ T_{\text{stop}})\,, \tag{3.50}$$

where $T_{\max}$ is a predefined maximum runtime and T_{stop} is the time at which a satisfactory value of the UFO cost is reached [**?**]. This means that the learning problem is not only about shaping the pulse amplitudes, but also about discovering sufficiently good controls within a finite horizon. The runtime penalty in the UFO objective and the stopping rule in Eq. (3.50) therefore work together to encourage short, high-quality trajectories.

An important implementation detail is that the RL agent in the paper uses two neural networks. The first, the *policy network*, maps the current state to a probability distribution over the next control action. The second, the *value network*, estimates the discounted future reward associated with the current state [**?**]. Denoting these schematically by

$$\pi_{\boldsymbol{\theta}}(a_n \mid s_n) \qquad \text{and} \qquad V_{\boldsymbol{\omega}}(s_n), \tag{3.51}$$

one sees that the control problem is implemented in actor–critic form: the policy network generates pulse proposals, while the value network evaluates their long-term quality and stabilizes training. The paper reports that both networks are fully connected with hidden-layer dimensions 64, 32, and 32, respectively [**?**]. While this architectural detail is not the conceptual core of the method, it is useful to report it in the thesis because it clarifies the scale and practical implementation of the RL agent.

From a control perspective, the most important conceptual point is that the environment state is not a low-dimensional hand-crafted summary such as "current pulse amplitude" or "elapsed time" alone. Instead, it contains information about the current simulated unitary gate. This is a strong design choice: it means the agent is trained to reason directly in terms of how much of the target operation has already been built, rather than purely in terms of local control variables. Such a state representation is well matched to gate synthesis, since the desired task is naturally expressed in terms of the evolving unitary transformation rather than a single state vector or observable.

The RL formulation of the paper can therefore be summarized as follows:

- the *state* encodes the current simulated gate U_n,
- the *action* specifies the continuous control parameters for the next time interval,
- the *environment* propagates the noisy Schrödinger dynamics,
- the *reward* is the negative UFO control cost,
- one *episode* corresponds to a complete pulse trajectory for a target gate.

This formulation is particularly natural for robust analog quantum control, because it preserves the sequential character of pulse generation while allowing the physical optimization objective to be used directly as the reward signal.

For the structure of the thesis, this subsection is the bridge between the physics of the UFO objective and the algorithmic machinery of policy optimization. The next paragraph will build directly on it by explaining why the paper specifically chooses a continuous-variable policy-gradient method trained with trust-region policy optimization (TRPO), rather than a simpler gradient-descent or value-based RL approach.

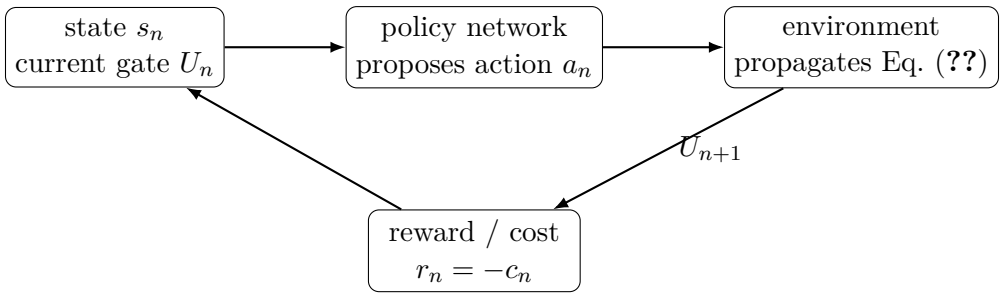


Figure 3.7: Paper-specific RL formulation. The state is built from the currently simulated gate, the policy proposes continuous control parameters for the next interval, the environment propagates the noisy quantum dynamics, and the resulting control cost is returned as reward.

3.3.2 Implementation of the policy and value networks

The previous chapter introduced the policy and value functions as abstract objects of actor–critic reinforcement learning. In the present implementation, these objects are realized through two separate fully connected feed-forward neural networks, one for the stochastic policy and one for the state-value function. Their role is not to replace the physical model of the control problem, which remains entirely specified by the multilevel gmon Hamiltonian and by the UFO cost, but to provide a trainable parametric representation of the control law and of the associated long-term return within the TRPO optimization loop.

The design of these networks follows the structure of the RL environment introduced in Sec. 3.3.1. At each decision step, the agent receives a state representation encoding the current status of the physical gate-construction process, and must output either a distribution over the next continuous control action or a scalar estimate of the expected future return from that state. Since both tasks are defined on fixed-dimensional vector inputs and no explicit recurrent memory is required beyond the state representation itself, a multilayer perceptron architecture is sufficient for the purposes of the present study.

State representation. The observation supplied to the agent is built from the current simulated propagator of the full truncated multilevel system. In the present implementation, each local superconducting mode is truncated to three levels, so that the total physical Hilbert-space dimension is

$$d_{\text{phys}} = 3^2 = 9. \tag{3.52}$$

At time step t, let $U_t \in \mathbb{C}^{9\times 9}$ denote the current full propagator. The state is then constructed by vectorizing this matrix, separating its real and imaginary parts, and appending a scalar encoding the normalized progress through the episode. In compact form,

$$s_t = \Big(\Re(\text{vec}(U_t)), \Im(\text{vec}(U_t)), t/T_{\max}\Big). \tag{3.53}$$

Since U_t has $9^2 = 81$ complex entries, the resulting observation dimension is

$$d_{\text{obs}} = 2 \cdot 81 + 1 = 163. \tag{3.54}$$

This representation is more informative than a purely time-based or control-based state encoding, because it exposes to the agent the currently accumulated physical action of the pulse sequence itself. In this sense, the policy does not act only on a clock variable, but on a compact summary of how much of the desired unitary evolution has already been realized in the full multilevel model.

Policy network. The policy is implemented as a stochastic continuous-action actor. More precisely, the policy network maps the current state s_t to the mean of a Gaussian distribution in a seven-dimensional raw-action space,

$$\mu_\theta(s_t) \in \mathbb{R}^7, \tag{3.55}$$

while the exploration scale is determined by a learned log-standard-deviation parameter

$$\log \sigma \in \mathbb{R}^7, \tag{3.56}$$

which is state-independent in the present implementation. The raw action is therefore sampled according to

$$\tilde{a}_t \sim \mathcal{N}(\mu_\theta(s_t), \mathrm{diag}(\sigma^2)), \tag{3.57}$$

and is subsequently mapped to the bounded action domain through a componentwise hyperbolic tangent,

$$a_t = \tanh(\tilde{a}_t). \tag{3.58}$$

The use of the squashing map in Eq. (3.58) is important because the physical control channels are amplitude-bounded. It allows the Gaussian policy to remain unconstrained in the raw space while ensuring that the actual action presented to the environment lies in the compact interval $[-1, 1]^7$.

The squashed action a_t is not yet the physical control vector entering the Hamiltonian. It must still be mapped to the seven control channels of the gmon model,

$$u_t = (g_t,\ \delta_{1,t}, \delta_{2,t},\ f_{1,t}, f_{2,t},\ \phi_{1,t}, \phi_{2,t}), \tag{3.59}$$

through a deterministic rescaling procedure. In the present code, the linear-amplitude channels $g_t, \delta_{1,t}, \delta_{2,t}, f_{1,t}, f_{2,t}$ are obtained by multiplying the corresponding action components by the maximum control amplitude, while the phase channels are mapped from $[-1, 1]$ to $[0, 2\pi]$. In this way, the policy network learns in a normalized action space, whereas the environment receives controls expressed directly in physical units.

Value network. The critic is implemented as a separate feed-forward network that outputs a single scalar,

$$V_\omega(s_t) \in \mathbb{R}, \tag{3.60}$$

which approximates the expected discounted future reward from the current state. As in standard actor–critic formulations, this estimate is used to construct advantage signals for the policy update and therefore plays a stabilizing role in the optimization. The value network is trained by regression against empirical returns collected from sampled rollouts, while the policy network is updated through the TRPO step described in Sec. 3.3.3. The two networks are thus optimized jointly but serve distinct purposes: the actor generates candidate control decisions, whereas the critic estimates their long-term usefulness.

Both the policy and the value networks use the same backbone architecture. Each is a multilayer perceptron with tanh activation functions between successive affine layers. The policy head is a linear map from the last hidden layer to the seven-dimensional mean vector in Eq. (3.55), while the critic head is a linear map from the same hidden representation to the scalar output in Eq. (3.60). This architecture matches the one reported in the original paper and was retained here in order to remain as close as possible to the reference implementation while avoiding additional architectural degrees of freedom in the empirical comparison [**?**].

It is worth emphasizing that the neural networks introduced here are embedded in a tightly constrained physical loop. The policy does not output an abstract machine-learning action unrelated to the device; it outputs the next candidate control values for the Hamiltonian channels

of the multilevel gmon model. Likewise, the value function is not trained against an arbitrary task-dependent label, but against returns generated from the UFO objective, which itself encodes gate fidelity, leakage suppression, boundary regularity, and runtime. For this reason, the actor–critic architecture should be interpreted as an optimization interface between reinforcement learning and the physical control model, rather than as an independent phenomenological model of the quantum dynamics.

TODO: fix this paragraph

3.3.3 Why the paper uses trust-region policy optimization

Having formulated the control problem as a continuous-variable reinforcement-learning task, the next question is which policy-optimization algorithm should be used. The reference paper chooses *trust-region policy optimization* (TRPO) rather than simpler direct gradient-descent or gradient-free search schemes [**?**, **?**]. This choice is not incidental. It reflects the specific structure of the quantum-control landscape considered in the paper: the action space is continuous, the reward depends on a high-dimensional pulse trajectory, the environment is stochastic due to simulated control noise, and the optimization landscape is expected to contain many imperfect local optima and saddle regions. In such a setting, policy updates that are too aggressive can destabilize learning, while updates that are too myopic may fail to exploit nonlocal regularities of good control trajectories.

The paper explicitly argues that applying second-order optimization ideas at the level of the *policy* is superior to simpler optimization strategies such as direct gradient descent or differential evolution on the control parameters themselves [**?**]. The intuition is that the policy neural network can encode regularities shared by many successful control trajectories, rather than optimizing each pulse as an unrelated object. In a high-dimensional continuous-control problem, this is potentially a major advantage: instead of performing local descent directly in the raw control space, the algorithm learns a structured distribution over control actions conditioned on the evolving gate state. Such a representation is particularly valuable when nearby target gates or nearby regions of the control landscape share common dynamical patterns.

At the mathematical level, TRPO updates the policy by maximizing a local surrogate objective while constraining the change in policy distribution. If $\boldsymbol{\theta}_{\text{old}}$ denotes the current policy parameters and $\boldsymbol{\theta}$ the updated ones, the standard TRPO step is based on the constrained problem

$$\max_{\boldsymbol{\theta}} \; \mathcal{L}_{\boldsymbol{\theta}_{\text{old}}}(\boldsymbol{\theta}) = \mathbb{E}_{(s,a)\sim\pi_{\boldsymbol{\theta}_{\text{old}}}} \left[\frac{\pi_{\boldsymbol{\theta}}(a \mid s)}{\pi_{\boldsymbol{\theta}_{\text{old}}}(a \mid s)} \, \hat{A}_{\boldsymbol{\theta}_{\text{old}}}(s,a) \right], \tag{3.61}$$

subject to the trust-region constraint

$$\mathbb{E}_{s\sim\pi_{\boldsymbol{\theta}_{\text{old}}}} \left[D_{\text{KL}}(\pi_{\boldsymbol{\theta}_{\text{old}}}(\,\cdot \mid s) \parallel \pi_{\boldsymbol{\theta}}(\,\cdot \mid s)) \right] \leq \delta_{\text{TR}}, \tag{3.62}$$

where $\hat{A}_{\boldsymbol{\theta}_{\text{old}}}(s,a)$ is an estimate of the advantage function and $\delta_{\text{TR}} > 0$ fixes the allowed size of the update in distribution space [**?**]. The significance of Eq. (3.62) is that it limits how much the policy is allowed to change in one step, thereby reducing the risk that a noisy gradient estimate produces a catastrophic deterioration in control quality.

This trust-region mechanism is especially well suited to the quantum-control problem studied in the paper. The reward depends on the UFO objective, which itself couples fidelity, leakage, boundary conditions, and runtime. Because these terms may compete with one another, the effective reward landscape can be highly irregular and sensitive to policy updates. A naive gradient step could easily move the policy to a region that explores qualitatively different pulse trajectories and thereby destroy previously learned good behavior. By restricting the policy variation through a KL divergence constraint, TRPO encourages *stable policy improvement*, which is particularly valuable when the control trajectory is long and the reward is influenced by stochastic perturbations in the Hamiltonian [**?**, **?**].

Another important reason for the choice of TRPO is that the paper is not simply optimizing one deterministic pulse vector. It is training a stochastic policy in a noisy environment. In this context, the value network estimates the expected discounted future reward, while the policy network proposes distributions over the next continuous control action. The paper reports that after sampling a batch of size

$$N_{\text{batch}} = 20000 \tag{3.63}$$

episodes, the policy network is updated so as to maximize expected discounted future reward *within the trusted region*, while the value network is updated to fit the new empirical returns [**?**]. Thus, the optimization is performed not step by step on a single control trajectory, but statistically over a large ensemble of sampled episodes. This batch-based learning is well matched to TRPO, whose trust-region update is specifically designed to use sampled trajectory information efficiently and robustly.

The paper also gives a conceptual argument for why TRPO should outperform direct optimization of pulse parameters. Direct gradient descent in very high-dimensional nonconvex landscapes is known to struggle with flat saddle regions and poorly conditioned directions, especially when gradients become small [**?**]. In the present control problem, these difficulties are exacerbated by leakage, stochastic control noise, and the fact that the control trajectory must satisfy multiple competing objectives simultaneously. TRPO addresses this at the policy level: rather than approximating or computing the full Hessian of the raw control landscape, it performs a model-free second-order policy update that is constrained by the geometry of the policy distribution itself. In this sense, the algorithm imports second-order stabilization ideas into a setting where direct control-Hessian methods would be prohibitively expensive.

There is also a more physical reason why a trust-region policy method is attractive here. Quantum control pulses are not arbitrary vectors; they are temporally correlated objects whose early and late segments interact through phase accumulation, transient leakage, and the final gate constraint. A useful optimizer must therefore capture *nonlocal structure* across the trajectory. The paper emphasizes exactly this point, arguing that the policy neural network can learn nonlocal regularities of effective control solutions, and that these regularities are difficult to exploit with simpler local optimization schemes [**?**]. In other words, TRPO is not chosen merely because it is a standard continuous-control RL algorithm; it is chosen because its policy-based formulation is compatible with the trajectory-level structure of analog quantum control.

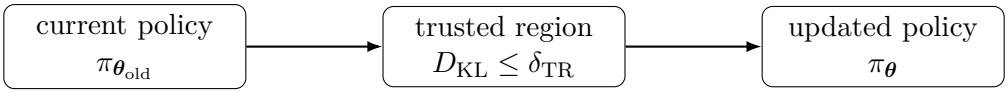
From the implementation viewpoint, the paper uses TRPO in an actor–critic architecture. Denoting the policy network by $\pi_{\boldsymbol{\theta}}(a \mid s)$ and the value network by $V_{\boldsymbol{\omega}}(s)$, the learning loop alternates between: **TODO:** DA SOSTITUIRE CON SNIPPET/PSEUDOCODE

1. sampling many episodes under the current stochastic policy,
2. estimating discounted future rewards and advantages,
3. updating the policy network by solving the trust-region step,
4. updating the value network to fit the newly sampled returns.

Although these are standard ingredients of modern policy-gradient RL, their role in the paper is highly specific: they provide a scalable optimization engine for a physically structured continuous-control problem in which leakage and stochastic robustness are part of the objective itself.

For the purposes of the thesis, the main message of this subsection is therefore clear. The paper uses TRPO because it combines three properties that are especially desirable in robust analog quantum control:

- it handles *continuous actions* naturally,
- it supports *stochastic policy learning* in noisy environments,



policy improvement is allowed only if the new action distribution
remains sufficiently close to the old one

Figure 3.8: Geometric intuition behind the TRPO update. Rather than taking an unconstrained gradient step, the policy is improved only within a trust region defined by a KL-divergence constraint, which stabilizes learning in continuous and stochastic control environments.

- it enforces *stable policy updates* through a trust-region constraint.

This makes it a particularly reasonable choice for the UFO optimization problem, where the policy must learn high-quality pulse trajectories in a multilevel superconducting system under realistic control noise.

In the structure of the thesis, this subsection should be read as the algorithmic complement of the previous one. The earlier subsection explained how the physics problem is cast as an RL environment; the present one explains why TRPO is the specific optimization engine chosen to solve it. The next subsection will complete the learning formulation by explaining how stochastic control noise is injected during training in order to promote robustness.

3.3.4 Training in a stochastic control environment

A distinctive feature of the reference paper is that robustness is not treated merely as a post-training diagnostic. Instead, stochastic control errors are inserted directly into the training environment, so that the reinforcement-learning agent is optimized under noisy conditions from the outset [**?**]. This is a conceptually important design choice. In a purely nominal training regime, the agent would learn control trajectories that are optimal only for the unperturbed Hamiltonian, and robustness would be assessed only afterward by testing how much the resulting performance degrades under perturbations. The paper adopts the opposite strategy: it places the agent in a stochastic environment during learning and thereby encourages the discovery of pulse sequences whose performance remains good across many noisy realizations.

The stochasticity is introduced at the level of the control amplitudes. At each discrete time step t_k, the ideal control Hamiltonian specified by the policy is modified by random fluctuations representing control noise. Denoting by $\hat{H}(t_k)$ the intended Hamiltonian and by $\delta\hat{H}(t_k)$ the perturbation, the noisy one-step propagation used during training takes the form

$$U_{k+1} = \exp\Big[-i\Delta t\,\big(\hat{H}(t_k) + \delta\hat{H}(t_k)\big)\Big]\,U_k. \tag{3.64}$$

Equation (3.64) is the stochastic counterpart of the noiseless environment update introduced in the previous subsection. The agent therefore does not interact with a single deterministic Hamiltonian trajectory, but with an ensemble of randomly perturbed trajectories generated from the same policy action sequence.

In the implementation described by the paper, the control perturbations are modeled as Gaussian fluctuations added to the control amplitudes [**?**]. Writing schematically the ideal control vector at time step k as

$$\boldsymbol{u}(t_k) = \big(g(t_k), \delta_1(t_k), \delta_2(t_k), f_1(t_k), f_2(t_k), \dots\big), \tag{3.65}$$

the noisy realization may be represented as

$$\boldsymbol{u}_{\text{noisy}}(t_k) = \boldsymbol{u}(t_k) + \boldsymbol{\xi}(t_k), \tag{3.66}$$

where

$$\boldsymbol{\xi}(t_k) \sim \mathcal{N}(\mathbf{0},\, \Sigma_\xi) \tag{3.67}$$

is a Gaussian random vector. In this way, the same deterministic policy can generate different realized control trajectories on different episodes, reflecting the fact that in an experimental setting the nominal pulse sent to the hardware is never reproduced perfectly.

From the viewpoint of reinforcement learning, this changes the optimization target in an essential way. The policy no longer maximizes the return of a single deterministic trajectory, but the *expected* return over both the trajectory distribution induced by the policy and the distribution of control perturbations:

$$J(\boldsymbol{\theta}) = \mathbb{E}_{\tau,\xi}\left[\sum_{k=0}^{N-1} \gamma^k r_k\right]. \tag{3.68}$$

Equivalently, if the reward is defined as the negative UFO cost, one may write

$$J(\boldsymbol{\theta}) = -\mathbb{E}_{\tau,\xi}\left[C(\chi, \beta, \mu, \kappa)\right]. \tag{3.69}$$

Thus, training in a stochastic environment directly transforms the control problem into a robust optimization problem: the agent is encouraged to find policies that perform well on average over noisy realizations rather than excelling only in the idealized nominal setting.

This is one of the most elegant aspects of the paper's methodology. Robustness is not introduced through a separate outer-loop procedure or through repeated experimental recalibration, but through the RL environment itself. The environment becomes a sampling mechanism for control uncertainty, and the policy-gradient update uses these noisy rollouts to improve the expected performance of the control strategy. In this sense, stochastic training acts as an implicit regularizer: pulse trajectories that are overly fragile under small perturbations will exhibit poor average return and will therefore be disfavored during learning.

The paper evaluates the success of this strategy through two key diagnostics. The first is the *average gate fidelity* under noisy control realizations,

$$\overline{F}_{\text{noisy}} = \mathbb{E}_\xi\left[F(U_\xi(T), U_{\text{target}})\right], \tag{3.70}$$

where $U_\xi(T)$ denotes the final propagator obtained under one sampled noisy trajectory. The second is the fidelity variance,

$$\mathrm{Var}_\xi[F] = \mathbb{E}_\xi\left[F_\xi^2\right] - \left(\mathbb{E}_\xi[F_\xi]\right)^2, \tag{3.71}$$

which quantifies the consistency of the gate performance across different noise realizations. These two quantities are complementary. A policy with high average fidelity but very large variance may still be experimentally undesirable, because it performs unreliably from shot to shot. By contrast, a policy with slightly lower mean fidelity but much lower variance may be more robust in practice.

The use of fidelity variance is particularly insightful. It reflects the fact that robustness should not be understood only as good *mean* performance, but also as small sensitivity to uncertainty. In other words, a robust control is not merely a control whose expected output is good; it is a control whose performance is also reproducible. This makes the paper's evaluation framework more physically meaningful than one based solely on nominal fidelity or even average fidelity alone [**?**]. It also explains why the paper emphasizes reductions in both average noisy gate error and fidelity variance when comparing the RL solutions to baseline methods.

A further conceptual point is that the noisy training environment is aligned with the physical error model of interest. The paper is not injecting arbitrary synthetic stochasticity for machine-learning convenience; it is perturbing the same control amplitudes that are subject

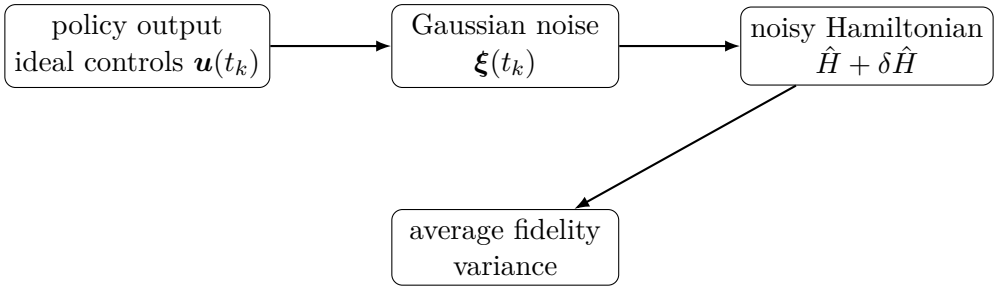


Figure 3.9: Training under stochastic control noise. The policy proposes ideal controls, Gaussian perturbations are added to produce a noisy Hamiltonian trajectory, and the resulting performance is evaluated statistically through average fidelity and fidelity variance.

to imperfections in actual superconducting hardware. Thus, the stochastic environment serves as a simple but physically interpretable proxy for experimental control uncertainty. This is important, because it distinguishes the methodology from generic noise-augmentation heuristics: the perturbations are chosen specifically to reflect the control channels that matter in the underlying Hamiltonian.

From the perspective of the thesis, this subsection is where robustness fully enters the learning formulation. Earlier subsections introduced robustness as a motivation and the UFO objective as a physically meaningful cost. The present subsection shows how the paper turns that idea into practice: the RL agent is trained on noisy Hamiltonian trajectories, and performance is judged by noisy average fidelity and fidelity variance. The next subsection will complete the framework by specifying the target family of two-qubit gates on which the method is trained and benchmarked.

3.4 Target gates, baselines, and original results

3.4.1 The target gate family $N(\alpha, \alpha, \gamma)$

Although the gmon Hamiltonian is fully controllable under the UFO framework, the reference paper does not benchmark the method on arbitrary two-qubit unitaries chosen at random. Instead, it focuses on a structured and physically important family of target gates,

$$N(\alpha, \alpha, \gamma) = \exp[i\,(\alpha\,\sigma_1^x\sigma_2^x + \alpha\,\sigma_1^y\sigma_2^y + \gamma\,\sigma_1^z\sigma_2^z)]\,, \tag{3.72}$$

where $\sigma_j^x, \sigma_j^y, \sigma_j^z$ are the Pauli operators acting on qubit j, and α, γ are continuous real parameters [**?**]. Equation (3.72) defines the gate family used throughout the numerical study of the paper and should therefore be treated as one of the key equations of the thesis.

From a structural viewpoint, Eq. (3.72) describes a family of nonlocal two-qubit operations generated by a combination of XX, YY, and ZZ interactions. The equal coefficients in front of the XX and YY terms are especially significant, because they reflect the exchange-type interaction that is naturally compatible with the effective XY-like entangling part of the gmon Hamiltonian. The $\gamma\,\sigma_1^z\sigma_2^z$ contribution then enriches the family by adding an Ising-like component. Thus, the target family is neither artificially simple nor completely generic: it is rich enough to contain many practically relevant entangling gates, while still being closely connected to the natural interaction structure of the hardware.

This hardware relevance is one of the main reasons the family is well chosen for benchmarking analog control. A completely arbitrary two-qubit unitary may require highly nontrivial synthesis and may obscure the relation between the target gate and the native couplings of the physical system. By contrast, the family $N(\alpha, \alpha, \gamma)$ is structured in a way that allows one to ask a much sharper question: to what extent can direct analog control on the gmon exploit its natural

interaction terms to realize this family more efficiently than standard gate synthesis? In other words, the target family is not just mathematically convenient; it is chosen so that the comparison between hardware-native analog control and compiled circuit-based implementation becomes especially meaningful [**?**].

The paper emphasizes that Eq. (3.72) includes, up to single-qubit rotations, several well-known and practically important two-qubit gates, including the SWAP, iSWAP, CNOT, and CZ gates, as well as the fermionic SWAP gate and Givens rotations [**?**]. This observation is important for the thesis because it shows that the benchmark is not restricted to a narrow toy example. On the contrary, the family spans operations that are central in quantum information processing and in quantum simulation. In particular, the inclusion of fermionic SWAP and Givens-type operations gives the study direct relevance to fermionic Hamiltonian simulation and quantum chemistry algorithms.

This last point deserves special emphasis. The paper explicitly motivates the family $N(\alpha, \alpha, \gamma)$ by its relevance to the simulation of many-electron systems [**?**]. Both fermionic SWAP operations and Givens rotations appear naturally in Jordan–Wigner-based mappings of fermionic problems onto qubit systems. Therefore, identifying analog control pulses that implement gates in this family more efficiently than their standard circuit-synthesis counterparts has potential applications that go well beyond a purely methodological control study. It connects the control problem to a concrete class of quantum algorithms where two-qubit gate efficiency is a major practical concern.

Another reason the family is a good benchmark is that it is continuously parametrized by α and γ. This allows the paper to study not just isolated target gates, but an entire manifold of related operations. In practical terms, this makes it possible to examine how gate runtime and control difficulty vary as one moves through the family, rather than reporting only a few isolated case studies. It also makes visible one of the interesting phenomena reported later in the paper, namely that reinforcement learning appears to benefit from transfer across nearby targets in parameter space [**?**]. A continuously parametrized gate family is therefore much more informative than a disconnected list of benchmark gates.

From the control-theoretic viewpoint, the family $N(\alpha, \alpha, \gamma)$ is also useful because it contains targets with varying degrees of alignment with the native hardware interactions. Some choices of (α, γ) are more naturally compatible with the exchange-like interaction already present in the gmon Hamiltonian, while others require a more intricate combination of local detuning, microwave control, and tunable coupling. This makes the family a good probe of *hardware efficiency*: it allows one to see which targets are especially natural for the device and which require more elaborate analog steering.

For the purposes of the thesis, the key role of Eq. (3.72) is therefore twofold. First, it defines the class of gates on which the UFO framework is trained and tested. Second, it provides a physically meaningful bridge between the mathematical control problem and applications in quantum simulation and quantum chemistry. The next subsection will build on this by explaining how the paper benchmarks its analog-control solutions against standard circuit-based baselines and which performance metrics are used in the comparison.

3.4.2 Baselines and comparison strategy

Once the target gate family $N(\alpha, \alpha, \gamma)$ has been specified, the next step is to determine how the performance of the proposed control method should be assessed. The reference paper uses a comparison strategy built around two distinct questions [**?**]. First, can the RL-based analog control produce target gates more *robustly* than alternative pulse-optimization methods under stochastic control noise? Second, can it realize those gates more *quickly* than their standard circuit-synthesis counterparts? These two questions lead naturally to two complementary classes of baselines: direct control-optimization baselines for robustness, and circuit-synthesis baselines for runtime.

The first baseline is a direct optimization of the control trajectory using the GRAPE method, implemented with the Adam optimizer, and an instance of Stochastic Gradient Descent (SGD) [**?**, **?**]. This baseline is important because it represents the most immediate non-RL alternative: one keeps the same physical control problem and the same general objective structure, but optimizes the control parameters directly rather than through a policy-gradient agent. In the logic of the paper, this baseline tests whether the use of a policy-based second-order learning method such as TRPO actually provides an advantage over simpler first-order optimization of the raw control variables.

The second baseline is a reinforcement-learning control solution trained in a *noise-free* environment. This comparison is conceptually very useful, because it isolates the effect of stochastic training. If the noise-optimized RL controller outperforms the noise-free RL controller under noisy deployment conditions, then the improvement can be attributed not merely to the use of reinforcement learning as such, but specifically to the fact that the agent was trained under stochastic control perturbations. The comparison between these two RL settings therefore answers the question: is robustness really being learned from the noisy environment, or is it simply a generic byproduct of the RL architecture?

The third and most structurally different baseline is the *optimal gate synthesis* benchmark for the target family $N(\alpha, \alpha, \gamma)$. The paper notes that this family admits an optimal decomposition into a minimum-depth circuit of arbitrary single-qubit rotations and CZ-type entangling gates, giving a depth-seven circuit containing three two-qubit gates and five single-qubit gates [**?**]. Using state-of-the-art experimental gate times, the paper sets

$$T_{1q} = 20 \text{ ns}, \qquad T_{\text{CNOT}} = 45 \text{ ns}, \tag{3.73}$$

so that the total runtime of the synthesized implementation is

$$T_{\text{syn}} \approx 215 \text{ ns}. \tag{3.74}$$

Equation (3.74) provides the reference against which the analog RL control solutions are compared in terms of runtime. This is a particularly meaningful baseline, because it addresses the core hardware-efficiency claim of the paper: namely, whether direct pulse-level optimization can outperform even an *optimal* discrete circuit decomposition for the same target gate family.

The robustness comparison is based on repeated sampling of noisy control realizations. Let $\tilde{\boldsymbol{c}}$ denote an optimized control trajectory, and let $\delta\tilde{\boldsymbol{c}}$ be a random perturbation sampled from a Gaussian control-noise model,

$$\delta\tilde{\boldsymbol{c}} \sim \mathcal{N}(0, \sigma_{\text{noise}}). \tag{3.75}$$

The paper then evaluates the final gate under the perturbed control $\tilde{\boldsymbol{c}} + \delta\tilde{\boldsymbol{c}}$ and computes the corresponding gate fidelity. The primary robustness metric is the average fidelity,

$$F_{\text{ave}} = \mathbb{E}_{\delta\tilde{\boldsymbol{c}} \sim \mathcal{N}(0,\sigma_{\text{noise}})} \left[F\big(U(\tilde{\boldsymbol{c}} + \delta\tilde{\boldsymbol{c}})\big) \right], \tag{3.76}$$

which measures the mean control performance under the chosen stochastic error model. This quantity is central because it asks not how good the pulse is in the idealized nominal setting, but how well it performs when the controls are randomly perturbed in the way expected experimentally.

The second robustness metric is the sampled fidelity variance,

$$\sigma^2_{\text{fidelity}} = \mathbb{E}_{\delta\tilde{\boldsymbol{c}} \sim \mathcal{N}(0,\sigma_{\text{noise}})} \left[\big(F\big(U(\tilde{\boldsymbol{c}} + \delta\tilde{\boldsymbol{c}})\big) - F_{\text{ave}} \big)^2 \right]. \tag{3.77}$$

Equation (3.77) complements the average fidelity by measuring the shot-to-shot stability of the control scheme under noise. This distinction is important. A control strategy can have a high mean fidelity but still be experimentally fragile if its output fluctuates strongly across

noise realizations. For this reason, the paper uses both Eqs. (3.76) and (3.77) as robustness diagnostics.

The Gaussian control-noise model itself is varied over a family of noise strengths,

$$\sigma_{\text{noise}} \in [0.1,\, 3.5]\ \text{MHz}, \tag{3.78}$$

with $\sigma_{\text{noise}} = 1$ MHz treated in the paper as a representative experimentally relevant scale [**?**]. This range is useful for two reasons. First, it allows the paper to test robustness across a continuum of increasingly severe perturbation strengths rather than at a single arbitrarily chosen noise value. Second, it makes the resulting fidelity curves and variance curves far more informative: one can see not only which method performs best at one point, but how rapidly each method degrades as the stochastic control error becomes stronger.

The overall comparison strategy can therefore be summarized as follows. For *robustness*, the paper compares three control schemes:

1. noise-optimized RL control,
2. noise-free RL control,
3. SGD-based control optimization.

For *runtime*, it compares the noise-optimized analog RL control against the optimal gate-synthesis baseline. This division is very sensible. The SGD and noise-free RL baselines answer whether the learning formulation and noisy training genuinely improve robust pulse optimization, while the gate-synthesis baseline answers whether analog control provides a hardware-efficiency advantage over standard circuit compilation.

One of the strengths of this comparison strategy is that it does not rely on a single metric. A weaker paper might have reported only nominal final infidelity, in which case it would have been unclear whether the method improved robustness, runtime, or both. Here, by contrast, each baseline is tied to a specific scientific question:

- **SGD baseline:** does policy-gradient RL outperform direct pulse optimization?
- **Noise-free RL baseline:** does stochastic training actually improve robustness?
- **Optimal gate synthesis baseline:** does analog control improve hardware efficiency?
- **Average fidelity and variance:** is the resulting control both accurate and stable under noise?

This makes the numerical results much easier to interpret and gives the paper a clearer argumentative structure.

For the purposes of the thesis, the present subsection should be read as the methodological setup for the results section. It tells the reader what is being compared, why those baselines are the relevant ones, and which metrics are used to judge success. The next subsection can then summarize the principal findings of the paper, now that the comparison framework has been made explicit.

In compact form, the evaluation logic of the paper may be summarized as

$$\text{performance} = (\text{runtime},\ F_{\text{ave}},\ \sigma^2_{\text{fidelity}}), \tag{3.79}$$

with each baseline probing a different aspect of this performance vector.

3.4.3 Main findings of the original paper

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With the physical model, the UFO objective, the reinforcement-learning formulation, and the target gate family all in place, the main findings of the reference paper can now be stated clearly. The paper reports that training a TRPO-based reinforcement-learning agent in a stochastic control environment yields analog gate controls that are simultaneously fast, accurate, and substantially more robust to Gaussian control noise than the baseline methods considered [**?**]. In other words, the central claim of the work is not merely that reinforcement learning can find workable controls, but that the combination of the UFO objective with noisy training leads to a qualitatively better balance between fidelity, robustness, and runtime than alternative approaches.

The first major result concerns robustness under stochastic control errors. For the family of Gaussian control-noise models considered in the paper, with variance

$$\sigma_{\text{noise}} \in [0.1,\, 3.5] \text{ MHz}, \tag{3.80}$$

the noise-optimized RL controller maintains an average fidelity in the range

$$\overline{F}_{\text{RL,noisy}} \in [99.5\%,\, 98\%], \tag{3.81}$$

across the tested noise interval [**?**]. In particular, at the representative value

$$\sigma_{\text{noise}} = 1 \text{ MHz}, \tag{3.82}$$

the paper states that the obtained control satisfies its robustness criterion

$$1 - \overline{F} \leq \epsilon_0, \qquad \epsilon_0 = 0.007. \tag{3.83}$$

These results are significant because they show that the optimized controls are not only high-fidelity in the nominal setting, but remain accurate under a broad family of experimentally relevant perturbations.

The second major result is comparative. When evaluated under the same Gaussian control-noise model, the noise-optimized RL solution outperforms both the noise-free RL solution and the SGD baseline. The paper reports up to

$$\text{one order of magnitude} \tag{3.84}$$

improvement in average gate infidelity relative to the RL controller trained without stochastic noise, and around

$$\text{two orders of magnitude} \tag{3.85}$$

improvement in average gate infidelity relative to the SGD baseline [**?**]. This is one of the most important conclusions of the paper, because it isolates the source of the improvement: the gain is not due only to the use of RL in general, but specifically to the combination of policy-gradient learning with a noisy training environment.

A third important finding concerns stability. The paper shows that the sampled fidelity variance of the noise-optimized RL solution is consistently about one order of magnitude smaller than that of the two comparison methods throughout the tested noise range [**?**]. Since fidelity variance measures the sensitivity of the gate performance to random perturbations, this means that the RL-trained control is not only more accurate on average, but also more reproducible from one noisy realization to another. This is a particularly meaningful notion of robustness from the viewpoint of experimental implementation: a control protocol whose average performance is good but whose shot-to-shot fluctuation is large is much less useful than a protocol with similar mean fidelity but significantly smaller variance.

The fourth major result concerns runtime. For the subfamily

$$N(\alpha, \alpha, \pi/2), \qquad \alpha \in [0, \pi], \tag{3.86}$$

the paper reports a factor-of-ten improvement in gate runtime relative to the optimal gate-synthesis baseline [**?**]. This is a striking claim, because it directly supports the paper's broader thesis that hardware-native analog control can outperform even an optimal discrete circuit decomposition when the target gate is well aligned with the natural interaction structure of the device. In the paper's interpretation, this speedup is not accidental: the gate family in Eq. (3.86) has a two-qubit entangling structure that is especially compatible with the native gmon Hamiltonian, and the RL optimizer is able to exploit this compatibility automatically.

The paper makes this physical point explicit by observing that the target unitary $N(\alpha, \alpha, \pi/2)$ can be rewritten so that its entangling part is directly realizable under the gmon Hamiltonian with

$$\delta_j(t) = 0, \qquad f_j(t) = 0, \qquad j \in \{1, 2\}, \tag{3.87}$$

that is, essentially through time evolution under the native exchange interaction alone [**?**]. This interpretation is extremely important for the thesis, because it shows that the learning agent is not merely producing numerically good pulses by black-box trial and error. Rather, it appears to detect and exploit nontrivial regularities linking the structure of the target unitary to the native interaction channels of the hardware. In this sense, one of the paper's strongest messages is that RL can serve not only as an optimizer, but also as a mechanism for uncovering hardware-efficient control strategies.

At the same time, the paper does not present its results as the final word on the problem. It notes, for example, that isolated peaks in the gate-time curves may be due to control singularities and that the hardness of the analog-control landscape in the presence of leakage and stochastic errors still requires further study [**?**]. This is useful for the thesis because it identifies a natural opening for extension: the paper establishes a compelling proof of concept, but it does not exhaust the broader research question of how robust RL-based control generalizes across noise models, target families, and hardware assumptions.

Taken together, the results of the paper may be summarized as follows:

$$\text{noise-optimized RL} \quad \Longrightarrow \quad \begin{cases} \text{higher average fidelity under noise,} \\ \text{lower fidelity variance,} \\ \text{shorter runtimes for hardware-aligned targets,} \end{cases} \tag{3.88}$$

relative to the baseline methods studied in the work. This compact summary captures the three pillars of the paper's contribution: robustness, stability, and hardware efficiency.

Chapter 4

Empirical assessment of the control framework

4.1 Experimental protocol and evaluation setup

All numerical experiments in this thesis are performed within the same effective control framework introduced in the previous section. The physical system is modeled as a pair of weakly anharmonic superconducting modes truncated to three local levels each, so that the full Hilbert space has dimension $d_{\text{phys}} = 3^2 = 9$, while the computational two-qubit subspace retains dimension $d_{\text{comp}} = 4$. The dynamics are generated by the effective gmon Hamiltonian introduced in Section 3.1.2, and are simulated in discrete time with timestep Δt over a finite control horizon T. At each time step, the control action specifies the seven-dimensional control vector

$$\boldsymbol{u}(t) = \big(g(t),\, \delta_1(t), \delta_2(t),\, f_1(t), f_2(t),\, \phi_1(t), \phi_2(t)\big), \tag{4.1}$$

consisting of a tunable coupling pulse, two local detuning controls, two microwave amplitudes, and two microwave phases. These controls are passed through a finite-bandwidth filter before being applied to the Hamiltonian, so that the simulated pulses remain consistent with the experimentally motivated smooth-control assumptions built into the framework. For every experiment, the target operation is taken from the gate family

$$N(\alpha, \alpha, \gamma) = \exp[i\,(\alpha\,\sigma_1^x\sigma_2^x + \alpha\,\sigma_1^y\sigma_2^y + \gamma\,\sigma_1^z\sigma_2^z)]\,, \tag{4.2}$$

and control quality is assessed through the same UFO cost structure introduced previously, namely a weighted combination of gate infidelity, leakage penalty, boundary penalty, and runtime. This uniform physical and numerical setup is essential for the interpretation of the experiments, because it ensures that all methods are compared within the same multilevel model, under the same control constraints, and against the same class of target gates.

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4.1.1 Compared methods

Within this common framework, four optimization and comparison schemes are considered.

- The first is a direct gradient-based baseline, implemented through the Adam optimizer, which searches directly over the filtered control trajectory for a fixed gate horizon.
- The second is a policy-gradient reinforcement-learning controller trained in the nominal environment, that is, without stochastic perturbations applied during training.
- The third is the noise-optimized TRPO controller, which is trained in a stochastic environment where Gaussian perturbations are injected into the control amplitudes and into the anharmonicity-related parameter entering the Hamiltonian.

- The fourth comparison is not a pulse-level optimizer but a circuit-level reference, namely the optimal gate-synthesis baseline used to benchmark runtime on the family $N(\alpha, \alpha, \gamma)$.

These four methods play different methodological roles. The Adam baseline tests whether a standard first-order optimization of the raw control variables is sufficient to obtain high-quality pulses. The nominal TRPO controller tests whether reinforcement learning is already competitive in the idealized deterministic setting. The noise-optimized TRPO controller tests whether robustness can be learned directly from stochastic training. Finally, the gate-synthesis baseline tests whether direct analog control offers a runtime advantage over a discrete compiled implementation of the same target gate. Taken together, these methods define a comparison protocol that is not centered on a single optimizer, but on a set of scientifically distinct questions concerning accuracy, robustness, and hardware efficiency.

4.1.2 Metrics

The performance of each control method is evaluated through a set of complementary metrics designed to probe different aspects of the control problem. At the most basic level, the implemented gate is assessed through its fidelity with respect to the target unitary. For a final physical propagator $U(T)$, projected onto the computational subspace, the gate fidelity is computed as

$$F_{\text{gate}} = \frac{1}{16} \left| \mathrm{Tr}\Big(U_{\text{proj}}^{\dagger}(T)\, U_{\text{target}} \Big) \right|^2 , \tag{4.3}$$

where $U_{\text{target}} = N(\alpha, \alpha, \gamma)$. This quantity is reported together with the total UFO cost, which combines fidelity, leakage, boundary penalties, and runtime into a single scalar objective. Since the framework is explicitly multilevel, nominal performance is not characterized by fidelity alone. For every optimized control plan, the simulation also returns the TSWT-based leakage estimate L_{tot}, its decomposition into boundary and integral contributions, the boundary penalty term, and the final runtime in nanoseconds. These quantities allow one to distinguish between a control that is merely accurate on the logical subspace and one that is also physically well behaved in the full truncated Hilbert space. In addition to nominal performance, robustness is evaluated statistically under sampled noisy control realizations. Given a control plan $\tilde{\boldsymbol{c}}$ and a Gaussian noise level σ_{noise}, the key robustness diagnostics are the average fidelity

$$\overline{F} = \mathbb{E}_{\delta\tilde{\boldsymbol{c}}}\Big[F(U(\tilde{\boldsymbol{c}} + \delta\tilde{\boldsymbol{c}})) \Big], \tag{4.4}$$

the corresponding average gate fidelity, and the fidelity variance

$$\mathrm{Var}(F) = \mathbb{E}_{\delta\tilde{\boldsymbol{c}}} \left[\left(F - \overline{F} \right)^2 \right]. \tag{4.5}$$

These metrics are essential because they distinguish mean robustness from shot-to-shot stability: a control scheme with high average fidelity but large variance remains experimentally fragile. Finally, for the runtime study, the relevant metric is the total control duration required to realize a target gate, compared against the optimal discrete gate-synthesis baseline. The resulting evaluation protocol is therefore explicitly multidimensional: it measures logical accuracy, leakage suppression, robustness under stochastic perturbations, and hardware efficiency rather than relying on a single scalar score.

4.1.3 Evaluation protocol

All experiments follow a common procedural logic. For a given target gate $N(\alpha, \alpha, \gamma)$, a control method is first trained or optimized under its own learning rule until either a prescribed iteration budget is exhausted or a termination criterion on the UFO cost is satisfied. The resulting best control plan is then stored and re-evaluated in a deterministic nominal setting in order to

extract its final cost, fidelity, leakage, and runtime under identical physical assumptions. For robustness studies, the same saved control plan is subsequently subjected to repeated noisy rollouts, in which Gaussian perturbations are independently sampled and injected into the control channels and into the anharmonicity-related parameter of the Hamiltonian. This Monte Carlo evaluation yields the empirical estimates entering Eqs. (4.4) and (4.5). In the runtime study, by contrast, the focus is not on statistical robustness but on how the optimized gate duration varies across the target family, especially along the subfamily $N(\alpha, \alpha, \pi/2)$, where analog control can be meaningfully compared with optimal gate synthesis. The same timestep, control bounds, bandwidth assumptions, and physical truncation are retained throughout, so that differences between methods can be attributed to the optimization strategy rather than to inconsistencies in the simulation model. This common evaluation protocol is particularly important for the interpretation of the results: it ensures that the subsequent comparisons are methodologically fair, while also making the distinction between nominal accuracy, noisy robustness, and runtime efficiency completely explicit.

4.2 Parameter-choice study

The empirical performance of the present control framework depends not only on the underlying physical model and on the definition of the UFO objective, but also on a number of methodological choices that must be fixed before a meaningful comparison can be carried out. This is especially true in the reinforcement-learning setting considered in this thesis, where optimization quality is affected simultaneously by the training configuration, by the expressive capacity of the policy and value networks, and by the scalarization adopted in the control cost. For this reason, parameter selection should not be treated as a minor implementation detail. On the contrary, it constitutes an explicit part of the experimental methodology, since different choices may alter convergence stability, final gate quality, robustness under noise, and even the apparent relative performance of the compared optimizers.

A further reason for discussing parameter choice carefully is that the reference framework of Niu *et al.* fixes several ingredients rather explicitly. In particular, the original study adopts a continuous-action TRPO agent with two fully connected neural networks of hidden sizes 64, 32, and 32, updates the policy after batches of 20000 episodes, injects Gaussian amplitude fluctuations with a standard deviation of 1 MHz during stochastic training, and uses the same UFO weight configuration $\chi = \beta = 10$, $\mu = 0.2$, $\kappa = 0.1$ across the target gates considered. The same paper also evaluates robustness on the representative gate $N(2.2, 2.2, \pi/2)$, which therefore provides a natural reference target for model-selection purposes in the present reproduction. These design choices are important because they define the baseline notion of a *fair comparison* with the original framework: if the thesis changes some of them, this should be done deliberately and justified explicitly, rather than implicitly through undocumented implementation drift.

At the same time, the goal of this section is not to claim a globally exhaustive hyperparameter optimization over the full design space. Such a program would be computationally prohibitive and would add little scientific value to the main questions of the thesis. The objective is instead more focused: to identify a parameter regime that is stable, reproducible, physically sensible, and sufficiently well aligned with the reference paper to support a credible reproduction study. In other words, the purpose of the parameter-choice stage is to reduce arbitrariness, not to optimize every numerical setting in isolation. This distinction is methodologically important. A thesis-level validation study should not depend on a hidden collection of ad hoc choices tuned separately for each experiment; it should rely on a small and transparent set of decisions that are fixed once and then used consistently throughout the empirical chapter.

For clarity, the parameter-selection analysis is therefore decomposed into three separate studies. The first concerns the *training hyperparameters*, such as the timestep, maximum horizon, trust-region radius, exploration scale, batch size, and termination criterion. These

parameters primarily affect optimization stability, sample efficiency, and computational cost. The second concerns the *neural-network architecture* used for the policy and value functions. This is treated separately because the expressive capacity of the actor–critic representation is not simply another training detail: it may materially influence the quality of the learned control policy and therefore must be distinguished from purely algorithmic settings. The third concerns the *UFO cost-function weights*, which play an even more special role. Unlike ordinary optimizer hyperparameters, these coefficients are part of the physical definition of the control objective itself, since they determine how gate infidelity, leakage, boundary regularity, and runtime are balanced against one another. For this reason, architecture choice and cost-function choice should not be conflated with the tuning of TRPO-specific parameters.

This separation is also useful from the viewpoint of interpretation. If network size and UFO weights were tuned simultaneously together with training hyperparameters, any later performance difference would become harder to attribute: an observed improvement could arise from a better optimizer configuration, from increased representational capacity, or from a different physical scalarization of the control problem. By isolating the three studies, the empirical discussion becomes much cleaner. The reader can first understand which choices are made for numerical stability, then which are made for representational adequacy, and finally which affect the physical tradeoff structure encoded in the reward. This is particularly important in the present thesis, because one of its goals is not only to reproduce the original framework, but also to clarify which parts of its performance are tied to the learning algorithm and which depend more directly on the structure of the cost function.

The remainder of this section is organized accordingly. Section 4.2.1 discusses the training hyperparameters and the practical trade-offs that motivate their final selection. The following subsection is devoted to the sweep over policy and value network sizes, with the aim of determining whether the paper architecture is already sufficient for the present implementation or whether a different capacity yields a more stable control regime. After that, a dedicated sensitivity study of the UFO coefficients is introduced, in order to separate the role of the physical objective from that of the optimizer itself. Finally, the section closes by fixing one final parameter configuration for the remainder of the chapter. Unless explicitly stated otherwise, all later experiments will be performed under that single configuration, so that the subsequent comparisons are not biased by experiment-specific retuning.

In this way, the parameter-choice study serves a double purpose. On the one hand, it provides the methodological foundation needed to interpret the numerical results that follow. On the other hand, it establishes the conditions under which the present work can be compared fairly with the original study of Niu *et al.*, whose reported framework combines a stochastic TRPO environment, a fixed actor–critic architecture, a representative robustness benchmark on $N(2.2, 2.2, \pi/2)$, and a uniform UFO scalarization across the tested gate family. The parameter-selection stage is therefore not ancillary to the empirical chapter, but the step that makes its later claims scientifically interpretable.

4.2.1 Training hyperparameters and practical trade-offs

A first layer of parameter selection concerns the training hyperparameters of the reinforcement-learning pipeline. These choices do not define the physical control objective itself, but they strongly affect optimization stability, sample efficiency, and computational cost. For this reason, they should be discussed separately from both the neural-network architecture and the UFO cost-function weights, which will be treated in the following subsections. The purpose of the present subsection is therefore not to perform an exhaustive hyperparameter optimization, but to identify a training regime that is sufficiently stable, reproducible, and well aligned with the reference framework to support the experiments reported later in this chapter.

The training-parameter study is anchored to a representative gate from the target family

$N(\alpha, \alpha, \gamma)$, namely

$$U_{\text{rep}} = N(2.2, 2.2, \pi/2). \tag{4.6}$$

This choice is natural for two reasons. First, it provides a nontrivial two-qubit target that is sufficiently far from the identity to make the control problem genuinely informative. Second, it matches the representative robustness benchmark used in the original paper, and therefore offers a convenient reference point for tuning the implementation while preserving comparability with the published framework [**?**]. In this way, the parameter-selection stage is not performed on an arbitrary auxiliary task, but on a gate that is already central to the main empirical narrative of the thesis.

Within this restricted setting, the main training parameters of interest are the timestep Δt, the maximum admissible gate duration $T_{\max}$, the trust-region radius `max_kl`, the initial exploration scale `init_log_std`, the batch size measured in episodes per policy update, the number of training iterations, and the terminal-cost threshold used for early stopping. These quantities affect different aspects of the optimization. The timestep controls the temporal resolution of the pulse and therefore the expressivity of the sequential decision problem. The maximum duration defines the horizon over which the policy is allowed to construct the gate. The trust-region radius determines how aggressively the policy can move in one TRPO update, while the initial log-standard deviation controls the width of the Gaussian policy and therefore the amount of exploration at the beginning of training. Finally, the batch size and iteration budget determine the statistical reliability of the policy-gradient estimate and the total wall-time cost of the experiment.

The timestep and maximum horizon involve the clearest physical–computational trade-off. A smaller timestep gives the policy access to a finer control mesh and therefore to a richer set of pulse shapes, which is desirable in a multilevel system where the balance between fidelity, leakage suppression, and runtime may depend sensitively on pulse modulation. However, this also increases the episode length and makes the optimization problem higher-dimensional and more expensive. Similarly, a larger value of $T_{\max}$ enlarges the space of reachable solutions, but it also makes exploration more difficult and weakens the hardware-efficiency motivation of the analog-control framework. In the present implementation, the default choice for the main runs is

$$\Delta t = 2\,\text{ns}, \qquad T_{\max} = 180\,\text{ns}, \tag{4.7}$$

which represents a practical compromise between temporal resolution and computational tractability. This mesh is fine enough to support nontrivial pulse shaping, while keeping the horizon short enough for repeated multilevel rollouts to remain feasible.

A second important trade-off concerns policy stability versus exploration. In the present codebase, the stochastic policy is Gaussian in the raw action space, and therefore the parameter `init_log_std` directly controls the initial dispersion of sampled controls. If this value is too large, the early policy explores too broadly and frequently pushes actions into the saturation regime after the squashing map, which makes learning noisy and unstable. If it is too small, the policy explores only weakly and may fail to discover useful pulse trajectories within a reasonable number of iterations. A similar balance appears in the TRPO parameter `max_kl`: large trust regions allow faster progress in principle, but also increase the risk of unstable policy jumps, whereas very small trust regions make learning conservative and slow. In the configuration adopted for the main experiments, these considerations lead to the nominal choices

$$\texttt{init_log_std} = -1.0, \qquad \texttt{max_kl} = 0.01, \tag{4.8}$$

together with the standard TRPO settings

$$\gamma_{\text{RL}} = 0.99, \qquad \lambda = 0.97, \qquad \texttt{cg_iters} = 10, \qquad \texttt{damping} = 0.1. \tag{4.9}$$

These values are not presented as globally optimal, but as a stable regime that proved adequate for the representative target and remained close to standard TRPO practice.

The stopping threshold and batch size introduce a further methodological compromise. A loose termination criterion reduces training time, but risks accepting controls that are only superficially satisfactory. A stricter threshold is scientifically preferable, since it forces the policy to search for genuinely low-cost solutions, but it is also harder to satisfy consistently within a fixed training budget. In the present study, the default early-stopping threshold is set to

$$C_{\text{term}} = 0.15, \tag{4.10}$$

which is intentionally demanding and therefore avoids premature convergence toward mediocre solutions. On the data-collection side, the reference paper uses updates based on 20000 episodes per batch [**?**]. In the present implementation, this value is retained for the main paper-matched runs, while lighter exploratory sweeps may use 10000 episodes per batch when computational cost must be reduced. In the same spirit, full comparison runs are allowed to continue for up to 100 policy iterations, whereas shorter exploratory searches may use much smaller iteration budgets.

The training-noise setting is fixed in direct continuity with the reference framework. During stochastic training, Gaussian control perturbations with standard deviation

$$\sigma_{\text{train}} = 1\,\text{MHz} \tag{4.11}$$

are injected into the relevant control channels, consistently with the noisy-environment protocol described in the original paper [**?**]. This choice is important not only because it preserves comparability with the reference results, but also because it defines the robust-training regime used later in the chapter. The broader robustness evaluation still spans a larger interval of noise strengths, but the value in Eq. (4.11) is the default operating point at which the RL controller is exposed to stochastic perturbations during learning.

Taken together, these considerations define the default training setup used in the remainder of the empirical chapter:

$$\Delta t = 2\,\text{ns}, \quad T_{\max} = 180\,\text{ns}, \quad \texttt{init_log_std} = -1.0, \quad \texttt{max_kl} = 0.01, \quad C_{\text{term}} = 0.15, \tag{4.12}$$

with $\gamma_{\text{RL}} = 0.99$, $\lambda = 0.97$, 10 conjugate-gradient iterations, damping 0.1, and a stochastic training noise of 1 MHz. Larger batches of 20000 episodes are used when staying as close as possible to the paper is important, while 10000-episode batches are reserved for lighter preliminary sweeps. This configuration is not claimed to be uniquely optimal, but it provides a coherent and transparent operating regime under which later comparisons can be interpreted without ambiguity.

It is worth stressing once more that the present subsection concerns only *training* hyperparameters. The architecture of the policy and value networks is deliberately held fixed here, so that the effect of optimization settings is not confounded with the representational capacity of the actor–critic model. Likewise, the UFO weights are not tuned in this subsection, because they are part of the physical scalarization of the control problem rather than ordinary optimizer settings. These two aspects are therefore deferred to the next subsections, where they can be analyzed on their own terms.

4.2.2 Neural-network size sweep for policy and value functions

In the original UFO framework, both the policy and value functions are represented by fully connected neural networks with three hidden layers of sizes 64, 32, and 32 [**?**]. Since the present implementation also relies on a continuous-action actor–critic TRPO pipeline, the architecture of these networks is not a negligible detail. The hidden-layer layout directly affects the representational capacity of the policy, the stability of the value-function approximation, the efficiency of exploration in the continuous control space, and ultimately the quality of the learned

gate-control strategy. For this reason, the choice of network size should be treated separately from the tuning of training hyperparameters and from the selection of the UFO cost weights.

The purpose of the present subsection is therefore to assess whether modest changes in network width and depth materially affect performance in the current implementation. Importantly, this study is not intended as an exhaustive architecture search over a large design space. Rather, it is a restricted sweep aimed at answering a narrower question: within a family of architectures of comparable scale, does increasing width or depth improve the learned balance between control quality, leakage suppression, and robustness? This question is especially relevant because the reference paper motivates the use of neural networks as compact representations of non-local regularities in successful control trajectories [**?**]. If that interpretation is correct, then architecture should matter, but not necessarily in a monotonic way: larger networks may provide more expressive power, yet also introduce harder optimization and greater seed sensitivity.

The sweep was performed on the same representative target gate used for the training-parameter study,

$$U_{\mathrm{rep}} = N(2.2, 2.2, \pi/2), \tag{4.13}$$

and under a fixed reduced-budget training protocol designed for model selection. More precisely, all runs used noise-optimized TRPO with Gaussian training noise of standard deviation 1 MHz, timestep $\Delta t = 2\,\mathrm{ns}$, maximum gate duration $T_{\max} = 180\,\mathrm{ns}$, trust-region radius `max_kl` $= 0.01$, initial exploration scale `init_log_std` $= -1.0$, termination threshold $C_{\mathrm{term}} = 0.4$, and the nominal UFO weights

$$(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1). \tag{4.14}$$

Each architecture was trained for 20 TRPO iterations with 5000 episodes per batch and evaluated over 3 random seeds. The tested hidden-layer layouts were

$$(64, 64, 64), \qquad (64, 64, 64, 32), \qquad (128, 64, 32), \qquad (128, 64, 32, 32). \tag{4.15}$$

It should be noted explicitly that the exact paper architecture $(64, 32, 32)$ was not part of this restricted sweep. Consequently, the present study should be interpreted as a comparison among nearby thesis-side alternatives rather than as a direct empirical rejection of the reference architecture.

To compare the candidate networks, several criteria were considered simultaneously. The primary selection metric was the best evaluation cost reached during training, since this is the quantity most directly aligned with the UFO objective. However, this scalar alone is not sufficient for a meaningful choice. A network could achieve a favorable total cost by trading leakage against fidelity or by settling into an unnecessarily long runtime. For this reason, the architecture comparison also tracked best evaluation fidelity, best evaluation leakage, effective evaluation runtime, and robustness under noisy deployment through the average fidelity at $\sigma = 1\,\mathrm{MHz}$. In this way, the architecture was not chosen solely by one scalar score, but by its overall position in the multi-objective trade-off relevant to the control problem.

The results of the restricted sweep show a clear but not entirely trivial pattern. Among the four tested candidates, the architecture

$$(64, 64, 64) \tag{4.16}$$

achieves the best overall mean evaluation cost,

$$\overline{C}_{\mathrm{eval}} = 0.3584, \tag{4.17}$$

and also the highest mean best-evaluation fidelity,

$$\overline{F}_{\mathrm{eval}} = 0.9941. \tag{4.18}$$

At the same time, it gives the highest mean robustness score at $\sigma = 1\,\mathrm{MHz}$,

$$\overline{F}_{\sigma=1\,\mathrm{MHz}} = 0.9637, \tag{4.19}$$

with a mean evaluation runtime of approximately

$$\overline{T}_{\mathrm{eval}} = 106.7\,\mathrm{ns}. \tag{4.20}$$

This is an important result because it indicates that, within the explored subset, moderate widening of the network improves the optimization outcome without requiring extra depth.

The deeper alternatives do not improve this picture. The architecture $(64, 64, 64, 32)$ performs slightly worse in all main aggregated metrics, with a higher mean evaluation cost and a lower mean robust fidelity. The architecture $(128, 64, 32)$ remains close in mean cost, but displays a somewhat less favorable robustness profile and larger seed-to-seed variability. The deepest tested model, $(128, 64, 32, 32)$, is particularly instructive: it achieves the *lowest mean leakage*,

$$\overline{L}_{\mathrm{eval}} = 3.56 \times 10^{-4}, \tag{4.21}$$

but at the price of the *worst* mean evaluation cost, the lowest mean fidelity, the longest mean runtime, and the weakest mean robustness among the tested candidates. This indicates that minimizing leakage in isolation is not an adequate architecture-selection principle. A more expressive or deeper network may learn a comparatively conservative control strategy with slightly cleaner multilevel confinement, yet still fail to produce the best overall balance among the terms entering the UFO objective.

This interpretation is reinforced by the qualitative structure of the trade-off plots. The leakage boxplot is useful because it shows that the lowest-leakage architecture is not the one with the best total control quality. The robustness boxplot shows that the 64-64-64 architecture is not only strong on average, but also competitive in terms of seed-level consistency. Finally, the Pareto-style scatter plot of leakage versus robustness gives perhaps the clearest visual summary: the architectures do not lie on a simple monotonic frontier where lower leakage automatically implies higher robustness. Instead, the sweep suggests that the best architecture is the one occupying the most favorable compromise region of the leakage–robustness plane, rather than the one that optimizes either axis independently.

A further methodological point should be emphasized. Because the sweep was carried out under a reduced training budget, namely 20 policy iterations and 5000 episodes per batch, the resulting ranking should be interpreted as a comparison of *practical trainability* under limited optimization time rather than as a proof of asymptotic superiority under full convergence. This distinction matters. A larger architecture may in principle catch up under a much longer training budget, but if it is consistently slower, noisier, or less reliable in the practical regime relevant to the rest of the thesis, then it is still a less attractive default choice. For the purposes of the present empirical chapter, the relevant question is therefore not which architecture might eventually win under unlimited compute, but which one provides the best and most stable control quality under the realistic budget used for the experiments.

On the basis of this restricted sweep, the architecture

$$(64, 64, 64) \tag{4.22}$$

is adopted as the default policy/value hidden-layer configuration for the subsequent thesis-side experiments. This choice is motivated by its best mean evaluation cost, best mean fidelity, and strongest average robustness among the tested candidates, together with a competitive runtime and acceptable leakage. At the same time, this conclusion should be phrased carefully. Since the exact paper architecture $(64, 32, 32)$ was not included in the tested subset, the present result does not establish that the reference layout is suboptimal in absolute terms. It only shows that,

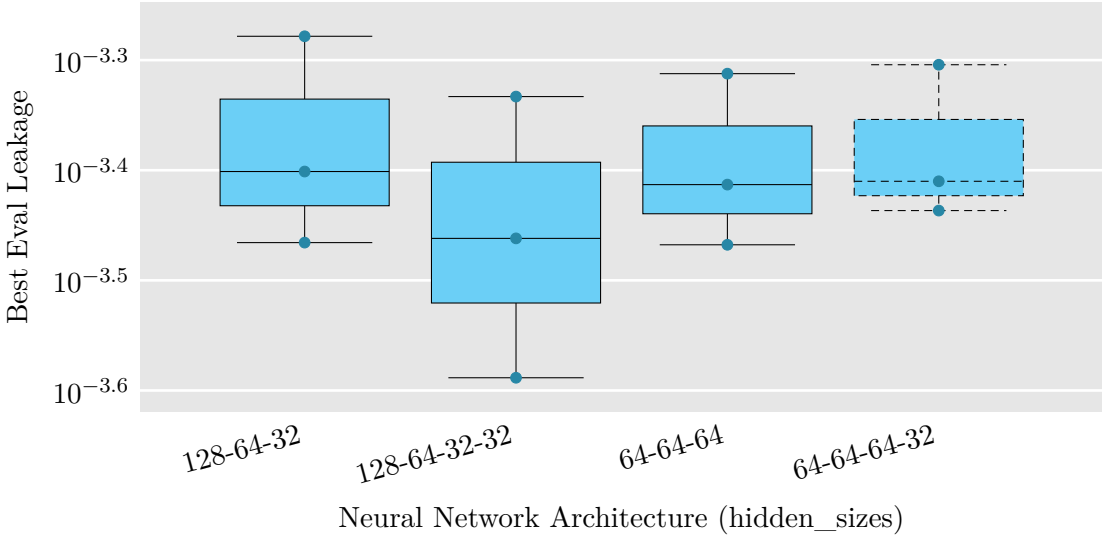


Figure 4.1: Recommended figure for the architecture sweep: use the boxplot already generated for *Best Eval Leakage across Architectures*. The horizontal axis should report the four tested hidden-layer layouts, while the vertical axis should show the best evaluation leakage on a scientific or logarithmic-friendly scale. Overlay the individual seed points, as in the current plot, so that both the central tendency and the seed-level spread remain visible. The visual purpose of the figure is to show clearly that the architecture with the smallest mean leakage is not the one with the best overall evaluation cost.

within the explored family of nearby alternatives, $(64, 64, 64)$ is the most effective architecture for the current implementation.

For this reason, the present subsection supports two distinct conclusions. First, architecture choice is not neutral: changes in width and depth do measurably affect the final control trade-off. Second, the dependence on architecture is not monotonic: increasing network size does not automatically improve control quality, and in fact the deepest tested model performs worst in the aggregated objective despite having the lowest mean leakage. This is precisely why the architecture study is useful. It prevents the later experiments from relying on an arbitrary actor–critic layout, while also showing that the dominant effects observed in the rest of the chapter are unlikely to be artifacts of an especially pathological network choice.

4.2.3 Sensitivity study of the UFO cost-function weights

The choice of the UFO coefficients deserves a separate study because, unlike ordinary optimizer hyperparameters, these quantities are part of the physical definition of the control problem itself. As mentioned in 3.2.3, the total objective is written as

$$C(\chi, \beta, \mu, \kappa) = \chi[1 - F(U(T))] + \beta L_{\text{tot}} + \mu P_{\text{bdry}} + \kappa T, \tag{4.23}$$

where χ controls the weight of gate infidelity, β penalizes accumulated leakage, μ enforces boundary regularity, and κ penalizes runtime [**?**]. The reference paper reports that these coefficients were selected by grid search and then fixed uniformly across the target gates considered, with the final choice

$$\chi = \beta = 10, \qquad \mu = 0.2, \qquad \kappa = 0.1. \tag{4.24}$$

For the purposes of the present thesis, this makes the weight study methodologically important in two complementary senses. On the one hand, it allows one to test whether the balance adopted in the paper is also reasonable in the present independent implementation. On the

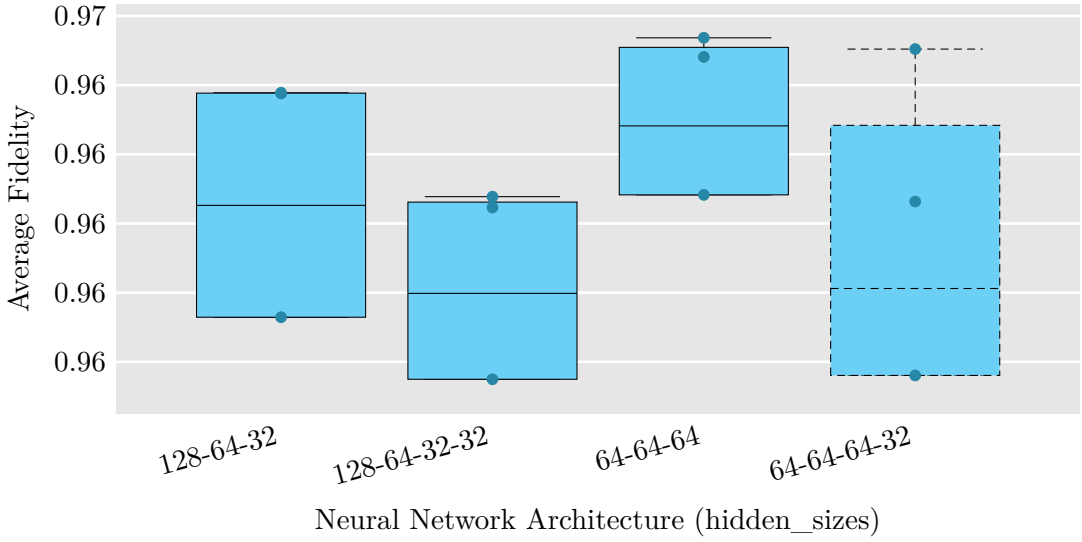


Figure 4.2: Recommended figure for the architecture sweep: use the boxplot already generated for *Robustness Fidelity across Architectures*. The horizontal axis should again report the four hidden-layer layouts, while the vertical axis should show the average fidelity under noisy evaluation at $\sigma = 1\,\mathrm{MHz}$. Overlay the individual seed values. This figure is meant to make the robustness ranking immediately visible and to show that the 64-64-64 architecture occupies the most favorable region in terms of average noisy fidelity.

other hand, it clarifies which aspects of the observed performance are due to the optimizer and which are instead induced directly by the physical scalarization of the control objective.

The sweep was performed on the same representative target gate used in the previous parameter-selection stages,

$$U_{\mathrm{rep}} = N(2.2, 2.2, \pi/2), \tag{4.25}$$

under a fixed reduced-budget noise-optimized TRPO configuration. More precisely, the training setup was

$$\Delta t = 2\,\mathrm{ns}, \qquad T_{\max} = 180\,\mathrm{ns}, \qquad \texttt{max_kl} = 0.005, \qquad \texttt{init_log_std} = -1.0, \tag{4.26}$$

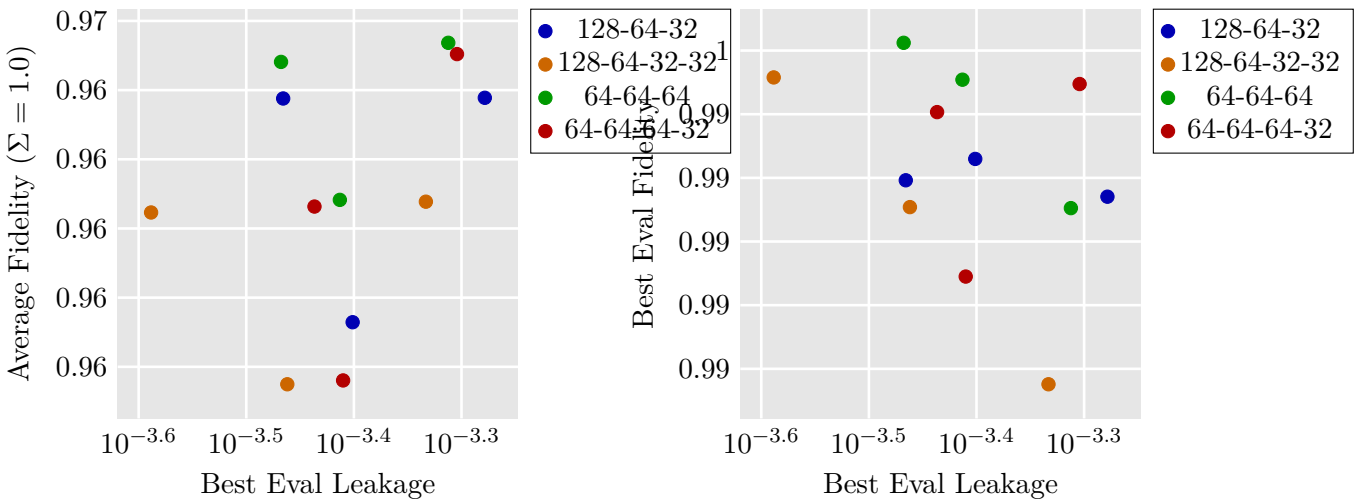
with 5 TRPO iterations, 10000 episodes per batch, 8 workers, termination threshold 0.15, Gaussian training noise of standard deviation 1 MHz, and the paper-matched network architecture 64-32-32. Two random seeds were used for each configuration. The explored coefficient ranges were

$$\chi \in \{5, 10, 20\}, \qquad \beta = 10, \qquad \mu \in \{0.1, 0.2, 0.4\}, \qquad \kappa \in \{0.05, 0.1, 0.2, 0.4\}, \tag{4.27}$$

for a total of $3 \times 1 \times 3 \times 4 = 36$ weight configurations and 72 runs once the two seeds are included.

The choice to keep β fixed deserves an explicit comment. Since the goal of the present experiment is to understand how the optimizer reacts to changing the balance between fidelity, boundary regularity, and runtime *relative* to leakage suppression, it is natural to treat β as a reference scale and vary the other coefficients around it. In this sense, the sweep should be interpreted primarily through the ratios χ/β, μ/β, and κ/β, rather than as a completely unconstrained four-dimensional search. This restriction also keeps the interpretation manageable: if all four coefficients were varied simultaneously, it would become much harder to isolate which part of the scalarization is actually responsible for a given change in behavior.

A second methodological point is even more important: the *best evaluation cost* is not an informative global selection criterion across this sweep. Indeed, different values of χ, μ, and κ define different scalarizations of the same physical control problem, so the numerical value of



(a) Robustness ($\sigma = 1$ MHz)

(b) Nominal Fidelity

Figure 4.3: Synthesis of the architecture sweep performance trade-offs. The scatter plots show (a) the average fidelity under noise versus evaluation leakage, and (b) the nominal evaluation fidelity versus evaluation leakage. Colors distinguish the tested hidden-layer layouts, with each point representing a separate seed. These figures highlight the multi-objective nature of the architecture selection: lower leakage does not always translate to higher nominal fidelity or better noise robustness.

$C(\chi, \beta, \mu, \kappa)$ is not directly comparable across configurations. This is not just a formal issue. In the present experiment, all runs terminate at the same evaluation time of 180 ns, so increasing κ contributes mainly a nearly rigid additive shift to the total cost without necessarily improving the learned control in any physically meaningful sense. For this reason, the search-script primary metric is retained only as an internal bookkeeping quantity, whereas the physically relevant comparison is based instead on nominal fidelity, best evaluation leakage, and robustness average fidelity at $\sigma = 1$ MHz.

Under this interpretation, the first clear result concerns the fidelity weight χ. The robustness boxplot in Fig. 4.4 shows that $\chi = 5$ leads to a visibly broader and lower-performing distribution, with several runs collapsing into a poor-robustness regime around 0.81–0.83. By contrast, $\chi = 10$ and $\chi = 20$ both yield substantially tighter and higher robustness distributions, concentrated around 0.92–0.95. This indicates that the low-fidelity-penalty regime underweights the logical-accuracy term too strongly relative to the other penalties. In other words, if χ is chosen too small compared with β, the optimizer is allowed to settle into controls that may be acceptable in terms of leakage or boundary behavior, yet remain clearly inferior in actual gate quality. Increasing χ to the paper scale already removes most of this failure mode, while a further increase to $\chi = 20$ produces only a marginal additional gain.

The behavior of the boundary weight μ is more subtle and, in some sense, more interesting. The sweep reveals a clear non-monotonic pattern. The intermediate value

$$\mu = 0.2 \tag{4.28}$$

produces the most favorable median trade-off, with high robustness and good nominal fidelity. Reducing the penalty to $\mu = 0.1$ remains viable, but leads on average to slightly larger leakage and a somewhat less favorable robustness profile. Increasing it to $\mu = 0.4$, however, makes the distribution strongly bimodal: for some values of χ it yields some of the best runs of the entire sweep, while for others it produces a catastrophic degradation in robustness. This is clearly visible in the robustness boxplot, where $\mu = 0.4$ generates the widest spread and the strongest

lower-tail failures. The physical interpretation is that boundary regularization is beneficial only up to a point. A moderate boundary penalty encourages smoother pulse endpoints and cleaner logical/Fock basis matching, but an overly strong one can overconstrain the optimizer and distort the balance with the fidelity term, thereby harming the overall control quality rather than improving it.

This interpretation is reinforced when leakage is examined separately. Figure 4.5 shows that larger μ tends, on average, to reduce the leakage proxy. In fact, the lowest-leakage region of the sweep is often associated with $\mu = 0.4$. However, this does *not* translate automatically into the best robust controller. The reason is precisely that leakage is only one component of the total physical objective. In several runs, the stronger boundary penalty helps suppress leakage but simultaneously worsens the broader fidelity–robustness trade-off. This is therefore another concrete example of the multi-objective nature of the UFO framework: minimizing one term of the cost more aggressively does not guarantee a better overall control solution.

The runtime coefficient κ has the weakest effect in the present restricted-budget sweep. This is visible both in the robustness sensitivity plot and in the leakage sensitivity plot, where the four tested values $\kappa \in \{0.05, 0.1, 0.2, 0.4\}$ produce distributions that are almost indistinguishable at the level of median performance. This weak dependence is not surprising once the evaluation protocol is taken into account. Since every run reaches the same best evaluation time of 180 ns, the runtime term does not act here as a strong discriminating mechanism on the learned stopping time, but mainly as a reweighting of the scalar objective. In other words, under this short and fixed-horizon model-selection regime, κ does not yet have enough room to reshape the control strategy as dramatically as χ and μ do. A stronger effect of κ is more likely to appear in the later runtime-family experiments, where gate duration is itself a central observable rather than an almost fixed boundary condition of the sweep.

The clearest global summary of the experiment is given by the Pareto-style leakage–robustness scatter in Fig. 4.6. Two clusters stand out. The first corresponds to a favorable region with robustness around 0.94 and leakage in the low-10^{-4} to mid-10^{-4} range. The second corresponds to a clearly worse region with robustness around 0.81–0.83, associated mainly with the underweighted-fidelity settings. The color coding by χ makes the main trend immediately visible: low χ is overrepresented in the poor-robustness cluster, whereas the paper-scale and larger-χ settings dominate the Pareto-favorable part of the plane. At the same time, the frontier is not determined by χ alone. The best region is occupied jointly by configurations near

$$(\chi, \mu) \approx (10, 0.2) \qquad \text{and} \qquad (\chi, \mu) \approx (20, 0.4), \tag{4.29}$$

with only weak dependence on κ within the explored range.

This last observation is crucial for the final choice. The numerically best average robustness is attained by the family of configurations centered around $(\chi, \beta, \mu) = (20, 10, 0.4)$, but the gain over the paper choice $(10, 10, 0.2)$ is extremely small. In the present sweep, the paper configuration remains essentially on the same Pareto-favorable frontier, with very similar robustness and leakage, while being directly aligned with the scalarization reported in the original study [**?**]. Since the improvement offered by the larger-χ, larger-μ corner is marginal and the present sweep is based on only two seeds and a very small training budget, there is no strong methodological reason to abandon the paper weights for the main experiments. On the contrary, keeping the reference choice

$$(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1) \tag{4.30}$$

is preferable for two reasons: it remains empirically near-optimal in the present implementation, and it preserves the cleanest possible fair-comparison protocol with the original UFO framework.

The main conclusion of this subsection is therefore twofold. First, the cost weights matter, and they matter in a physically interpretable way: too small a fidelity weight degrades robustness substantially, while an excessively strong boundary penalty can reduce leakage at the price of destabilizing the broader control trade-off. Second, the paper scalarization is not only historically

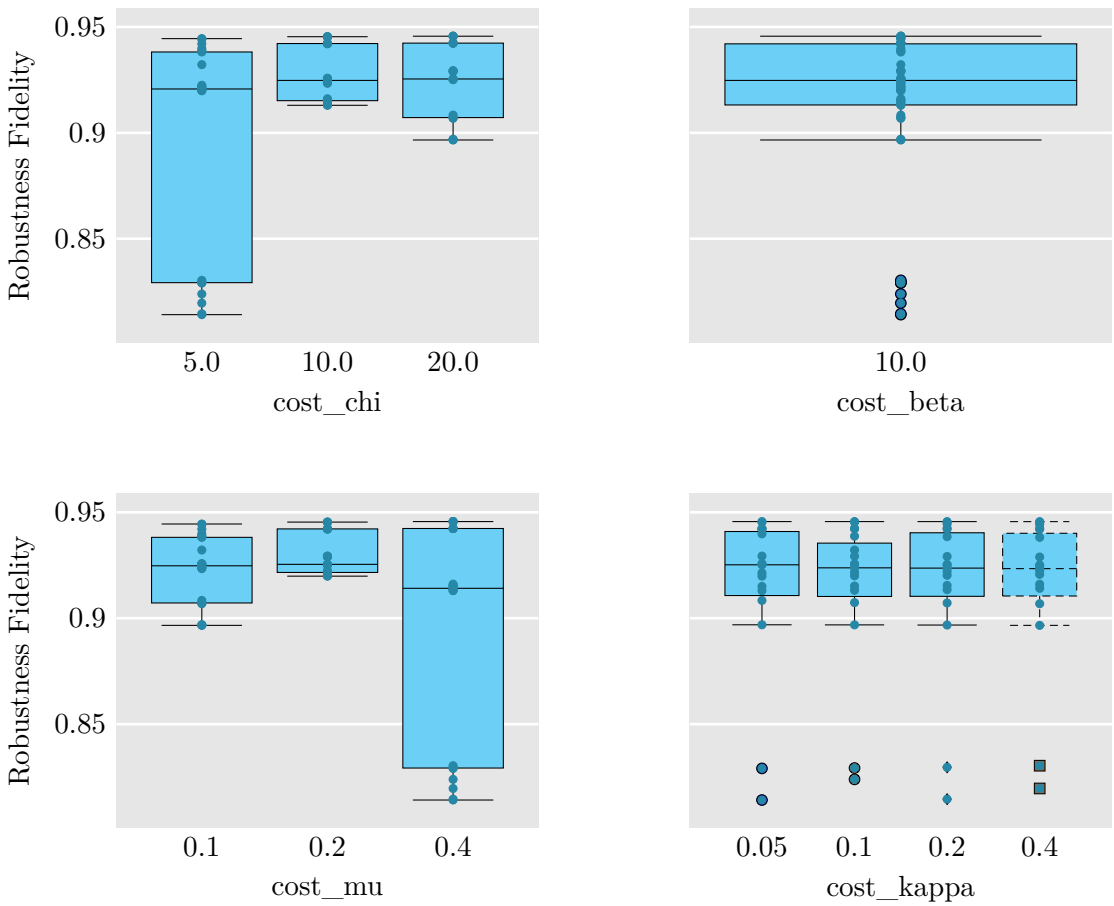


Figure 4.4: Recommended main sensitivity figure for the UFO-weight sweep: use the four-panel boxplot already generated for *Sensitivity of Robustness Fidelity to Cost Parameters*. The four panels should report the dependence of the noisy average fidelity at $\sigma = 1\,\mathrm{MHz}$ on χ, β, μ, and κ. Since β is fixed in the present experiment, the corresponding panel may either be kept as a visual reminder of the fixed reference scale or omitted in the final thesis version to avoid redundancy. The purpose of the figure is to show immediately that χ and μ produce the strongest variation in robustness, while κ has only a weak effect under the present fixed-horizon model-selection regime.

motivated, but also empirically reasonable in the present independent implementation. For this reason, the remainder of the empirical chapter will retain the reference UFO coefficients in Eq. (4.30), unless explicitly stated otherwise.

4.2.4 Final parameter configuration for the remainder of the chapter

In light of the parameter studies discussed above, a single reference configuration is now fixed for the remainder of the empirical chapter. The purpose of this final choice is not to claim global optimality over all possible numerical settings, but to define a stable and transparent operating regime under which the subsequent experiments can be compared consistently.

For the training hyperparameters, the default setup is

$$\Delta t = 2\,\mathrm{ns}, \qquad T_{\max} = 180\,\mathrm{ns}, \qquad \texttt{init_log_std} = -1.0, \qquad \texttt{max_kl} = 0.01, \qquad C_{\mathrm{term}} = 0.15, \tag{4.31}$$

with $\gamma_{\mathrm{RL}} = 0.99$, $\lambda = 0.97$, damping 0.1, and 10 conjugate-gradient iterations. Unless stated otherwise, the noise-optimized runs are trained with Gaussian control noise of standard deviation

$$\sigma_{\mathrm{train}} = 1\,\mathrm{MHz}, \tag{4.32}$$

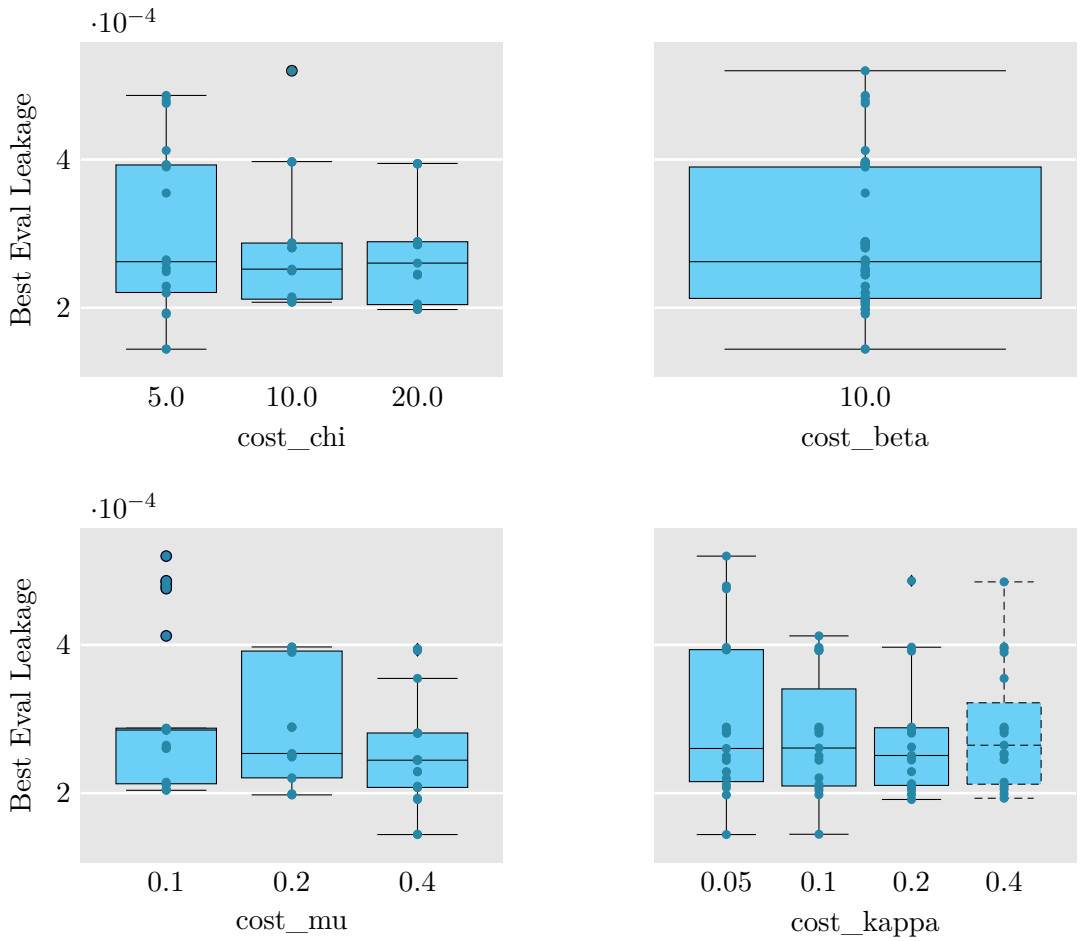


Figure 4.5: Recommended companion figure for the UFO-weight sweep: use the four-panel boxplot already generated for *Sensitivity of Best Eval Leakage to Cost Parameters*. The panels should report the dependence of best evaluation leakage on χ, β, μ, and κ, again with the possibility of suppressing the β panel in the final version since β is not varied. This figure is useful because it complements the robustness plot and shows that configurations with lower leakage are not always those with the best overall robustness. In particular, it helps visualize the non-monotonic role of μ, which may suppress leakage while still worsening the broader control trade-off when chosen too aggressively.

in direct continuity with the reference framework [**?**].

For the actor–critic representation, the architecture sweep suggests that network choice is not neutral, but also that improvements are not monotonic with increasing width or depth. Since the restricted sweep identified

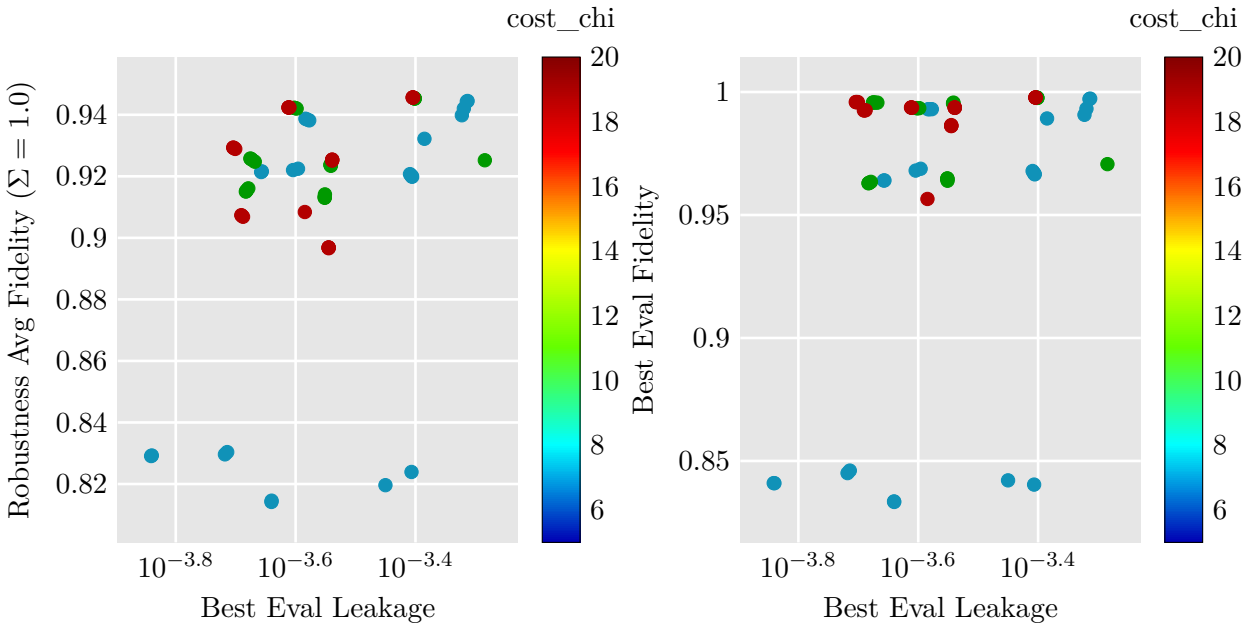
$$(64, 64, 64) \tag{4.33}$$

as the most favorable architecture among the tested alternatives, this layout is adopted as the default thesis-side configuration for the subsequent experiments.

Finally, for the UFO scalarization, the sensitivity study shows that the absolute value of the total cost is not directly comparable across different weight configurations, and that the most meaningful comparison must instead be based on fidelity, leakage, and robustness. Under this interpretation, the reference-paper choice

$$(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1) \tag{4.34}$$

remains very close to the best trade-off region observed in the present sweep. For this reason,



(a) Robustness ($\sigma = 1$ MHz)

(b) Nominal Fidelity

Figure 4.6: Synthesis of the UFO-weight sweep trade-offs. The scatter plots show (a) robustness versus leakage and (b) nominal fidelity versus leakage, with colors encoding the value of χ. This visual representation confirms that the paper choice $(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1)$ lies on or near the Pareto-optimal frontier for both objectives, justifying its selection as the reference configuration despite not holding the numerical record for every single metric.

and in order to preserve the cleanest possible comparison with the original UFO framework, these values are retained as the default cost-function weights in the remainder of the chapter [**?**].

Unless explicitly stated otherwise, all later experiments therefore use the combined reference configuration defined by Eqs. (4.31)–(4.34). This choice ensures that the empirical comparisons reported in the following sections are not affected by experiment-specific retuning, and that the results remain directly interpretable within the methodological logic established by the reference paper.

4.3 Single-target control optimization

4.3.1 Why this target is representative

As a representative nominal benchmark, this study first focuses on the target gate

$$U_{\text{target}} = N(2.2, 2.2, \pi/2), \tag{4.35}$$

which belongs to the two-qubit family $N(\alpha, \alpha, \gamma)$ introduced in the previous section. This choice is motivated by three considerations.

- First, it is a genuinely nontrivial target: it is neither close to the identity nor reducible to a particularly simple limiting case, and therefore provides a meaningful test of the optimizer's ability to coordinate entangling dynamics, local control, and leakage suppression within the full multilevel model.
- Second, it is directly aligned with the original benchmark logic of the reference paper, which uses the same gate family as the main testbed for robustness and runtime analyses [**?**]. For

this reason, selecting a target of the form $N(\alpha, \alpha, \pi/2)$ allows the present study to remain as close as possible to the original physical and algorithmic setting while still isolating one concrete case for detailed discussion.

- Third, fixing a single representative gate provides a clean comparison between optimization strategies before moving to broader parameter sweeps: by holding the target fixed, differences in final fidelity, leakage, runtime, and training behavior can be attributed more directly to the optimizer itself rather than to variations in target-gate difficulty across the family.

In this sense, Eq. (4.35) serves as the natural entry point for the empirical analysis, since it is sufficiently rich to be physically informative while remaining simple enough to support a transparent method-by-method comparison.

4.3.2 Adam baseline results

As a first non-RL reference, the representative target

$$U_{\text{target}} = N(2.2, 2.2, \pi/2) \tag{4.36}$$

was optimized by direct gradient-based pulse search using Adam, that is, through a baseline in the spirit of standard direct quantum optimal-control methods acting directly in control space rather than through policy learning [**?**, **?**]. The optimization was performed on the same multilevel gmon model and with the same UFO-inspired objective introduced in the previous sections, using a time discretization of $\Delta t = 2\,\text{ns}$, 400 Adam iterations for each candidate horizon, and a sweep over gate durations in the interval $T \in [40, 180]\,\text{ns}$ with 10 ns spacing. In order to keep the baseline aligned with the robustness-oriented setting of the paper, the optimization was carried out with the same nominal UFO weights $(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1)$ and with train-time Gaussian control noise of standard deviation 1 MHz, while the final metrics discussed in this subsection are recomputed in the nominal simulator.[1]

Under this nominal reevaluation, the best Adam solution is obtained at

$$T^{\star}_{\text{Adam}} = 70\,\text{ns}, \tag{4.37}$$

for which the final cost is

$$C^{\text{nom}}_{\text{Adam}} = 0.1576, \tag{4.38}$$

with nominal fidelity

$$F^{\text{nom}}_{\text{Adam}} = 0.99662, \tag{4.39}$$

total leakage

$$L^{\text{tot}}_{\text{Adam}} = 2.82 \times 10^{-4}, \tag{4.40}$$

and boundary contribution

$$P^{\text{Adam}}_{\text{bdry}} = 4.28 \times 10^{-3}. \tag{4.41}$$

These values already indicate that direct pulse optimization is capable of finding a high-quality control for the representative target, with gate fidelity comfortably above 99% and leakage kept in the 10^{-4} regime, which is precisely the scale expected from the leakage-aware control philosophy of the original UFO framework [**?**]. At the same time, the result is scientifically useful not only because it is "good" but because the horizon sweep reveals *why* it is good and where the main tradeoffs arise.

[1] This distinction is important in the present implementation. The internal optimizer summary selects $T = 60\,\text{ns}$ as the minimum of the noisy optimization objective, whereas the deterministic post-training reevaluation identifies $T = 70\,\text{ns}$ as the best nominal solution. Since the purpose of the present subsection is to establish the *nominal* single-target benchmark, the 70 ns solution is taken as the representative Adam baseline in the discussion below.

The dependence on the gate duration is particularly instructive. Very short horizons, such as 40 ns and 50 ns, perform poorly: the nominal fidelities drop to 0.7255 and 0.8818, while the corresponding boundary penalties remain very large, respectively 0.399 and 0.357. By contrast, the leakage itself stays in the same 10^{-4} range throughout this short-time regime. This means that the main failure mode of the ultra-short solutions is *not* an isolated explosion of leakage alone, but the inability to satisfy simultaneously the accuracy and boundary requirements of the UFO objective under a strongly time-compressed pulse. This behavior is fully consistent with the speed–leakage and speed–smoothness tension discussed in Section 2.2.3: in weakly anharmonic superconducting systems, shortening the pulse too aggressively broadens the effective control spectrum and makes it more difficult to maintain both clean logical evolution and well-behaved pulse boundaries [**?**, **?**]. The sharp transition from 60 ns to 70 ns is especially revealing in this respect. Over this interval, the fidelity increases from 0.9821 to 0.9966, the boundary term collapses from 9.75×10^{-2} to 4.28×10^{-3}, and the total nominal cost drops from 0.3806 to 0.1576, whereas the leakage decreases only moderately. This indicates that, near the optimum, the decisive improvement is not merely "less", but the fact that the pulse becomes sufficiently long to realize the target unitary while also respecting the boundary structure required by the UFO cost.

The long-horizon side of the sweep is equally informative. Beyond approximately 70–90 ns, the fidelity remains uniformly high and varies only weakly; for example, at 150 ns the Adam baseline reaches

$$F_{\text{Adam}}^{\text{nom}}(150\,\text{ns}) = 0.99708, \tag{4.42}$$

slightly above the 70 ns value, while the leakage decreases further to 6.82×10^{-5}. Nevertheless, the total cost rises again to 0.2851. This is an important point for the logic of the thesis. If one were to judge the control only by fidelity, the longer pulse would appear at least competitive, and perhaps even slightly preferable. However, once the full UFO objective is taken seriously, the improvement in fidelity and leakage is too small to compensate for the increased runtime. In other words, the Adam sweep gives a direct numerical illustration of the multi-objective nature of the framework introduced in Section 3.2.3: the best solution is not the one with the absolute highest fidelity, nor the one with the absolute smallest leakage, but the one that best balances accuracy, leakage suppression, boundary regularity, and total gate time. This is precisely the type of behavior one would expect in realistic superconducting-qubit control, where gate design is not a single-objective search but a compromise between physically competing desiderata [**?**, **?**].

A further useful benchmark emerges when this nominal Adam runtime is compared to the circuit-based reference used in the original paper. For the family $N(\alpha, \alpha, \gamma)$, Niu *et al.* estimate an optimal synthesized implementation time of approximately 215 ns, obtained from a depth-seven decomposition with three entangling gates and five single-qubit gates [**?**]. Relative to this reference, the representative Adam solution at 70 ns is already faster by a factor of about

$$\frac{215\,\text{ns}}{70\,\text{ns}} \approx 3.1, \tag{4.43}$$

even before introducing policy optimization or transfer learning. This does *not* yet prove the stronger runtime claims of the RL framework across the whole gate family, but it is a useful early indication that direct analog optimization on the present hardware model can already exploit structure unavailable to a fixed compiled-gate implementation.

Although the systematic robustness study is deferred to Section **??**, the Adam baseline also provides a first quantitative reference point in the noisy setting. For the control plan selected by the noisy training objective at 60 ns, the robustness evaluation at $\sigma_{\text{noise}} = 1\,\text{MHz}$ yields an average fidelity of 0.9686, an average gate fidelity of 0.9607, and a fidelity variance of 1.75×10^{-4}. These values are already respectable for a direct gradient baseline, but they also make clear why a stronger robustness-oriented method is desirable: they set the scale that the nominal and noise-optimized TRPO controllers will later be asked to surpass. In this sense, the Adam

Figure 4.7: Adam baseline on the representative target $N(2.2, 2.2, \pi/2)$ as a function of the candidate gate horizon. A clean and informative layout is a three-panel figure: **(a)** nominal total cost versus horizon, with the minimum at 70 ns highlighted and the optimizer-selected 60 ns point marked separately; **(b)** nominal fidelity and total leakage versus horizon, shown either with dual vertical axes or as two stacked subplots; **(c)** boundary penalty and time penalty versus horizon. The purpose of the figure is to show visually that the nominal optimum at 70 ns is produced by the simultaneous collapse of the boundary term and the sharp increase in fidelity, whereas the longer-horizon region exhibits fidelity saturation and only marginal further leakage reduction.

Figure 4.8: Preliminary family-wide Adam sweep for the gates $N(\alpha, \alpha, \pi/2)$, trained from scratch for each α. A visually effective design is a three-row figure: **top:** best selected horizon versus α as a step plot, showing the plateaus at 40, 50, 60, 70, and 80 ns; **middle:** best nominal fidelity and best nominal cost versus α; **bottom:** noisy average fidelity at $\sigma_{\text{noise}} = 1\,\text{MHz}$ and the associated fidelity variance versus α. This figure should make explicit that Adam performance is strongly non-uniform across the gate family, with an easier central region and significantly more difficult edge regions.

baseline performs exactly the role it should in the empirical chapter: it is not a strawman, but a meaningful non-RL benchmark that captures the main structure of the control problem while leaving clear room for improvement in robustness and consistency under stochastic perturbations.

A preliminary family-wide Adam sweep with fixed $\gamma = \pi/2$ further reinforces this interpretation. Even without transfer learning, the performance of the direct optimizer is markedly non-uniform across the gate family $N(\alpha, \alpha, \pi/2)$. The best results cluster around the central region $\alpha \approx 1.4$–1.7, where the optimizer consistently selects 40 ns as the best horizon, reaches nominal fidelities above 0.9987, and retains noisy average fidelities around 0.987–0.989 with very small fidelity variance of order 10^{-5}. By contrast, near the edges of the interval, i.e. for $\alpha \approx 0$ or $\alpha \approx \pi$, the best horizon shifts to 70 ns, the nominal fidelity drops to about 0.94, and the noisy average fidelity falls to about 0.936–0.938, with variance increasing by roughly one order of magnitude. This shows that the difficulty of the gate family is highly structured already at the level of direct gradient optimization. Not all members of $N(\alpha, \alpha, \pi/2)$ are equally easy to realize within the same bandwidth-limited multilevel control ansatz, and this fact will later be important when interpreting the runtime curves obtained with RL and transfer learning. Put differently, the Adam sweep provides a useful "zero" baseline for distinguishing between intrinsic gate-family geometry and genuine algorithmic gains due to policy optimization.

4.3.3 Nominal TRPO results

The second benchmark on the representative target

$$U_{\text{target}} = N(2.2, 2.2, \pi/2) \tag{4.44}$$

is obtained with the TRPO-based reinforcement-learning controller introduced in Chapter 3, but trained in the *nominal* environment, i.e. without stochastic perturbations injected during training. In the present implementation, this corresponds to setting `noise_optimized = false`, so that the policy is optimized against the deterministic multilevel gmon dynamics discussed in Section 3.3.1, even though the codebase retains the same general training interface used later for the noisy runs. The controller is trained with the same physical model and UFO weights used in the Adam baseline, namely $(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1)$, a time discretization $\Delta t = 2\,\text{ns}$, a maximum admissible gate time $T_{\max} = 180\,\text{ns}$, and a termination threshold $C_{\text{term}} = 0.3$. On the algorithmic side, the nominal TRPO run uses batches of 5000 episodes, 115 policy updates,

trust-region radius `max_kl` $= 0.01$, discount factor $\gamma_{\mathrm{RL}} = 0.99$, GAE parameter $\lambda = 0.97$, and fully connected policy/value networks with hidden sizes 64-32-32, consistent with the architecture described by Niu *et al.* [**?**].

Under deterministic evaluation, the best policy found in this nominal setting achieves

$$C_{\mathrm{TRPO}}^{\mathrm{nom}} = 0.2982, \qquad F_{\mathrm{TRPO}}^{\mathrm{nom}} = 0.99415, \qquad L_{\mathrm{TRPO}}^{\mathrm{nom}} = 1.60 \times 10^{-3}, \tag{4.45}$$

with an effective gate duration

$$T_{\mathrm{TRPO}}^{\mathrm{nom}} = 116\,\mathrm{ns}. \tag{4.46}$$

The same deterministic rollout also reports

$$C_{\min}^{\mathrm{TRPO}} = 0.2982 \qquad \text{at} \qquad t_{\min}^{\mathrm{TRPO}} = 116\,\mathrm{ns}, \tag{4.47}$$

which indicates that, unlike the Adam baseline where the horizon is fixed externally and selected by sweep, the nominal TRPO controller learns the control trajectory inside a larger admissible window and reaches its best stopping point well before the hard upper bound of 180 ns. This is an important structural difference between the two methods: Adam solves a family of fixed-horizon pulse-optimization problems, whereas TRPO explores a variable-length sequential control strategy under a cost-based stopping criterion. In that sense, the nominal TRPO result should not be interpreted simply as "another" point, but as the output of a policy that co-optimizes pulse shape and effective stopping time within the same RL formulation [**?**].

From the nominal-performance viewpoint, this result is already strong in absolute terms. A deterministic fidelity above 99.4% on the full multilevel model confirms that the policy-gradient formulation is able to learn a nontrivial high-quality analog control for the representative target. Moreover, the learned runtime of 116 ns remains substantially below the 215 ns synthesized-gate reference quoted in the original paper for the optimal circuit decomposition of the same target family [**?**]. In other words, even before introducing noise-aware training, the nominal TRPO controller already supports the more general thesis-level claim that direct analog control can exploit hardware-native structure to outperform standard compiled implementations in runtime. The gain is, however, more moderate than in the strongest analog-control cases emphasized by the paper: for the present target, the reduction is approximately

$$\frac{215\,\mathrm{ns}}{116\,\mathrm{ns}} \approx 1.85, \tag{4.48}$$

which is significant, but far from the order-of-magnitude speedups later reported for the noise-optimized runtime sweeps across the broader gate family [**?**].

At the same time, when this nominal TRPO solution is compared with the Adam baseline of the previous subsection, an instructive picture emerges. The best nominal Adam solution at 70 ns achieved a lower total cost, slightly higher fidelity, significantly smaller leakage, and a shorter runtime. More explicitly, TRPO is worse than Adam on this single deterministic target both in total cost (0.2982 vs. 0.1576) and in gate duration (116 ns vs. 70 ns), while its leakage is larger by roughly a factor of

$$\frac{1.60 \times 10^{-3}}{2.82 \times 10^{-4}} \approx 5.7. \tag{4.49}$$

The fidelity difference is much smaller in absolute terms, but still present:

$$F_{\mathrm{Adam}}^{\mathrm{nom}} - F_{\mathrm{TRPO}}^{\mathrm{nom}} \approx 2.47 \times 10^{-3}. \tag{4.50}$$

This comparison is scientifically important, because it prevents an over-interpretation of the RL method. On the representative clean target, nominal TRPO does *not* dominate a strong direct pulse optimizer. Instead, the deterministic single-target benchmark suggests that, when the model is known accurately and the objective is smooth and fully differentiable, direct gradient-based optimization remains highly competitive, and in the present case superior, in the purely

Figure 4.9: Suggested training-curve figure for the nominal TRPO run on $N(2.2, 2.2, \pi/2)$. A clean layout is a three-panel figure: **(a)** evaluation cost versus policy iteration, with the best checkpoint highlighted; **(b)** evaluation fidelity and evaluation leakage versus iteration, either on dual axes or in two aligned stacked panels; **(c)** evaluation gate time T and minimum-cost time $t_{\min}$ versus iteration. The key visual message should be that the controller rapidly reaches a high-fidelity regime, but the later part of training is mainly a refinement of the leakage/runtime compromise rather than a dramatic improvement in fidelity alone.

nominal regime. This is broadly consistent with the standard optimal-control literature, where direct gradient methods such as GRAPE are known to be particularly effective on fixed-target closed-system pulse-design problems [**?**, **?**].

This observation is not a weakness of the RL framework; rather, it clarifies what the RL framework should be expected to contribute. The original paper does not primarily claim that TRPO is universally better than direct optimization on every nominal benchmark. Its stronger claim is instead that a policy-based method trained in the appropriate stochastic environment can improve *robustness* and, in some hardware-aligned families, *runtime efficiency* relative to simpler baselines [**?**]. The present nominal-clean result is therefore valuable precisely because it isolates what TRPO does *without* the noise-aware component: it learns a high-fidelity analog gate and a nontrivial variable-time control strategy, but it does not yet exhibit an intrinsic nominal advantage over a well-tuned direct optimizer on this representative deterministic target. This makes the next subsection conceptually much stronger, because any improvement observed there can be attributed more specifically to stochastic training rather than to the mere use of reinforcement learning.

A second useful insight concerns the qualitative role of the UFO scalarization. The nominal TRPO controller reaches a reasonable deterministic fidelity, but it does so at a larger runtime and with substantially more leakage than the best Adam solution. This suggests that, in the clean setting, the policy has found a more conservative but less sharply optimized compromise in the multi-objective landscape. Put differently, the RL controller appears able to discover acceptable controls within the admissible region of the UFO cost, but in the absence of explicit noise pressure it has less incentive to carve out the particularly efficient low-leakage solution that a direct horizon-conditioned optimizer can reach on the same target. This interpretation is fully consistent with the control-theoretic discussion in Sections 2.3.2, 2.4.1, and 2.4.3: policy optimization is especially attractive when sequential structure, uncertainty, and robustness matter, but it need not be the most aggressive optimizer of a single deterministic pulse-shaping problem.

Finally, the nominal TRPO experiment also motivates a careful reading of training curves. Because the controller is evaluated after every policy update and the code stores both the current evaluation cost and the time $t_{\min}$ at which the minimum rollout cost is reached, the most informative training visualization is not simply fidelity versus iteration. A better representation is one that jointly tracks the evolution of evaluation cost, fidelity, leakage, and learned stopping time. In the present case, such a plot would show that the policy quickly enters a high-fidelity regime but continues to trade off leakage and runtime within the admissible 180 ns window, eventually settling on a best deterministic solution around the 116 ns scale. This makes the nominal TRPO subsection a natural bridge between the fixed-horizon analysis of the Adam baseline and the explicitly robustness-oriented analysis of the next subsection.

4.3.4 Comparative discussion

Taken together, the Adam and nominal-TRPO results clarify an important point about the representative single-target problem: the reinforcement-learning formulation is already *competi-*

tive in the nominal setting, but it is not yet *superior* to a strong direct pulse optimizer on this deterministic benchmark. This distinction is essential for the interpretation of the remainder of the chapter. The nominal TRPO controller reaches a high-quality solution for

$$U_{\text{target}} = N(2.2, 2.2, \pi/2), \tag{4.51}$$

with fidelity above 99.4% and runtime well below the synthesized-gate reference of about 215 ns discussed in the original paper [**?**]. However, when the same target is solved by direct gradient-based optimization with a horizon sweep, the Adam baseline achieves a clearly better nominal compromise among the quantities entering the UFO objective.

Using the best deterministic solutions identified in the previous two subsections, the comparison can be summarized as follows. **TODO:** Tabella riassuntiva con i risultati di Adam e TRPO al posto di questo

$$T^{\star}_{\text{Adam}} = 70\,\text{ns}, \qquad C^{\text{nom}}_{\text{Adam}} = 0.1576, \qquad F^{\text{nom}}_{\text{Adam}} = 0.99662, \qquad L^{\text{tot}}_{\text{Adam}} = 2.82 \times 10^{-4}, \tag{4.52}$$

whereas the nominal TRPO controller yields

$$T^{\text{nom}}_{\text{TRPO}} = 116\,\text{ns}, \qquad C^{\text{nom}}_{\text{TRPO}} = 0.2982, \qquad F^{\text{nom}}_{\text{TRPO}} = 0.99415, \qquad L^{\text{tot}}_{\text{TRPO}} = 1.60 \times 10^{-3}. \tag{4.53}$$

Therefore, on this benchmark, Adam improves over nominal TRPO simultaneously in total cost, fidelity, leakage, and runtime. Quantitatively, the runtime ratio is

$$\frac{T^{\text{nom}}_{\text{TRPO}}}{T^{\star}_{\text{Adam}}} \approx 1.66, \tag{4.54}$$

the leakage ratio is

$$\frac{L^{\text{tot}}_{\text{TRPO}}}{L^{\text{tot}}_{\text{Adam}}} \approx 5.7, \tag{4.55}$$

and the fidelity gap is

$$F^{\text{nom}}_{\text{Adam}} - F^{\text{nom}}_{\text{TRPO}} \approx 2.47 \times 10^{-3}. \tag{4.56}$$

Although this fidelity difference is numerically modest, the cost and leakage differences are not. In particular, the much larger leakage of nominal TRPO means that the RL controller, in the clean setting, finds a control that is acceptable and clearly nontrivial, but not as sharply optimized in the multilevel sense as the best pulse found by direct gradient descent.

This outcome is physically and methodologically plausible. As discussed in Sections 2.3.2 and 2.4.3, direct gradient-based quantum optimal-control methods are particularly strong when the problem is fully model-based, deterministic, and differentiable, and when the objective concerns a fixed target gate rather than a distribution of related tasks [**?**, **?**]. The present single-target benchmark lies precisely in that regime. Adam is allowed to optimize directly over the pulse parameters for each fixed horizon, and the horizon sweep then selects the best compromise afterward. TRPO, by contrast, optimizes a stochastic policy over sequential actions under a termination mechanism and must learn both the control law and the effective stopping structure within a single actor–critic pipeline. That extra flexibility is potentially valuable, but it also makes the nominal optimization problem less direct. For this reason, it is not surprising that, on one clean target, the direct optimizer can still produce the sharper deterministic solution.

At the same time, the comparison should not be misread as evidence against the RL formulation. First, nominal TRPO still performs well in absolute terms: a fidelity of 0.99415 at 116 ns remains a strong analog-control result, and it already supports the broader thesis claim that direct pulse-level optimization can beat the standard synthesized-gate runtime on realistic multilevel hardware [**?**]. Second, and more importantly, the nominal single-target benchmark is not where the main conceptual advantage of the RL framework is expected to appear. The discussion of robustness in Sections 2.3.3, 2.4.1, and 3.3.4 already suggested that policy-based

learning becomes especially attractive when the control problem is sequential, stochastic, and evaluated over ensembles of noisy trajectories rather than over one deterministic rollout. The present result is fully consistent with that logic: nominal TRPO does not dominate Adam on the clean benchmark, which means that any later advantage of the RL approach under noise cannot be dismissed as a trivial consequence of "RL" better at everything. Instead, it will have to be interpreted as a specific gain produced by the stochastic training environment and by the ability of the policy to internalize robustness-oriented regularities across trajectories, exactly as claimed in the original UFO paper [**?**].

A second important lesson concerns the role of the UFO scalarization itself. The single-target comparison makes clear that excellent nominal fidelity alone is not sufficient for a control to be considered preferable. The Adam and nominal-TRPO solutions both lie in a high-fidelity regime, yet they differ substantially in leakage and runtime. This confirms, in a concrete numerical setting, the conceptual point developed earlier in Sections 2.1.2 and 3.2.3: the relevant notion of success is not the maximization of one metric in isolation, but the identification of a balanced solution in a multi-objective landscape. In the representative target studied here, the direct optimizer reaches a better balance than nominal TRPO. The RL controller seems to settle into a broader, more conservative basin of acceptable solutions, whereas Adam succeeds in driving the pulse closer to the sharper low-runtime, low-leakage region of the landscape. This difference in optimization behavior will become important later, because noisy training may change which basin is actually preferable once robustness is included explicitly in the evaluation.

From the viewpoint of the overall chapter structure, this comparison also provides a clean logical transition to the next subsection. The question raised by the nominal benchmark is now sharply formulated: if direct gradient optimization already performs better on the deterministic single-target problem, why should one prefer the RL framework at all? The answer, if the main thesis of the reference paper is correct, must come from the behavior under stochastic perturbations. In other words, the present subsection establishes the necessary baseline against which the robustness claims of the framework can be judged. Adam is not a weak reference, but a strong nominal competitor. Therefore, if the noise-optimized RL controller later surpasses it in average fidelity and fidelity variance under control noise, that improvement will constitute meaningful evidence that robustness is not merely a side effect of analog optimization, but a property that is genuinely learned from the stochastic training environment [**?**].

4.4 Robustness under stochastic control noise

4.4.1 Noise model

To assess robustness in a physically meaningful and methodologically consistent way, all controllers considered in this section are evaluated under the same stochastic control-error model, following the general logic of the original UFO framework [**?**]. At the discrete time-step level, the nominal control trajectory is not implemented exactly, but is perturbed by additive Gaussian amplitude fluctuations. In the notation of the control vector introduced in Eq. (4.1), this means that the realized control at time step t_k takes the schematic form

$$\boldsymbol{u}^{(\mathrm{noisy})}(t_k) = \boldsymbol{u}(t_k) + \boldsymbol{\xi}(t_k), \tag{4.57}$$

where the noise term $\boldsymbol{\xi}(t_k)$ is sampled independently at each time step from a zero-mean Gaussian distribution. In line with both the original paper and the present implementation, these fluctuations are applied not only to the directly actuated control amplitudes, namely the coupling pulse $g(t)$, the detuning amplitudes $\delta_j(t)$, and the microwave amplitudes $f_j(t)$, but also to the anharmonicity-related parameter η, which enters the Hamiltonian as part of the multilevel device model [**?**]. This point is important: the noise model is not restricted to imperfect pulse delivery alone, but also probes sensitivity to uncertainty in a physically relevant Hamiltonian parameter.

Placeholder

Figure 4.10: Suggested comparison figure for the representative target $N(2.2, 2.2, \pi/2)$. A visually effective layout is a four-panel summary: **(a)** grouped bars for total nominal cost; **(b)** grouped bars for nominal fidelity; **(c)** grouped bars for total leakage on a logarithmic vertical scale; **(d)** grouped bars for runtime in ns. Only two methods should be shown here: Adam baseline and nominal TRPO. The key visual message should be that nominal TRPO is fully competitive in absolute terms, but Adam dominates on this deterministic single-target benchmark across all four reported metrics.

At the Hamiltonian level, each noisy rollout is therefore generated by replacing the nominal control-dependent evolution with a perturbed one-step propagator,

$$U_{k+1} = \exp\Big[-i\Delta t\,\big(\hat{H}(t_k) + \delta\hat{H}(t_k)\big)\Big]\,U_k, \tag{4.58}$$

where $\delta\hat{H}(t_k)$ collects the Gaussian perturbations induced by the noisy realization of the control parameters and of η. The overall noise strength is parametrized by a standard deviation σ_{noise}, which is varied systematically in the robustness study below. In this way, robustness is not inferred from a single perturbed trajectory, but from a family of stochastic control channels of increasing severity.

A further methodological distinction must be made between *training noise* and *evaluation noise*. For the Adam baseline and the nominal TRPO controller, optimization is performed in the deterministic environment, and robustness is assessed only *a posteriori* through noisy Monte Carlo rollouts. By contrast, for the noise-optimized TRPO agent, the same Gaussian perturbation model is already activated during training, so that the policy is learned in a stochastic environment rather than in a nominal one. This distinction is central for the interpretation of the results: if the noise-optimized TRPO controller later exhibits both higher average fidelity and lower fidelity variance under noisy evaluation, then the improvement cannot be attributed to reinforcement learning in general, but specifically to the fact that robustness has been learned from exposure to the stochastic environment itself [**?**].

The noisy performance of a given controller is then characterized statistically by repeated sampling. For each value of σ_{noise}, a fixed optimized control plan is executed over many independent noisy realizations, and the resulting empirical distribution of final fidelities is used to estimate both the average fidelity and the fidelity variance. In other words, the robustness metrics reported in the next paragraphs are not single-shot quantities, but Monte Carlo estimates over an ensemble of noisy trajectories,

$$\overline{F}(\sigma_{\text{noise}}) = \mathbb{E}_{\delta\boldsymbol{u}\sim\mathcal{N}(0,\sigma_{\text{noise}})}\left[F\big(U(\boldsymbol{u}+\delta\boldsymbol{u})\big)\right], \tag{4.59}$$

$$\overline{F}(\sigma_{\text{noise}}) = \mathbb{E}_{\delta\boldsymbol{u},\,\delta\eta}\left[F\big(U(\boldsymbol{u}+\delta\boldsymbol{u};\,\eta+\delta\eta)\big)\right], \tag{4.60}$$

Placeholder

Figure 4.11: Average fidelity $\overline{F}(\sigma_{\text{noise}})$ as a function of the Gaussian noise strength for the target gate $N(2.2, 2.2, \pi/2)$, comparing the Adam baseline, the nominal TRPO controller, and the noise-optimized TRPO controller. The main panel should display the full range $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz. An inset is recommended to zoom into the low-noise regime (for example $\sigma_{\text{noise}} \in [0.1, 1.2]$ MHz), where all methods remain close to unit fidelity and the relative ordering is less visible on the full scale. If a smoothing procedure is used for visual readability, the caption should explicitly state that smoothing is purely graphical and that the underlying robustness metrics are computed from Monte Carlo samples as defined in Eqs. (4.60)–(4.61).

and

$$\mathrm{Var}(F)(\sigma_{\text{noise}}) = \mathbb{E}_{\delta \boldsymbol{u}} \left[\left(F - \overline{F} \right)^2 \right]. \tag{4.61}$$

This evaluation protocol is particularly well suited to the present thesis, because it directly probes the central claim of the UFO framework: a useful control strategy should not only achieve high nominal fidelity, but should remain accurate and stable under realistic stochastic perturbations of the multilevel superconducting-qubit dynamics.

4.4.2 Average fidelity versus noise strength.

The main robustness comparison is reported in Fig. 4.11, where the average fidelity $\overline{F}(\sigma_{\text{noise}})$ is plotted as a function of the noise strength for the three control strategies considered in this study: the Adam baseline, the nominal TRPO controller, and the TRPO controller trained directly in the stochastic environment. Several observations emerge clearly.

First, all three controllers exhibit the expected monotonic degradation as the noise amplitude increases, which is consistent with the stochastic control model introduced in Sec. 4.4.1: stronger Gaussian perturbations induce a progressively larger mismatch between the intended control channel and the realized one, and therefore reduce the average fidelity of the implemented gate. This behaviour is fully aligned with the robustness logic of the original UFO framework, where the relevant question is not whether fidelity decreases under noise (which it must), but rather how slowly this degradation occurs for different training strategies [**?**]. The present curves therefore provide a direct operational measure of robustness in the sense of Eqs. (4.60) and (4.61).

Second, the noise-optimized TRPO controller defines the upper envelope of the comparison across essentially the entire explored range of σ_{noise}. In other words, once the perturbation level moves away from the near-deterministic regime, the controller trained under stochastic actuation preserves a larger average fidelity than both the nominal TRPO solution and the Adam baseline. This is the central empirical result of the subsection. It shows that exposure to the noisy environment during training does not merely produce a controller that works well at the single training point $\sigma_{\text{train}} = 1$ MHz, but rather yields a control policy whose performance

Placeholder

Figure 4.12: Fidelity variance as a function of the Gaussian noise strength for the target gate $N(2.2, 2.2, \pi/2)$, comparing the Adam baseline, the nominal TRPO controller, and the noise-optimized TRPO controller. For this figure, a logarithmic scale on the vertical axis is recommended, since the most informative differences appear in the low-variance regime and then widen progressively as σ_{noise} increases. If a smoothed curve is used for readability, the unsmoothed Monte Carlo estimates should remain visible in a faint form or be reported in the appendix. Each point is obtained from repeated noisy realizations of the same optimized control plan, consistently with the robustness protocol of Sec. 4.4.1.

degrades more gracefully over the broader noise family considered in evaluation, exactly in the spirit of the generalization claim advanced in the original paper [**?**].

Third, the low-noise end of the plot is already informative. The three methods start from different nominal operating points: the noise-optimized TRPO controller has the highest nominal fidelity (0.9987648850 at 90 ns), the nominal TRPO controller follows (0.9941508786 at 116 ns), while the Adam baseline starts lower (0.9820504069 at 60 ns) This ordering matters for the interpretation of the robustness curves, because the superiority of the noise-optimized controller under finite σ_{noise} is not obtained by sacrificing nominal quality. On the contrary, in this experiment the same controller that is most resilient under stochastic perturbations is also the one with the highest nominal fidelity among the three compared solutions

Fourth, the separation between the three curves becomes more pronounced as the noise strength increases. This widening gap is more important than the absolute numerical values at any single point: it indicates that the difference between the methods is not a constant offset inherited from the deterministic optimum, but a genuine difference in sensitivity to control perturbations. In particular, the nominal TRPO controller deteriorates faster than the noise-optimized one, while the Adam baseline remains consistently below the latter and, in the intermediate-to-high noise region, does not recover the same robustness trend. This is qualitatively consistent with the evidence reported in the original UFO study, where RL training under stochastic actuation produced the best average-fidelity curve across the tested Gaussian-noise family, outperforming both noise-free RL training and the gradient-based baseline [**?**].

Finally, from the perspective of the thesis narrative, Fig. 4.11 should be read together with the previous nominal optimization results rather than in isolation. The earlier paragraphs established that TRPO is already competitive as a nominal optimizer for the representative gate $N(2.2, 2.2, \pi/2)$. The present figure sharpens that conclusion: once robustness becomes the target quantity, the decisive advantage is no longer TRPO *per se*, but TRPO trained in the *correct environment*. That is, the crucial ingredient is not simply policy-gradient optimization, but policy-gradient optimization under explicit stochastic exposure. This is precisely the mechanism that the UFO framework claims to exploit, and the average-fidelity curves provide the first direct validation of that claim in the present implementation [**?**].

4.4.3 Fidelity variance versus noise strength

A complementary perspective on robustness is provided by Fig. 4.12, which reports the fidelity variance as a function of the noise strength. While the average-fidelity curve quantifies the mean degradation induced by stochastic control errors, the variance probes a different and equally important aspect, namely the *shot-to-shot stability* of the control strategy. A controller may retain a relatively good average fidelity and yet remain operationally fragile if its performance fluctuates strongly across different noisy realizations. In this sense, the fidelity variance measures the sensitivity of the optimized control to the Gaussian perturbations introduced in Sec. 4.4.1, and therefore provides a natural statistical indicator of robustness under repeated experimental execution [**?**]. This interpretation is fully consistent with the evaluation protocol adopted here, where the variance is estimated from repeated Monte Carlo realizations of the same optimized control plan at fixed σ_{noise}.

The first qualitative feature of the figure is that the fidelity variance increases for all three methods as the noise strength grows. This is expected: as the Gaussian perturbations become larger, the distribution of implemented noisy channels broadens, and different realizations of the same nominal pulse generate increasingly dispersed final fidelities. However, the three methods do not broaden at the same rate. The noise-optimized TRPO controller consistently exhibits the lowest variance across the explored range, the Adam baseline occupies an intermediate position, and the nominal TRPO controller displays the largest fluctuations, with a visibly steeper growth in the moderate-to-high noise regime. The ordering is therefore fully consistent with the one observed in the average-fidelity plot: the controller trained in the stochastic environment is not only the one with the best mean noisy performance, but also the one whose performance is least dispersed across repeated noisy realizations.

This result is particularly important for the interpretation of the framework. The comparison between the two TRPO agents is especially clean, since the nominal and noise-optimized controllers share the same policy-gradient machinery, the same physical model, and the same UFO objective, and differ primarily in the environment used during training. The lower variance of the noise-optimized policy therefore cannot be attributed to reinforcement learning in general, but must be understood as a consequence of stochastic training itself. In other words, exposure to noisy dynamics during optimization does not simply improve the mean fidelity under perturbations; it also produces a control policy whose final performance is statistically more stable from one noisy realization to another. This is precisely the type of effect one would expect if the policy had internalized control strategies that are intrinsically less sensitive to random actuation errors.

The comparison with the Adam baseline is also instructive. Although Adam remains clearly more stable than the nominal TRPO controller, it is still systematically outperformed by the noise-optimized TRPO agent in variance across the explored noise range. This means that the advantage of the stochastic RL controller is not limited to average performance alone. Rather, the same method that yields the highest $\overline{F}(\sigma_{\text{noise}})$ also suppresses the growth of shot-to-shot fluctuations most effectively. From the viewpoint of practical analog quantum control, this is a particularly strong robustness signal, because it shows that the learned controller is not only more accurate on average, but also more reproducible under repeated noisy implementations.

Taken together with Fig. 4.11, the variance curve therefore strengthens the main conclusion of this subsection. The noise-optimized TRPO controller occupies the most favorable overall robustness regime among the methods considered here: it preserves the highest average fidelity while simultaneously maintaining the lowest fidelity variance. The nominal TRPO controller, by contrast, is inferior in both metrics, which shows that policy optimization alone is not sufficient to guarantee robust behavior. What matters is training the policy in the appropriate stochastic environment. In this sense, the variance analysis provides direct additional evidence for the central claim of the UFO framework, namely that robustness against stochastic control errors is not merely evaluated after training, but can be actively learned through exposure to the relevant

noise model during optimization [**?**].

4.4.4 Interpretation

Taken together, the average-fidelity and fidelity-variance curves provide the clearest empirical validation of the central robustness claim of the present study. The key question is not whether all controllers degrade under stochastic control perturbations, which is unavoidable, but whether training in a noisy environment produces a control strategy that degrades *more gracefully* than one trained only in the nominal setting. From this viewpoint, the most informative comparison is the one between the two TRPO agents. Since the nominal and noise-optimized controllers share the same policy architecture, the same physical model, and the same UFO objective, their different behavior under noisy evaluation must be attributed primarily to the environment used during training. The fact that the noise-optimized policy achieves both higher average fidelity and lower fidelity variance across the explored range of σ_{noise} therefore shows that robustness is not a generic byproduct of reinforcement learning alone, but a property that is effectively learned through stochastic training [**?**].

This point is especially important in light of the nominal single-target comparison discussed earlier. In the deterministic setting, TRPO was shown to be competitive but not superior to the Adam baseline on the representative gate $N(2.2, 2.2, \pi/2)$. In that regime, the use of reinforcement learning by itself did not guarantee the best pulse. The robustness analysis changes this picture in a qualitatively meaningful way. Once performance is evaluated over an ensemble of noisy realizations rather than on a single nominal rollout, the controller trained in the stochastic environment becomes the strongest overall solution: it preserves the highest average fidelity while also exhibiting the smallest shot-to-shot fluctuations. This is precisely the setting in which the RL formulation becomes most compelling, since the policy is no longer judged only by how well it fits one deterministic control trajectory, but by how reliably it performs under a physically relevant family of perturbations.

The comparison with the Adam baseline reinforces this interpretation. Adam remains a meaningful and nontrivial reference, and its performance confirms that direct gradient-based optimization can still produce strong controls in this multilevel setting. However, the present results indicate that the best robust operating point is no longer attained by the direct optimizer once stochastic perturbations are taken into account. Instead, the noise-optimized TRPO controller dominates Adam in both of the robustness metrics considered here: it maintains a higher mean fidelity under noise and, at the same time, suppresses the variance more effectively. This is a particularly strong result, because it shows that the stochastic RL controller does not improve one aspect of robustness at the expense of another. Rather, it improves both the *accuracy* and the *stability* of the noisy implementation.

From a control-theoretic perspective, this suggests that the policy trained in the noisy environment has internalized pulse strategies that are intrinsically less sensitive to random perturbations of the control channels and of the anharmonicity-related Hamiltonian parameter. In other words, the learned policy appears to avoid control trajectories that are excessively fragile to small stochastic deviations, and instead converges toward solutions that remain effective across a broader ensemble of realizations. This interpretation is fully consistent with the original logic of the UFO framework, where robustness is not treated as a post hoc diagnostic, but as a property to be incorporated directly into the optimization environment [**?**].

A further important consequence is that robustness should be understood here as a genuinely multidimensional notion. The average fidelity and the fidelity variance do not probe the same aspect of performance, and neither of them is sufficient on its own. A practically useful controller should remain both accurate *and* statistically stable under repeated noisy executions. The present results show that the noise-optimized TRPO controller is the only method among the three considered that consistently occupies the most favorable region with respect to both criteria. For this reason, the evidence presented in this subsection supports a stronger conclusion than a

simple ranking of optimizers: it supports the claim that exposure to the appropriate stochastic environment during training changes the qualitative character of the learned control policy.

For the broader structure of the chapter, this subsection therefore plays a decisive interpretive role. The single-target nominal benchmark established that all methods can produce meaningful high-quality analog controls, but did not by itself justify the additional machinery of noisy-environment reinforcement learning. The robustness study does. It shows that, once the control problem is evaluated in the physically relevant regime of stochastic actuation errors, the decisive advantage of the UFO-inspired RL framework lies not merely in pulse optimization, but in the ability to *learn robustness* directly from the training environment. In this sense, the present reproduction supports one of the main scientific messages of the original paper: robustness against stochastic control errors is not an accidental emergent feature of analog control, but a trainable property that can be significantly improved by embedding the relevant perturbation model into the optimization loop [**?**].

4.5 Runtime analysis on the gate family $N(\alpha, \alpha, \pi/2)$

4.5.1 Runtime sweep protocol

To study hardware efficiency beyond the single-target setting, the analysis is extended from the representative gate $N(2.2, 2.2, \pi/2)$ to the full one-parameter family

$$U_{\text{target}}(\alpha) = N(\alpha, \alpha, \pi/2), \qquad \alpha \in [0, \pi]. \tag{4.62}$$

The purpose of this subsection is therefore different from that of the robustness study in section 4.4.1–4.4.6. There, the central question was whether stochastic training improves the stability of the learned controller under noisy perturbations. Here, by contrast, the relevant question is whether direct analog control can realize different gates in the family $N(\alpha, \alpha, \pi/2)$ with runtimes that are competitive with, or even shorter than, those associated with standard discrete gate synthesis. This makes the present sweep a family-level efficiency test rather than a single-target robustness benchmark, in direct continuity with one of the two main empirical claims of the original UFO paper [**?**].

The primary runtime experiment is implemented as a *curriculum TRPO sweep*. More precisely, for fixed $\gamma = \pi/2$, the controller is trained sequentially over increasing values of α, using the best checkpoint obtained at one value of α as the initialization for the next one. In the present code, the parameter grid is generated from $\alpha_{\min} = 0$ to $\alpha_{\max} = \pi$ with step size $\Delta\alpha = 0.1$, while each phase is trained for at most 100 TRPO iterations, with 20,000 episodes per batch, timestep $\Delta t = 2\,\text{ns}$, maximum allowed duration $T_{\max} = 600\,\text{ns}$, and advance criterion based primarily on an evaluation-cost threshold of 0.15. Operationally, this means that the runtime sweep is not performed by retraining each target completely from scratch, but by exploiting continuity in target space: nearby gates in the family are assumed to admit partially transferable control representations, so that the policy learned for one α provides a useful warm start for the next one. This choice is strongly aligned with the policy-based logic of the framework, since TRPO is specifically designed to perform stable updates of continuous stochastic policies under a KL-constrained trust-region step [**?**].

Alongside this curriculum-based RL sweep, a second experiment is carried out with the Adam baseline in order to probe the same gate family without transfer learning. In that case, each target is optimized independently from scratch, and the best runtime is selected through a horizon sweep over a discrete set of candidate durations. In the implementation used here, the tested horizons are

$$T \in \{60, 90, 120, 150, 180, 210, 240\}\,\text{ns}, \tag{4.63}$$

with 400 Adam iterations per horizon. The role of this second sweep is methodologically useful: it separates the effect of analog controllability across the gate family from the additional benefit

Placeholder

that may arise from curriculum transfer. In other words, the Adam sweep provides a "from" family-level baseline for how homogeneous or inhomogeneous the analog optimization problem already is, whereas the curriculum TRPO sweep tests whether neighboring gates can be traversed more efficiently by reusing previously learned control structure.

The final ingredient of the protocol is the comparison with the optimal discrete gate-synthesis baseline discussed earlier in Sec. 3.4.2. Following the benchmark logic of the reference paper, the relevant reference time is the estimated synthesized runtime

$$T_{\mathrm{syn}} \approx 215\,\mathrm{ns}, \tag{4.64}$$

obtained from the optimal decomposition of $N(\alpha, \alpha, \gamma)$ into single-qubit rotations and entangling gates, together with state-of-the-art gate-time estimates for those primitives [**?**]. The runtime curves reported below should therefore be interpreted against this fixed hardware-efficiency reference: for each α, the central quantity of interest is whether the optimized analog controller lies below, near, or above T_{syn}. This makes the sweep directly relevant to the broader physical question motivating the whole section, namely whether direct analog control on the gmon can exploit the native interaction structure of the device to outperform standard compiled implementations over a nontrivial family of two-qubit targets.

4.5.2 Runtime curve

Figure **??** reveals that the runtime profile along the family $N(\alpha, \alpha, \pi/2)$ is highly structured rather than monotonic. Instead of a smooth increase or decrease with α, the curriculum sweep exhibits a sequence of distinct regimes: an initial bootstrap region at very small α, a broad central basin of comparatively fast solutions, a sharp minimum at $\alpha \approx \pi/2$, and a final degradation region as α approaches π. This already suggests that the time required to implement the target gate is strongly dependent on how favorably the target aligns with the native controllable dynamics of the gmon system, rather than being set by a uniform compilation overhead across the whole family.

A first important point is that the $\alpha = 0$ phase should be interpreted separately from the rest of the curve. In the present curriculum protocol, this is the only point that is not warm-started from a previous nearby target, and therefore it acts as a genuine cold-start initialization of the whole sweep. Consistently with that role, it is also the weakest point of the experiment: the final deterministic fidelity remains only $F \approx 0.558$, the runtime saturates at 180 ns, the leakage is still 4.8×10^{-3}, and the corresponding $\sigma = 1\,\mathrm{MHz}$ average fidelity is only about 0.613. For this reason, $\alpha = 0$ is operationally informative, but it should not be overinterpreted as representative of the family-level runtime trend. The more meaningful structure of the curve emerges once the curriculum has entered its transferable regime, starting already from $\alpha \approx 0.157$.

Beyond this bootstrap point, the curve shows that the family is not uniformly difficult. A first reasonably good region appears already for $\alpha \approx 0.157$ and 0.314, where the controller

reaches runtimes of 142 ns and 140 ns, with nominal fidelities above 0.988 and robust average fidelities around 0.951 and 0.953, respectively. This indicates that the curriculum begins to exploit cross-target continuity quite early. However, the next interval, roughly $\alpha \in [0.47, 0.79]$, becomes harder again: the final runtime returns to the maximum observed value of 180 ns, while the final fidelities drop to the 0.948–0.954 range and the $\sigma = 1$ MHz average fidelity falls to about 0.906–0.918. This is not merely a runtime effect: the same region also exhibits visibly larger leakage and boundary-cost levels than the fast central basin, which suggests that the controller is using less clean pulse shapes and paying a higher dynamical price to reach the target.

The most striking feature of the figure is the central runtime valley around $\alpha \in [1.1, 2.4]$, where the controller becomes substantially faster while maintaining very high quality. In this interval, several targets are solved in the 76–140 ns range, and the best part of the basin lies roughly between $\alpha \approx 1.884$ and $\alpha \approx 2.355$, where the observed runtimes are 108 ns, 106 ns, 110 ns, and 118 ns, all with nominal fidelity above 0.982 and robust average fidelity at $\sigma = 1$ MHz still around 0.954–0.959. This is a genuinely important result: the runtime reduction is not obtained by sacrificing accuracy, since the same part of the curve also remains in the high-fidelity and high-robustness regime. The curve therefore indicates a broad family region in which fast analog implementations are not isolated accidents, but repeatable solutions.

At the center of the family, the point $\alpha = \pi/2$ stands out as an extreme minimum. Here the controller terminates after only 2 ns, with essentially unit nominal fidelity ($F \approx 0.999999$), negligible leakage ($\sim 4 \times 10^{-10}$), extremely small boundary cost, and average noisy fidelity still equal to 0.9994 at $\sigma = 1$ MHz. In addition, the corresponding phase requires only a single curriculum iteration, making it not only the fastest gate in the sweep but also the easiest one to learn. This is not a generic numerical fluctuation: it is a singular point in the empirical landscape, and the sharpness of the dip strongly suggests a special dynamical compatibility between the target gate at $\alpha = \pi/2$ and the native interaction structure of the device. The physical interpretation of this point will be discussed in detail in Paragraph 4, but already at the level of the curve it is clear that the runtime landscape contains a privileged native-alignment point rather than a smooth continuum of equally hard targets.

After this central minimum, the runtime rises again but not immediately catastrophically. The interval $\alpha \in [1.727, 2.669]$ still forms a favorable region: runtimes remain between 124 ns and 144 ns, fidelities stay in the 0.982–0.992 range, and robustness remains high. Only when the sweep enters the final high-α edge, roughly $\alpha \gtrsim 2.826$, does the curve degrade more clearly. In that last region, the runtime saturates again at 180 ns, while nominal fidelity falls to 0.922–0.940 and robust average fidelity to 0.887–0.902. This suggests that the favorable transfer structure exploited by the curriculum over the central basin becomes progressively weaker as the target is pushed toward the far end of the family.

A further nontrivial insight emerges when the final runtime is compared with the *minimum-cost time* reached within the same phase. For many of the well-converged points, the two coincide, which indicates that the controller not only discovers a good time scale but also stabilizes it as the final solution. However, in several difficult regions this coincidence breaks down. For example, around $\alpha \approx 0.471$ and 0.628, the final runtime is 180 ns even though the lowest-cost intermediate trajectory is reached already around 142–146 ns; similarly, near $\alpha \approx 2.826$, 2.983, and π, the minimum-cost times lie around 152–158 ns while the final rollout still ends at 180 ns. This indicates that in the hard regions the agent can transiently visit faster candidate solutions, but does not consistently stabilize them under the full optimization criterion. In other words, the difficult parts of the runtime curve are not simply regions where short solutions do not exist at all; rather, they are regions where short solutions appear to be less stable, less robust, or harder to preserve throughout training.

The same reading is supported by the auxiliary diagnostics shown in Figs. **??** and **??**. The fastest phases are also the ones with the cleanest constraint profile: lower leakage, lower boundary cost, and, in general, better agreement between nominal and noisy fidelity. Conversely, the

slowest phases are precisely those where leakage and boundary penalties rise again. This coupling between runtime and constraint satisfaction is crucial, because it shows that the runtime curve is not an isolated geometric observable; it is embedded in the same control trade-off already discussed for the single-target case. Short runtimes are achieved when the controller can simultaneously realize the target, suppress leakage, and keep the pulse endpoints under control. Long runtimes reappear when one or more of these constraints become harder to satisfy at the same time.

Overall, the runtime curve therefore conveys a richer message than a simple pointwise comparison of gate durations. It shows that the family $N(\alpha, \alpha, \pi/2)$ contains a broad favorable analog-control basin, centered around a particularly privileged point at $\alpha = \pi/2$, but also flanked by regions where the controller either needs longer trajectories or fails to stabilize the shorter ones it transiently discovers. This makes the curve especially informative for the rest of the subsection: it is not only a record of achieved runtimes, but a map of where the analog-control landscape is intrinsically easy, where it is fragile, and where the curriculum transfer mechanism is most and least effective.

4.5.3 Comparison with gate synthesis

A central question raised by the runtime sweep is whether the analog-control solutions found by the curriculum-trained TRPO agent are merely competitive with standard discrete compilation, or whether they actually outperform it in a meaningful way. To answer this, the natural reference is the optimal gate-synthesis decomposition of the family $N(\alpha, \alpha, \gamma)$, discussed in the original UFO paper on the basis of the Vatan–Williams two-qubit synthesis result [**?**, **?**]. In that benchmark, the target unitary is decomposed into a minimal-depth circuit over arbitrary one-qubit rotations and entangling gates, yielding an estimated runtime of approximately 215 ns for the $N(\alpha, \alpha, \gamma)$ family under the hardware timing assumptions adopted in [**?**]. Although this is not a pulse-level optimization in the exact same analog-control space used in the present work, it is still the correct compilation baseline for assessing whether direct continuous control provides a genuine temporal advantage over standard discrete synthesis.

Under that reference, the present runtime sweep shows that the analog-control strategy is not just occasionally better: across essentially the entire nontrivial family $N(\alpha, \alpha, \pi/2)$, it remains below the gate-synthesis benchmark. Even the difficult edge points of the sweep, where the learned controller saturates at 180 ns, still remain faster than the synthesis estimate by about 35 ns, i.e. by roughly 16%. This already matters, because those are precisely the hardest regions of the landscape, where nominal fidelity and noisy fidelity both deteriorate and where the curriculum no longer finds the clean short-time solutions available in the central basin. In other words, the analog approach preserves a runtime advantage even where the optimization problem becomes visibly more difficult.

The advantage becomes much more substantial in the broad central part of the family. Around $\alpha \in [1.73, 2.36]$, the final learned runtimes lie between 106 ns and 124 ns, which corresponds to a reduction of about 42% to 51% relative to the 215 ns gate-synthesis reference. In this regime, the speedup is therefore not marginal but systematic. Importantly, this reduction does not come with a collapse in control quality: the same points remain in the high-fidelity regime ($F \approx 0.982$–0.986) and maintain robust average fidelity near 0.955–0.959 at $\sigma = 1\,\mathrm{MHz}$. The comparison is therefore qualitatively stronger than a bare runtime statement: for a substantial subset of the gate family, analog control is simultaneously faster, accurate, and reasonably robust, rather than merely trading correctness for speed.

An even stronger discrepancy appears at selected favorable points of the landscape. For instance, at $\alpha \approx 1.10$ the learned runtime drops to 76 ns, corresponding to a speedup of almost a factor 3 over the synthesis baseline, while still retaining nominal fidelity above 0.98 and robust average fidelity above 0.96. The singular point $\alpha = \pi/2$ is more extreme still: the controller terminates after 2 ns, with essentially unit nominal fidelity and negligible leakage. This value

should not be treated as representative of the entire family, because it reflects a very special alignment between target and native Hamiltonian structure; nevertheless, it is conceptually important. It demonstrates that, at least for some targets, the compilation overhead implicit in discrete gate synthesis is not a fundamental lower bound on execution time. Rather, it is an artifact of expressing the target through a universal discrete gate alphabet instead of exploiting the analog structure of the hardware directly.

This distinction is crucial for interpreting the result correctly. The discrete synthesis benchmark is "optimal" within a circuit model based on a prescribed elementary gate library, whereas the analog controller optimizes directly in the continuous Hamiltonian-control space. The two are therefore solving related but not identical optimization problems. For this reason, the present comparison should not be read as a statement that the analog controller has beaten an absolute quantum speed limit. What it does show, however, is that once the hardware admits a rich native interaction and sufficiently flexible pulse shaping, the shortest experimentally meaningful implementation of a target gate need not coincide with its shortest decomposition over a discrete universal gate set. From the point of view of near-term hardware efficiency, this is precisely the relevant notion of improvement.

The comparison also clarifies where the gain is modest and where it is dramatic. In the hard regions near the two edges of the sweep, the analog controller is only moderately faster than gate synthesis, and the runtime gap narrows precisely when leakage, boundary penalties, and robustness degradation become more pronounced. In the central basin, by contrast, the gap widens sharply. This indicates that the benefit of analog control is strongly target-dependent: it is largest when the target gate can be realized in a way that is geometrically and dynamically aligned with the native couplings of the physical device, and smallest when extra control resources are needed to compensate for a less favorable target structure. Thus, the runtime comparison is not only a numerical benchmark, but also a diagnostic of how much compilation overhead can be removed by working directly at the Hamiltonian level.

Overall, the results support the same qualitative conclusion reported in the original UFO work, but in a more fine-grained family-resolved form. The analog-control solutions found here are broadly faster than the standard gate-synthesis reference, often by a large margin, and the strongest gains occur exactly in the regions where the control landscape appears most compatible with the underlying gmon interaction. This confirms that the runtime advantage of analog control is real, not anecdotal: for this gate family, direct pulse-level optimization can indeed surpass standard discrete synthesis as a practically relevant notion of implementation efficiency.

4.5.4 Physical interpretation

The runtime landscape observed in Fig. **??** is physically meaningful, and should not be read as a purely numerical peculiarity of the optimizer. The key point is that the target family considered here, $N(\alpha, \alpha, \pi/2)$, is not arbitrary with respect to the underlying gmon Hamiltonian. As discussed in the system-model section, the native entangling interaction is generated primarily by the exchange-like coupling term $g(t)\,(\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2)$, whose effective action on the computational subspace is naturally aligned with the $XX{+}YY$ sector of the two-qubit interaction. Consequently, when the desired gate requires a balance of nonlocal content that can be synthesized largely along this native direction, the controller does not need to "fight" the hardware: it can exploit the intrinsic entangling channel directly, while using the remaining controls mainly to shape phases, suppress leakage, and satisfy the boundary constraints. In precisely this regime, shorter runtimes are not only empirically observed, but physically expected. This is fully consistent with the original motivation of the UFO framework, which emphasizes that analog control becomes especially advantageous when the target gate is geometrically close to the entangling structure naturally generated by the device Hamiltonian [**?**].

This interpretation also explains why the best-performing region of the sweep is concentrated in the middle of the α-interval rather than at its boundaries. In the central basin, the optimizer

can realize the required nonlocal action with relatively modest auxiliary control effort: the pulse does not need large corrective excursions in the local channels, the boundary term remains moderate, and the TSWT leakage proxy stays under control. From a dynamical viewpoint, the gate can be implemented by letting the system evolve along a direction that is already available in hardware, rather than by repeatedly compensating unwanted motion through more elaborate control shaping. The shorter runtimes found in this region are therefore plausibly associated with smoother and more "natural" trajectories in control space. This interpretation is reinforced by the fact that the same α-window also displays strong nominal fidelities and good noisy fidelities at $\sigma = 1\,\mathrm{MHz}$: the region that is faster is also the region that is easier to control reliably. In that sense, temporal efficiency and robustness are not independent observations here, but two manifestations of the same underlying alignment between target geometry and native device physics.

By contrast, the growth of runtime toward the two edges of the family is also physically interpretable. As α moves away from the favorable central window, the desired gate becomes less directly compatible with the entangling evolution that the gmon hardware provides most naturally. The controller must then rely more heavily on additional detuning and microwave components to steer the dynamics toward the target while simultaneously suppressing population transfer outside the computational subspace. This extra compensation effort has two immediate consequences. First, it tends to lengthen the control sequence, because more time is needed to accumulate the correct effective nonlocal action without violating the leakage and boundary penalties. Second, it makes the solution more fragile, since longer and more structured pulses expose the system for a greater duration to both coherent modeling imperfections and stochastic perturbations. The qualitative correlation observed across the sweep (higher runtimes, slightly worse robust fidelities, and reduced margin to the gate-synthesis baseline near the edges) is therefore exactly what one would expect if those target gates were dynamically less aligned with the native Hamiltonian.

The special behavior near $\alpha = \pi/2$ should be interpreted in the same spirit, but with additional caution. The extremely short runtime found there suggests that this point may correspond to a particularly favorable symmetry or local-equivalence configuration within the gate family, so that the desired transformation can be reached with very little pulse duration once the controller is placed in the appropriate dynamical sector. This does not mean that $2\,\mathrm{ns}$ should be taken as a generic realizability statement for arbitrary two-qubit gates, nor that the analog controller has saturated a universal quantum speed limit. Rather, it indicates that within a restricted family and on a hardware platform with a strongly structured native interaction, isolated gates may exist whose implementation cost collapses because the target is exceptionally well matched to the underlying entangling generator. For the purposes of the present thesis, the important conclusion is therefore not the extremal value itself, but the broader structural pattern around it: the runtime minima occur where physical compatibility is highest.

Taken together, these observations support the main conceptual message of this subsection. The runtime advantage of analog control over discrete synthesis is largest not in a random subset of the target family, but precisely where the target gate is best aligned with the natural interaction channels of the superconducting device. This makes the result much stronger than a purely algorithmic benchmark. It shows that the improvement is rooted in the physics of the platform: direct pulse-level optimization can remove part of the compilation overhead associated with universal gate decomposition because it exploits the Hamiltonian that the hardware already implements natively. Conversely, when that alignment deteriorates, the runtime advantage shrinks, the controller must work harder to enforce fidelity and leakage constraints, and the solutions become less efficient. The runtime sweep therefore provides a concrete dynamical interpretation of why analog control can outperform standard gate synthesis on this family: it is not merely optimizing faster, but optimizing along the natural directions of the device itself.

4.6 Overall assessment of the validated framework

4.6.1 Summary of empirical evidence

4.6.2 Agreement and discrepancies with the original paper

Chapter 5

Conclusion

5.1 Summary of the thesis

5.2 Limitations and future work